

Solver Specification and Verification Report:

`highOrderManufacturedFDAImplicitJfNK`

Manufactured monodomain-like PDE–ODE solver with high-order finite volumes, Picard, diagonal- I_{ion} , and JFNK nonlinear coupling

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Abstract

This report documents the numerical formulation implemented in `highOrderManufacturedFDAImplicitJfNK`. The solver is a method of manufactured solutions (MMS) analogue of the electro-activation solver: it solves a monodomain-like reaction–diffusion PDE for the transmembrane potential V_m , evolves local state ODEs for (u_1, u_2, u_3) , evaluates a nonlinear ionic-current surrogate $I_{\text{ion}}(V_m, u_1, u_2, u_3)$, and allows the same nonlinear coupling strategies used in the physiological solver: `Picard`, `diagonalIion`, and `JFNK`. The PDE can be discretised with the standard cell-centred finite-volume operator or with a high-order Local Regression Estimators (LRE) reconstruction. The nonlinear source can be evaluated at cell centres or integrated from cell quadrature points, where the state ODEs are also advanced. Time integration is a θ method with `backwardEuler` ($\theta = 1$) or `crankNicolson` ($\theta = 1/2$), and the mass matrix can be `lumped` or high-order `consistent`. The results section is data driven: tables are generated from the current tutorial outputs and the plotted figures can be regenerated with the supplied MATLAB script.

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1 Purpose and notation

The MMS problem is not a physiological ionic model. Its role is to provide an exact solution for a PDE–ODE–source structure that is deliberately close to the electro-activation solver. If the same finite-volume operator, same source quadrature, same state ODE integration, same θ time discretisation, and same nonlinear coupling are used here and in the electro solver, then the observed MMS convergence gives evidence for the numerical order of those shared building blocks.

Symbol	Meaning	Location in solver
$\Omega \subset \mathbb{R}^d$	computational domain, $d = 1, 2, 3$	OpenFOAM mesh
K_i	finite-volume cell	<code>mesh.cells()</code>
$ K_i $	cell volume	<code>mesh.V()</code>
$V_m(\mathbf{x}, t)$	manufactured transmembrane potential	<code>Vm</code>
$\mathbf{u} = (u_1, u_2, u_3)$	manufactured state variables	<code>u1,u2,u3</code>
$I_{\text{ion}}(V_m, \mathbf{u})$	nonlinear source surrogate	<code>Iion</code>
χ and C_m	monodomain scaling constants	<code>cardiacProperties</code>
$\mathbf{D}$	conductivity tensor	<code>conductivity</code>
$\mathbf{M}$	mass matrix	<code>M</code>
$\mathbf{K}$	diffusion/stiffness matrix	<code>K</code>
$\mathbf{L} = (\chi C_m)^{-1} \mathbf{K}$	scaled diffusion operator	<code>L</code>
$\mathbf{A}, \mathbf{B}$	implicit θ -scheme matrices	<code>AImplicit, BImplicit</code>
$\mathbf{F}(\mathbf{x})$	nonlinear residual for voltage candidate $\mathbf{x}$	<code>residualFor</code>

The cell-centred vector of voltages at time t^n is denoted $\mathbf{V}^n \in \mathbb{R}^{N_c}$. A nonlinear iteration inside one time step uses the superscript k , e.g. $\mathbf{V}^k$ is a candidate for $\mathbf{V}^{n+1}$. Quadrature point q in cell K_i is denoted $\mathbf{x}_{iq}$ with positive weight ω_{iq} .

2 Governing equations

2.1 Electrophysiology motivation

The physiological monodomain equation can be written, in one common convention, as

$$\chi C_m \frac{\partial V_m}{\partial t} = \nabla \cdot (\mathbf{D} \nabla V_m) - \chi C_m I_{\text{ion}}(V_m, \mathbf{s}) + I_{\text{ext}}, \quad (1)$$

where $\mathbf{s}$ are ionic states and I_{ion} is often stored as a voltage rate with units V s^{-1} . Dividing by χC_m gives

$$\frac{\partial V_m}{\partial t} = \frac{1}{\chi C_m} \nabla \cdot (\mathbf{D} \nabla V_m) - I_{\text{ion}}(V_m, \mathbf{s}) + \frac{I_{\text{ext}}}{\chi C_m}. \quad (2)$$

The manufactured solver replaces the physiological states by three smooth MMS states and replaces the cell-model ionic current by a constructed nonlinear function. The purpose is not to reproduce cellular electrophysiology, but to exercise the same numerical responsibilities: diffusion, local ODEs, nonlinear source, quadrature, implicit time stepping, and nonlinear iteration.

2.2 Manufactured PDE–ODE system

The solver advances

$$\frac{\partial V_m}{\partial t} = \frac{1}{\chi C_m} \nabla \cdot (\mathbf{D} \nabla V_m) - I_{\text{ion}}(V_m, u_1, u_2, u_3), \quad (3)$$

$$\dot{u}_1 = (u_1 + u_3 - V_m)^2 u_2^2 + \frac{1}{2} (u_1 + u_3 - V_m) u_2^2 (V_m - u_3), \quad (4)$$

$$\dot{u}_2 = -(u_1 + u_3 - V_m) u_2^3, \quad (5)$$

$$\dot{u}_3 = 0. \quad (6)$$

The nonlinear ionic-current surrogate is

$$I_{\text{ion}}(V_m, u_1, u_2, u_3) = -\frac{1}{2} (u_1 + u_3 - V_m) u_2^2 (V_m - u_3) + \frac{\beta}{\chi C_m} (V_m - u_3). \quad (7)$$

The PDE source used in the matrix right-hand side is

$$S(V_m, \mathbf{u}) = -I_{\text{ion}}(V_m, \mathbf{u}). \quad (8)$$

Thus the term called `Iion` in the output is the nonlinear manufactured current, while the term passed into the voltage equation is its negative.

2.3 Exact solution and the role of β

The exact fields are built from two smooth spatial functions F and G :

$$V_m^*(\mathbf{x}, t) = \sqrt{1+t} F(\mathbf{x}), \quad (9)$$

$$u_1^*(\mathbf{x}, t) = (1+t)G(\mathbf{x}) + V_m^*(\mathbf{x}, t), \quad (10)$$

$$u_2^*(\mathbf{x}, t) = \frac{1}{(1+t)\sqrt{G(\mathbf{x})}}, \quad (11)$$

$$u_3^*(\mathbf{x}, t) = 0. \quad (12)$$

The solver uses

$$F = \begin{cases} \cos(\pi x), & d = 1, \\ \cos(\pi x) \cos(2\pi y), & d = 2, \\ \cos(\pi x) \cos(2\pi y) \cos(3\pi z), & d = 3, \end{cases} \quad G = \begin{cases} 1+x, & d = 1, \\ 1+xy^2, & d = 2, \\ 1+xy^2z^3, & d = 3. \end{cases} \quad (13)$$

The constant β closes the analytical balance between the Laplacian of F and the conductivity tensor. For a diagonal constant conductivity sampled from `conductivity`[0],

$$\beta = \begin{cases} -\pi^2 D_{xx}, & d = 1, \\ -\pi^2 (D_{xx} + 4D_{yy}), & d = 2, \\ -\pi^2 (D_{xx} + 4D_{yy} + 9D_{zz}), & d = 3. \end{cases} \quad (14)$$

This choice makes the manufactured source compatible with the exact trigonometric mode under the selected conductivity tensor. Boundary values for V_m , u_1 , u_2 , and u_3 are overwritten with the exact solution where appropriate, so the verification error is a discretisation error rather than a boundary-data error.

3 Spatial discretisation

3.1 Finite-volume integral form

Integrating (3) over a control volume K_i gives

$$\int_{K_i} \frac{\partial V_m}{\partial t} dV = \frac{1}{\chi C_m} \int_{\partial K_i} \mathbf{n} \cdot \mathbf{D} \nabla V_m dA + \int_{K_i} S(V_m, \mathbf{u}) dV. \quad (15)$$

After spatial discretisation the semi-discrete problem has the matrix form

$$\mathbf{M} \frac{d\mathbf{V}}{dt} = \mathbf{L}\mathbf{V} + \mathbf{S}(\mathbf{V}, \mathbf{u}), \quad \mathbf{L} = \frac{1}{\chi C_m} \mathbf{K}. \quad (16)$$

3.2 Standard cell-centred operator

When `useHighOrder_Vm=false`, the solver assembles the standard cell-centred finite-volume diffusion operator. For an internal face f between cells P and N , the orthogonal part is of the form

$$\int_f \mathbf{n} \cdot \mathbf{D} \nabla V_m dA \approx a_f (V_N - V_P), \quad (17)$$

where a_f contains the face area, face-normal direction, conductivity tensor, and distance between cell centres. Boundary faces use the OpenFOAM patch type. `zeroGradient` and `empty` patches give no-flux contributions; fixed value patches contribute known boundary terms.

For smooth solutions on regular orthogonal meshes, this standard finite-volume operator is second-order accurate in space for diffusion. On skewed or non-orthogonal meshes the effective order depends on the mesh quality and the non-orthogonal treatment.

3.3 High-order LRE operator

When `useHighOrder_Vm=true`, the solver uses Local Regression Estimators (LRE) stencils. Around each cell centre $\mathbf{x}_i$, a polynomial reconstruction of degree N is fitted from neighbouring cell values:

$$V_m(\mathbf{x}_i + \mathbf{r}) \approx V_i + \nabla V_i \cdot \mathbf{r} + \frac{1}{2} \mathbf{r}^T H_i \mathbf{r} + \frac{1}{6} T_i[\mathbf{r}, \mathbf{r}, \mathbf{r}] + \dots \quad (18)$$

The dictionary entries `N`, `Nn`, `k`, and `maxStencilSize` control the polynomial degree, target number of stencil neighbours, radial weight exponent, and maximum stencil size. The solver can evaluate reconstructed values, gradients, Hessians, and third-order tensors, depending on the selected polynomial order.

The high-order stiffness matrix is assembled by evaluating reconstructed gradients at face quadrature points:

$$\int_f \mathbf{n} \cdot \mathbf{D} \nabla V_m dA \approx \sum_{q \in f} \omega_{fq} \mathbf{n}_f \cdot \mathbf{D}_f \nabla_h V_m(\mathbf{x}_{fq}). \quad (19)$$

The resulting operator is sparse but has a wider stencil than the standard finite-volume Laplacian. For smooth fields, good stencils, sufficiently accurate quadrature, and non-degenerate meshes, the expected spatial consistency improves with the LRE polynomial degree. In practice the observed order is the minimum of: reconstruction order, quadrature order, time-integration order, ODE tolerance floor, nonlinear residual floor, and mesh/stencil conditioning. If the error is controlled only by the interpolation/reconstruction of V_m , then the expected behaviour is the usual polynomial one: linear reconstruction (p1) gives second-order accuracy, quadratic reconstruction (p2) gives third-order accuracy, and cubic reconstruction (p3) gives fourth-order accuracy.

3.4 Lumped and consistent mass matrices

The lumped mass matrix is diagonal:

$$\mathbf{M}_{ij}^{\text{lump}} = |K_i| \delta_{ij}. \quad (20)$$

It is cheap, robust, positive, and coincides with the usual finite-volume mass for one unknown per cell. If high-order V_m is disabled, the solver uses this diagonal mass even if the input asks for `consistent`, because there is no high-order reconstruction basis to integrate.

The `consistent` mass matrix is available when `useHighOrderVm=true` and is assembled with cell quadrature and the LRE reconstruction weights:

$$\mathbf{M}_{ij}^{\text{cons}} \approx \sum_K \sum_{q \in K} \omega_{Kq} \phi_i(\mathbf{x}_{Kq}) \phi_j(\mathbf{x}_{Kq}), \quad (21)$$

where ϕ_i are the reconstruction-induced basis functions. This better matches the high-order spatial approximation, but it is sparse rather than diagonal and increases the cost and conditioning sensitivity of the implicit solve.

3.5 High-order integration of I_{ion} and states

The source integral can be approximated either by a cell-centred rule or by quadrature. Since $S = -I_{\text{ion}}$, when `useHighOrderIion=false`,

$$S_i = \int_{K_i} S dV = - \int_{K_i} I_{\text{ion}} dV \approx -I_{\text{ion}}(V_i, u_{1,i}, u_{2,i}, u_{3,i}) |K_i|. \quad (22)$$

When `useHighOrderIion=true`, states are stored and advanced at dedicated cell quadrature points and the source integral is approximated as

$$S_i = \int_{K_i} S dV = - \int_{K_i} I_{\text{ion}} dV \approx - \sum_{q \in K_i} \omega_{iq} I_{\text{ion}}(V_{mh}(\mathbf{x}_{iq}), u_{1,iq}, u_{2,iq}, u_{3,iq}). \quad (23)$$

4 Time integration

4.1 The θ method

Let $\Delta t = t^{n+1} - t^n$. The solver maps `backwardEuler` to $\theta = 1$ and `crankNicolson` to $\theta = 1/2$. Applying the θ method to (16) gives

$$\mathbf{M} \frac{\mathbf{V}^{n+1} - \mathbf{V}^n}{\Delta t} = \mathbf{L} (\theta \mathbf{V}^{n+1} + (1 - \theta) \mathbf{V}^n) + \theta \mathbf{S}^{n+1} + (1 - \theta) \mathbf{S}^n. \quad (24)$$

Equivalently,

$$\mathbf{A} \mathbf{V}^{n+1} = \mathbf{B} \mathbf{V}^n + \theta \mathbf{S}^{n+1} + (1 - \theta) \mathbf{S}^n, \quad (25)$$

$$\mathbf{A} = \frac{1}{\Delta t} \mathbf{M} - \theta \mathbf{L}, \quad \mathbf{B} = \frac{1}{\Delta t} \mathbf{M} + (1 - \theta) \mathbf{L}. \quad (26)$$

`backwardEuler` is first order in time and strongly damping. It is usually the most robust choice for stiff or poorly converged nonlinear problems. `crankNicolson` is second order in time for smooth solutions and converged nonlinear/source terms, but is less dissipative. Its observed order can drop if the nonlinear iterations, state ODE tolerance, or spatial error dominate.

4.2 ODE states inside one PDE time step

For a voltage candidate $V_m^{n+1,k}$, the state ODEs are advanced from the same time-step-start states $\mathbf{u}^n$. Inside the ODE integrator, voltage is interpolated linearly between the old and candidate values:

$$V_m(\tau) = (1 - \alpha)V_m^n + \alpha V_m^{n+1,k}, \quad \alpha = \frac{\tau}{\Delta t}, \quad 0 \leq \tau \leq \Delta t. \quad (27)$$

The available state solvers are:

<code>stateODESolver</code>	Method	Nominal order
Euler or <code>forwardEuler</code>	explicit Euler	1
RK4	classical Runge–Kutta	4
RKF45 or <code>RKT45</code>	adaptive embedded Fehlberg pair	4/5

For RKF45, the fifth-order candidate is accepted and the fourth-order embedded estimate supplies the local error indicator

$$e = \max_j \frac{|u_j^{(5)} - u_j^{(4)}|}{\text{stateODEAbsTol} + \text{stateODERelTol} \max(|u_j^{(5)}|, |u_j^{(4)}|)}. \quad (28)$$

A sub-step is accepted if $e \leq 1$ or if the minimum step has been reached; the next sub-step is scaled by a safety factor proportional to $e^{-1/4}$. In convergence studies, the ODE tolerances must be tight enough that the state integration error does not mask the spatial or PDE time-integration order.

5 Nonlinear coupled solve

5.1 Residual and diagnostics

For any voltage candidate $\mathbf{x}$, the solver evaluates the corresponding state fields and source by calling `evaluateNonlinearFields`. This means: reconstruct V_m to I_{ion} quadrature points if required, reset/evolve states from t^n to t^{n+1} using $\mathbf{x}$, compute I_{ion} , and assemble the source. The coupled voltage residual is

$$\mathbf{F}(\mathbf{x}) = \mathbf{A}\mathbf{x} - [\mathbf{B}\mathbf{V}^n + \theta (\mathbf{S}(\mathbf{x}) + \mathbf{b}^{n+1}) + (1 - \theta) (\mathbf{S}^n + \mathbf{b}^n)], \quad (29)$$

where $\mathbf{b}^n$ and $\mathbf{b}^{n+1}$ are known boundary/source contributions from the manufactured boundary treatment.

The solver records several relative diagnostics:

$$R_{\text{cpl}} = \frac{\|\mathbf{F}(\mathbf{x})\|_2}{\|\mathbf{r}(\mathbf{x})\|_2 + \epsilon}, \quad R_{V_m} = \frac{\|\mathbf{V}^{k+1} - \mathbf{V}^k\|_2}{\|\mathbf{V}^k\|_2 + \epsilon}, \quad (30)$$

$$R_{I_{\text{ion}}} = \frac{\|\mathbf{I}_{\text{ion}}^{k+1} - \mathbf{I}_{\text{ion}}^k\|_2}{\|\mathbf{I}_{\text{ion}}^k\|_2 + \epsilon}, \quad R_{u_j} = \frac{\|\mathbf{u}_j^{k+1} - \mathbf{u}_j^k\|_2}{\|\mathbf{u}_j^k\|_2 + \epsilon}. \quad (31)$$

Here ϵ is the OpenFOAM `SMALL` safeguard. When high-order I_{ion} quadrature is active, state residuals are measured at the quadrature points; otherwise they are measured at cell centres. The maximum state residual is

$$R_{\mathbf{u}} = \max(R_{u_1}, R_{u_2}, R_{u_3}). \quad (32)$$

The residual history is written to `postProcessing/highOrderManufacturedFDAImplicitJfNK/nonlinearResiduals.dat` when `writeNonlinearResiduals=true`.

5.2 Picard fixed-point method

Picard treats the nonlinear source as lagged during each linear voltage solve. Given $\mathbf{V}^k$, the solver computes $\mathbf{S}^k = \mathbf{S}(\mathbf{V}^k, \mathbf{u}(\mathbf{V}^k))$ and then solves

$$\mathbf{A}\hat{\mathbf{V}}^{k+1} = \mathbf{B}\mathbf{V}^n + \theta(\mathbf{S}^k + \mathbf{b}^{n+1}) + (1 - \theta)(\mathbf{S}^n + \mathbf{b}^n). \quad (33)$$

The relaxed update is

$$\mathbf{V}^{k+1} = \mathbf{V}^k + \omega \left(\hat{\mathbf{V}}^{k+1} - \mathbf{V}^k \right), \quad \omega = \text{nonlinearRelaxation}. \quad (34)$$

Convergence requires at least `minNonlinearIterations`; in the Picard branch the active criteria are $R_{V_m} \leq \text{nonlinearVmTolerance}$ and all state residuals below `nonlinearStatesTolerance`. If Picard exhausts the iteration budget, the current implementation accepts the last Picard iterate and prints a warning, matching the legacy `highOrderManufacturedFDAImplicit` behaviour.

Algorithm 1 Picard nonlinear coupling

Input: Old solution $\mathbf{V}^n, \mathbf{u}^n$, source $\mathbf{S}^n$

- 1: Build $\mathbf{A}$ and $\mathbf{B}$ from $\mathbf{M}, \mathbf{L}, \theta$, and Δt
- 2: Choose initial guess $\mathbf{V}^0$ from $\mathbf{V}^n$ or extrapolation
- 3: **for** $k = 0, \dots, N_{\text{nl}} - 1$ **do**
- 4: Evaluate states and source $\mathbf{S}^k$ from $\mathbf{V}^k$
- 5: Solve $\mathbf{A}\hat{\mathbf{V}}^{k+1} = \mathbf{r}(\mathbf{S}^k)$
- 6: Relax $\mathbf{V}^{k+1} = \mathbf{V}^k + \omega(\hat{\mathbf{V}}^{k+1} - \mathbf{V}^k)$
- 7: Re-evaluate/update states and source for diagnostics
- 8: Compute $R_{V_m}, R_{u_1}, R_{u_2}, R_{u_3}, R_{I_{\text{ion}}}$
- 9: **if** minimum iterations reached and Picard tolerances hold **then**
- 10: accept and exit loop
- 11: **end if**
- 12: **end for**

Output: Last Picard iterate is accepted; a warning is printed if not converged

5.3 Diagonal- I_{ion} linearisation

`diagonalIion` keeps the same scalar voltage unknowns but adds an approximate local derivative of the nonlinear source to the matrix. A forward finite difference is used:

$$d_i \approx \frac{S_i(\mathbf{V}^k + \varepsilon \mathbf{1}) - S_i(\mathbf{V}^k)}{\varepsilon}, \quad \varepsilon = \text{diagonalIionEpsilon}. \quad (35)$$

The linearised source is

$$S_i(\mathbf{V}^{k+1}) \approx S_i(\mathbf{V}^k) + d_i \left(V_i^{k+1} - V_i^k \right). \quad (36)$$

Moving the unknown part to the left gives

$$(\mathbf{A} - \theta \text{diag}(\mathbf{d})) \hat{\mathbf{V}}^{k+1} = \mathbf{r}(\mathbf{S}^k) - \theta \mathbf{d} \odot \mathbf{V}^k, \quad (37)$$

where $\odot$ is componentwise multiplication and

$$\mathbf{r}(\mathbf{S}^k) = \mathbf{B}\mathbf{V}^n + \theta \left(\mathbf{S}^k + \mathbf{b}^{n+1} \right) + (1 - \theta) (\mathbf{S}^n + \mathbf{b}^n). \quad (38)$$

This is the same right-hand side used by the Picard voltage solve; the `diagonalIion` branch only moves the local linearised source contribution to the left-hand side. This is not a full Newton Jacobian because it ignores off-diagonal source coupling introduced by reconstruction and ignores explicit state sensitivities beyond what the finite-difference source evaluation captures locally. It is often more implicit than Picard while remaining cheaper and simpler than JFNK.

Algorithm 2 diagonalIion nonlinear coupling

Input: Old solution $V^n, \mathbf{u}^n$, source S^n

```
1: Choose initial guess  $V^0$ 
2: for  $k = 0, \dots, N_{\text{nl}} - 1$  do
3:   Evaluate  $S^k$  and  $I_{\text{ion}}^k$  from  $V^k$ 
4:   Perturb  $V_m$  by diagonalIionEpsilon and re-evaluate source
5:   Form  $d_i \approx \partial S_i / \partial V_i$  by finite difference
6:   Solve  $(\mathbf{A} - \theta \text{diag}(\mathbf{d})) \hat{V}^{k+1} = \mathbf{r}(S^k) - \theta \mathbf{d} \odot V^k$ 
7:   Relax the voltage update
8:   Re-evaluate source/states at the relaxed voltage
9:   Check coupled, voltage,  $I_{\text{ion}}$ , and optional state criteria
10:  if minimum iterations reached and diagonal tolerances hold then
11:    accept and exit loop
12:  end if
13: end for
14: if not converged and nonlinearAcceptUnconverged=false then
15:   roll back  $V_m, I_{\text{ion}}$ , and states to the beginning of the time step
16: end if
```

5.4 Jacobian-free Newton–Krylov

JFNK seeks a zero of (29):

$$\mathbf{F}(\mathbf{x}) = 0. \quad (39)$$

Newton’s method would solve

$$\mathbf{J}(\mathbf{x}_k) \delta_k = -\mathbf{F}(\mathbf{x}_k), \quad \mathbf{x}_{k+1} = \mathbf{x}_k + \alpha \delta_k, \quad (40)$$

but the full Jacobian is not assembled. Instead, GMRES only needs products $\mathbf{J}\mathbf{v}$, approximated by the finite difference

$$\mathbf{J}(\mathbf{x})\mathbf{v} \approx \frac{\mathbf{F}(\mathbf{x} + \epsilon \mathbf{v}) - \mathbf{F}(\mathbf{x})}{\epsilon}, \quad \epsilon = \text{JfnkEpsilon} \frac{1 + \|\mathbf{x}\|_2}{\|\mathbf{v}\|_2 + \epsilon_{\text{mach}}}. \quad (41)$$

Each GMRES matrix-vector product therefore performs a full nonlinear residual evaluation: ODE solve, source evaluation, source quadrature if active, and matrix-vector application. This is the main cost of JFNK.

The inner linear solve uses restarted GMRES(m), where $m = \text{JfnkMaxKrylovIterations}$ and the number of restarts is `JfnkMaxRestarts`. The Arnoldi basis is orthogonalised with modified Gram–Schmidt, and the least-squares projected problem is solved with Eigen’s column-pivoted Householder QR. GMRES stops when its relative residual is below `JfnkLinearTolerance` or when the restart budget is exhausted.

5.5 Armijo line search and rollback

After GMRES proposes a Newton correction δ_k , the solver optionally uses Armijo backtracking. Starting from $\alpha = \text{nonlinearRelaxation}$, it accepts the trial step if

$$\|\mathbf{F}(\mathbf{x}_k + \alpha \delta_k)\|_2^2 \leq (1 - 2c\alpha) \|\mathbf{F}(\mathbf{x}_k)\|_2^2, \quad c = \text{JfnkArmijoC}. \quad (42)$$

If the condition fails, α is halved until either the condition is met, `JfnkLineSearchMaxIter` is reached, or `JfnkLineSearchAlphaMin` is reached. This globalises Newton by discouraging steps that increase the nonlinear residual.

For JFNK and `diagonalIion`, if the nonlinear solver reaches the maximum number of iterations without convergence, the default behaviour is a rollback: V_m, I_{ion} , and the state fields are restored to the beginning of the time step. Setting `nonlinearAcceptUnconverged=true` disables this rollback and accepts the last iterate. Picard is the exception: to preserve the legacy manufactured implicit behaviour, an unconverged Picard iteration accepts the last Picard iterate with a warning.

Algorithm 3 Jacobian-free Newton–Krylov nonlinear coupling

Input: Old solution $V^n, \mathbf{u}^n$, old source S^n

```
1: Build initial guess  $\mathbf{x}_0$  from  $V^n$  or extrapolation
2: for  $k = 0, \dots, N_{\text{nl}} - 1$  do
3:   Evaluate  $F(\mathbf{x}_k)$ , states, source, and  $I_{\text{ion}}$ 
4:   if minimum iterations reached and nonlinear tolerances hold then
5:     accept  $\mathbf{x}_k$  and exit loop
6:   end if
7:   Define matrix-free product  $\mathbf{v} \mapsto [F(\mathbf{x}_k + \epsilon \mathbf{v}) - F(\mathbf{x}_k)]/\epsilon$ 
8:   Use restarted GMRES to approximately solve  $\mathbf{J}\delta_k = -F(\mathbf{x}_k)$ 
9:   if jfnkLineSearch=true then
10:    Set  $\alpha \leftarrow \text{nonlinearRelaxation}$ 
11:    while Armijo condition (42) fails and budget remains do
12:       $\alpha \leftarrow \max(\alpha/2, \text{jfnkLineSearchAlphaMin})$ 
13:    end while
14:  else
15:     $\alpha \leftarrow \text{nonlinearRelaxation}$ 
16:  end if
17:  Update  $\mathbf{x}_{k+1} = \mathbf{x}_k + \alpha \delta_k$ 
18:  Re-evaluate states/source at  $\mathbf{x}_{k+1}$  and write residual diagnostics
19:  if minimum iterations reached and nonlinear tolerances hold then
20:    accept  $\mathbf{x}_{k+1}$  and exit loop
21:  end if
22: end for
23: if not converged and nonlinearAcceptUnconverged=false then
24:   roll back  $V_m, I_{\text{ion}}$ , and states to their time-step-start values
25: end if
```

6 Theoretical accuracy and practical error floors

For a smooth manufactured solution and exact nonlinear solves, the intended orders are:

Component	Expected order	Main caveat
standard FV diffusion	second order in space on regular orthogonal meshes	mesh quality
LRE high-order reconstruction	increases with polynomial degree N	stencil quality and quadrature
backwardEuler	first order in time	numerical damping
crankNicolson	second order in time	requires converged nonlinear/source terms
Euler states	first order in time	rarely appropriate for high-order studies
RK4 states	fourth order for smooth ODEs	fixed step over PDE Δt
RKF45 states	adaptive fourth/fifth-order pair	tolerance sets error floor
Picard/JFNK/diagonal	nonlinear iteration, not spatial order by itself	residual tolerance must be small enough

The measured MMS order is the order of the entire chain. It can be limited by the largest of the following errors: spatial reconstruction, mass matrix approximation, source quadrature, PDE time stepping, state ODE integration, linear solver tolerance, nonlinear residual tolerance, and floating-point or conditioning effects. In particular, a high-order spatial method will not show its nominal order unless Δt , ODE tolerances, and nonlinear tolerances are sufficiently tight.

7 Solver controls

The numerical controls are read from `constant/spatialIntegrationProperties` and `constant/timeIntegrationProperties`. The tutorial `run_convergence.py` writes these dictionaries.

Control	Meaning
<code>useHighOrder_Vm</code>	Selects standard FV or LRE high-order diffusion/mass reconstruction for V_m .
<code>useHighOrder_Iion</code>	Selects cell-centred source/state update or quadrature-point source/state update.
<code>implicitScheme</code>	<code>backwardEuler</code> or <code>crankNicolson</code> ; sets θ .
<code>massMatrix</code>	<code>lumped</code> diagonal FV mass or LRE consistent mass.
<code>linearSolver</code>	Eigen sparse solver for non-JFNK voltage systems: <code>BiCGSTAB</code> or <code>SparseLU</code> .
<code>tolerance, maxIterations</code>	Linear PDE solver stopping tolerance and iteration cap.
<code>nonlinearMethod</code>	<code>Picard</code> , <code>diagonalIion</code> , or <code>JFNK</code> .
<code>nonlinearIterations</code>	Maximum nonlinear iterations per PDE time step.
<code>minNonlinearIterations</code>	Minimum nonlinear iterations before convergence can be accepted.
<code>nonlinearTolerance</code>	Coupled residual tolerance for JFNK/diagonal branches.
<code>nonlinearVmTolerance</code>	Relative L_2 tolerance for the voltage update.
<code>nonlinearStatesTolerance</code>	Relative L_2 tolerance for state updates.
<code>nonlinearRequireStatesConvergence</code>	If true, state residuals are part of the JFNK/diagonal stopping test.
<code>nonlinearAcceptUnconverged</code>	If false, JFNK/diagonal roll back on nonlinear failure.
<code>nonlinearIionTolerance</code>	Relative L_2 tolerance for I_{ion} changes.
<code>nonlinearRelaxation</code>	Picard relaxation or initial Newton step length.
<code>jfnkMaxKrylovIterations</code>	GMRES restart length m .
<code>jfnkMaxRestarts</code>	Number of restarted GMRES cycles.
<code>jfnkLinearTolerance</code>	GMRES relative residual tolerance.
<code>jfnkEpsilon</code>	Scaling of the finite-difference Jacobian-vector perturbation.
<code>jfnkInitGuessOrder</code>	Constant or linear extrapolation initial guess.
<code>jfnkInitGuessVmMin/Max</code>	Clamp for extrapolated guess and optionally for ODE input.
<code>jfnkClampODEInput</code>	If true, clamps voltage passed to state ODE evaluations.
<code>jfnkLineSearch,</code> <code>jfnkLineSearchMaxIter,</code> <code>jfnkLineSearchAlphaMin,</code> <code>jfnkArmijoC</code>	Armijo backtracking controls.
<code>diagonalIionEpsilon</code>	Finite-difference perturbation for diagonal source derivative.
<code>stateODESolver</code>	<code>RKF45</code> , <code>RKT45</code> , <code>RK4</code> , or <code>Euler</code> .
<code>stateODEInitialStep,</code> <code>stateODEAbsTol, stateODERelTol,</code> <code>stateODEMaxSteps</code>	Adaptive ODE step and tolerance controls.
<code>writeNonlinearResiduals</code>	Writes per-iteration nonlinear residual diagnostics.

8 Output diagnostics

At the final time, the solver writes cell-centred fields, manufactured exact fields, error fields, reconstructed/quadrature diagnostics when available, and a text summary. The summary includes mesh size, dimension, Δt , time scheme, mass matrix, high-order settings for V_m and I_{ion} , selected nonlinear method, relevant tolerances, ODE controls, nonlinear convergence statistics, wall-clock timings, and peak resident memory. These diagnostics are intended to make each convergence point auditable: the numerical method used to generate a data point is recorded alongside the error data.

9 Error Measures and Convergence Rates

The solver writes cell-centred volume-weighted L_1 , L_2 , and L_∞ errors for V_m , u_1 , and u_2 . For a generic field $q \in \{V_m, u_1, u_2\}$, let q_i be the numerical value stored at the centre of cell K_i , $q^*(\mathbf{x}, t)$ the manufactured exact solution, and

$$e_i(q) = q_i - q^*(\mathbf{C}_i, t), \quad |\Omega_h| = \sum_i |K_i|. \quad (43)$$

The cell-centred errors written by the solver are

$$E_1^C(q) = \frac{1}{|\Omega_h|} \sum_i |K_i| |e_i(q)|, \quad (44)$$

$$E_2^C(q) = \left[\frac{1}{|\Omega_h|} \sum_i |K_i| e_i(q)^2 \right]^{1/2}, \quad (45)$$

$$E_\infty^C(q) = \max_i |e_i(q)|. \quad (46)$$

These are volume-weighted mean/RMS errors, not unnormalised integral norms.

The solver also writes reconstructed quadrature errors, denoted by the superscript G . Let $\mathbf{x}_{iq}$ be a quadrature point in cell K_i and let ω_{iq} be its normalised quadrature weight, so that $m_{iq} = |K_i| \omega_{iq}$ is the physical measure associated with that point. For voltage, q_{iq}^G is the LRE/Taylor reconstructed voltage $V_{m,h}(\mathbf{x}_{iq})$; for the states, when high-order ionic integration is active, q_{iq}^G is the state advanced directly at the ionic quadrature point. The quadrature-point error is

$$e_{iq}^G(q) = q_{iq}^G - q^*(\mathbf{x}_{iq}, t), \quad |\Omega_h|_G = \sum_i \sum_q m_{iq}. \quad (47)$$

The corresponding quadrature errors are

$$E_1^G(q) = \frac{1}{|\Omega_h|_G} \sum_i \sum_q m_{iq} |e_{iq}^G(q)|, \quad (48)$$

$$E_2^G(q) = \left[\frac{1}{|\Omega_h|_G} \sum_i \sum_q m_{iq} (e_{iq}^G(q))^2 \right]^{1/2}, \quad (49)$$

$$E_\infty^G(q) = \max_{i,q} |e_{iq}^G(q)|. \quad (50)$$

If the corresponding high-order diagnostic is inactive, the report maps the cell-centred errors into the same output columns.

For the quadrature diagnostics, the exact-solution RMS norm and the relative percentage error are

$$\|q^*\|_{2,G} = \left[\frac{1}{|\Omega_h|_G} \sum_i \sum_q m_{iq} (q^*(\mathbf{x}_{iq}, t))^2 \right]^{1/2}, \quad E_{\text{rel},2}^G(q) = 100 \frac{E_2^G(q)}{\max(\|q^*\|_{2,G}, \text{SMALL})}. \quad (51)$$

Pairwise rates use consecutive mesh sizes,

$$p_{N_1, N_2} = \frac{\log(E_{N_1}/E_{N_2})}{\log(h_{N_1}/h_{N_2})}, \quad h_N = 1/N. \quad (52)$$

The convergence tables below recompute log-log fits directly from the `errors\_vs\_N` files, using both the full $N = 10$ – 100 mesh range and the finer $N = 70$ – 100 range.

10 Results

This section collects the current convergence results, tables, and figure references. The numerical results are generated from the tutorial output files; figures can be regenerated by running `plot_convergence.m` in this report directory. Each plot legend reports the fitted orders over $N = 10$ – 100 and $N = 70$ – 100 plus the accumulated runtime and peak-RSS cost of the corresponding mesh sweep. Missing figure boxes indicate plots that have not yet been generated from the latest data.

10.1 Two-dimensional manufactured problem

The tables in this subsection are generated from `/home/pablo/OpenFOAM/pablo-v2312/run/tutorials_electro/highOrderManufacturedFDAImplicitJfNK/results/example0/2D/`. For each nonlinear method and time step, one table reports the observed orders for all available high-order configurations, time schemes, and mass matrices. Each table is followed by the corresponding six-panel convergence figures, one figure for each scheme/mass block.

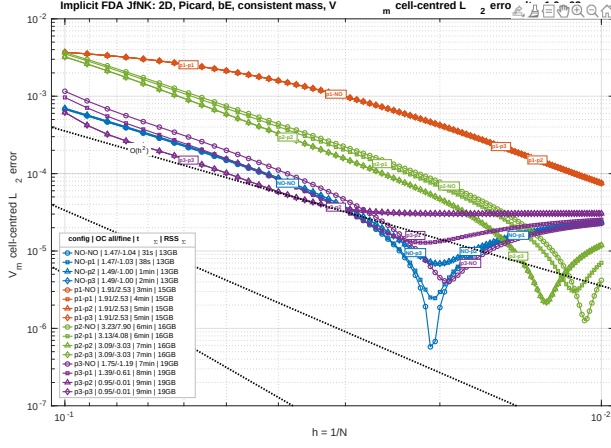
10.1.1 Picard

$$\Delta t = 10^{-2}$$

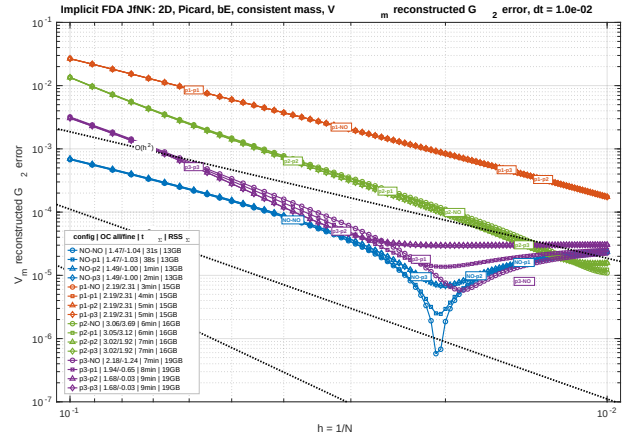
Table 2: 2D manufactured results for Picard with $\Delta t = 10^{-2}$. The first six convergence columns are log–log fits over meshes $N = 10$ – 100 ; the next six columns are fits over $N = 70$ – 100 . The time column is the sum of the reported wall-clock total over all mesh runs in the row, and the memory column is the sum of the reported peak resident memory in MB over the same mesh runs. The corresponding convergence figures are Figs. 1, 2, 3, 4.

				Fit $N = 10$ –100						Fit $N = 70$ –100							
V_m	I_{ion}	Scheme	Mass	V_m	V_m^G	u_1	u_1^G	u_2	u_2^G	V_m	V_m^G	u_1	u_1^G	u_2	u_2^G	t_{Σ} [s]	RSS $_{\Sigma}$ [MB]
NO	NO	bE	cons.	1.473	1.473	1.349	1.349	1.349	1.349	-1.042	-1.042	-0.923	-0.923	-0.923	-0.923	31.423	13399.5
NO	p1	bE	cons.	1.470	1.470	2.008	2.177	2.118	2.181	-1.035	-1.035	2.029	2.429	2.397	2.429	38.111	13505.7
NO	p2	bE	cons.	1.486	1.486	2.003	2.054	2.058	2.056	-0.995	-0.995	2.015	2.137	2.203	2.138	76.687	13717.7
NO	p3	bE	cons.	1.486	1.486	2.003	2.064	2.058	2.067	-0.995	-0.995	2.015	2.162	2.203	2.164	100.48	13819.4
p1	NO	bE	cons.	1.911	2.188	1.671	1.671	1.681	1.681	2.531	2.308	2.347	2.347	2.351	2.351	197.75	15196.
p1	p1	bE	cons.	1.911	2.188	2.007	2.219	1.915	2.226	2.531	2.308	2.033	2.426	2.388	2.430	231.25	15283.4
p1	p2	bE	cons.	1.912	2.188	2.003	2.218	2.008	2.216	2.531	2.308	2.016	2.251	2.242	2.249	275.72	15469.0
p1	p3	bE	cons.	1.912	2.188	2.003	2.235	2.008	2.234	2.531	2.308	2.016	2.278	2.242	2.277	288.72	15616.3
p2	NO	bE	cons.	3.232	3.056	3.225	3.225	3.225	3.225	7.896	3.694	5.447	5.447	5.484	5.484	348.43	16252.7
p2	p1	bE	cons.	3.125	3.053	2.013	3.035	2.179	3.034	4.083	3.116	2.032	2.760	2.090	2.783	353.42	16297.2
p2	p2	bE	cons.	3.086	3.017	2.008	3.032	2.158	3.029	-3.034	1.918	2.016	2.489	2.061	2.502	407.80	16515.7
p2	p3	bE	cons.	3.086	3.017	2.009	3.028	2.158	3.026	-3.034	1.918	2.017	2.501	2.061	2.513	404.62	16615.0
p3	NO	bE	cons.	1.749	2.178	1.650	1.650	1.654	1.654	-1.193	-1.240	-1.051	-1.051	-1.052	-1.052	427.29	18992.8
p3	p1	bE	cons.	1.394	1.937	2.011	1.801	1.972	1.804	-0.612	-0.646	2.031	-0.582	1.754	-0.590	477.08	19100.9
p3	p2	bE	cons.	0.948	1.680	2.004	1.999	1.994	1.987	-0.005	-0.028	2.016	-0.045	1.894	-0.048	547.39	19242.6
p3	p3	bE	cons.	0.948	1.680	2.004	1.986	1.995	1.974	-0.005	-0.028	2.016	-0.046	1.894	-0.048	553.29	19361.6
NO	NO	bE	lump.	1.473	1.473	1.349	1.349	1.349	1.349	-1.042	-1.042	-0.923	-0.923	-0.923	-0.923	31.976	13394.9
NO	p1	bE	lump.	1.470	1.470	2.008	2.177	2.118	2.181	-1.035	-1.035	2.029	2.429	2.397	2.429	38.417	13502.6
NO	p2	bE	lump.	1.486	1.486	2.003	2.054	2.058	2.056	-0.995	-0.995	2.015	2.137	2.203	2.138	78.079	13715.3
NO	p3	bE	lump.	1.486	1.486	2.003	2.064	2.058	2.067	-0.995	-0.995	2.015	2.162	2.203	2.164	94.982	13828.3
p1	NO	bE	lump.	1.911	2.188	1.671	1.671	1.681	1.681	2.531	2.308	2.347	2.347	2.351	2.351	231.17	14709.9
p1	p1	bE	lump.	1.911	2.188	2.007	2.219	1.915	2.226	2.531	2.308	2.033	2.426	2.388	2.430	234.88	15120.5
p1	p2	bE	lump.	1.912	2.188	2.003	2.218	2.008	2.216	2.531	2.308	2.016	2.251	2.242	2.249	275.39	15282.1
p1	p3	bE	lump.	1.912	2.188	2.003	2.235	2.008	2.234	2.531	2.308	2.016	2.278	2.242	2.277	267.72	15392.8
p2	NO	bE	lump.	3.227	3.058	3.176	3.176	3.177	3.177	4.728	3.144	1.952	1.952	1.998	1.998	348.27	15150.4
p2	p1	bE	lump.	3.068	3.037	2.013	3.014	2.163	3.014	0.954	2.507	2.032	2.316	2.042	2.351	359.75	16071.6
p2	p2	bE	lump.	2.922	2.970	2.008	3.011	2.147	3.010	-5.003	1.425	2.016	2.298	2.038	2.313	401.72	16213.2
p2	p3	bE	lump.	2.922	2.970	2.008	3.008	2.147	3.007	-5.003	1.425	2.017	2.314	2.038	2.329	426.32	16380.9
p3	NO	bE	lump.	1.324	1.925	1.267	1.267	1.268	1.268	-0.661	-0.697	-0.592	-0.592	-0.593	-0.593	439.09	17235.8
p3	p1	bE	lump.	1.102	1.774	2.010	1.651	1.956	1.649	-0.256	-0.284	2.031	-0.254	1.717	-0.260	479.94	18709.0
p3	p2	bE	lump.	1.018	1.647	2.004	1.942	1.986	1.929	0.212	0.192	2.016	0.164	1.873	0.162	541.27	18851.2
p3	p3	bE	lump.	1.018	1.647	2.004	1.929	1.986	1.917	0.212	0.192	2.016	0.163	1.873	0.162	544.97	18996.4
NO	NO	cN	cons.	1.992	1.992	2.135	2.135	2.134	2.134	1.985	1.985	2.500	2.500	2.497	2.497	22.791	13420.4
NO	p1	cN	cons.	1.992	1.992	2.000	2.057	2.005	2.061	1.985	1.985	2.002	2.031	2.023	2.031	31.672	13520.9
NO	p2	cN	cons.	1.992	1.992	1.999	2.015	2.002	2.017	1.986	1.986	2.001	2.010	2.011	2.010	59.517	13734.6
NO	p3	cN	cons.	1.992	1.992	1.999	2.019	2.002	2.021	1.986	1.986	2.001	2.011	2.011	2.011	77.638	13880.1
p1	NO	cN	cons.	1.911	2.146	1.616	1.616	1.626	1.626	2.612	2.168	2.381	2.381	2.387	2.387	155.47	15198.7
p1	p1	cN	cons.	1.911	2.146	1.998	2.154	1.838	2.160	2.612	2.168	2.007	2.316	2.258	2.316	186.26	15261.4
p1	p2	cN	cons.	1.912	2.146	1.999	2.182	1.951	2.181	2.612	2.168	2.002	2.161	2.105	2.157	210.03	15466.2
p1	p3	cN	cons.	1.912	2.146	1.999	2.196	1.951	2.195	2.612	2.168	2.002	2.177	2.105	2.173	230.94	15611.4
p2	NO	cN	cons.	2.127	2.567	2.176	2.176	2.176	2.176	2.016	2.179	2.139	2.139	2.139	2.139	264.47	16187.6
p2	p1	cN	cons.	2.149	2.608	2.004	2.716	2.109	2.720	2.019	2.210	2.002	2.416	2.034	2.422	272.53	16270.9
p2	p2	cN	cons.	2.212	2.708	2.004	2.942	2.118	2.941	2.026	2.303	2.001	2.735	2.020	2.738	311.82	16485.4

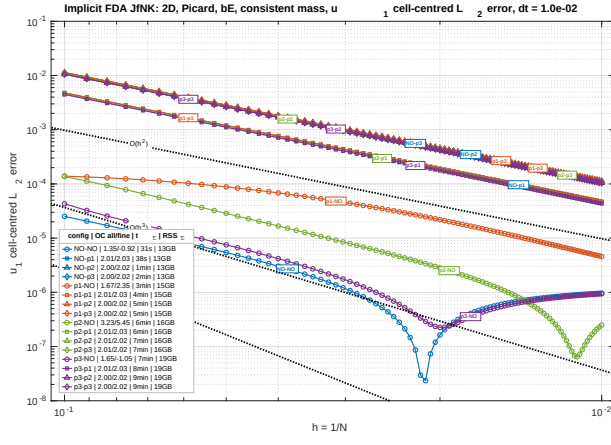
V_m	I_{ion}	Scheme	Mass	Fit $N = 10-100$						Fit $N = 70-100$						t_Σ [s]	RSS $_\Sigma$ [MB]
				V_m	V_m^G	u_1	u_1^G	u_2	u_2^G	V_m	V_m^G	u_1	u_1^G	u_2	u_2^G		
p2	p3	cN	cons.	2.212	2.708	2.004	2.940	2.118	2.940	2.026	2.303	2.001	2.739	2.020	2.743	311.49	16626.9
p3	NO	cN	cons.	2.121	2.481	2.256	2.256	2.258	2.258	1.999	2.037	2.472	2.472	2.469	2.469	320.46	19007.1
p3	p1	cN	cons.	2.181	2.631	2.002	2.762	1.992	2.758	2.003	2.056	2.002	2.661	2.008	2.651	354.98	19077.1
p3	p2	cN	cons.	3.520	3.971	2.000	3.744	1.999	3.748	4.171	4.124	2.001	1.457	2.006	1.395	401.38	19271.3
p3	p3	cN	cons.	3.520	3.971	2.000	3.738	1.999	3.742	4.171	4.124	2.001	1.419	2.006	1.354	407.52	19384.7
NO	NO	cN	lump.	1.992	1.992	2.135	2.135	2.134	2.134	1.985	1.985	2.500	2.500	2.497	2.497	23.037	13415.4
NO	p1	cN	lump.	1.992	1.992	2.000	2.057	2.005	2.061	1.985	1.985	2.002	2.031	2.023	2.031	30.294	13518.1
NO	p2	cN	lump.	1.992	1.992	1.999	2.015	2.002	2.017	1.986	1.986	2.001	2.010	2.011	2.010	59.660	13738.4
NO	p3	cN	lump.	1.992	1.992	1.999	2.019	2.002	2.021	1.986	1.986	2.001	2.011	2.011	2.011	78.779	13872.0
p1	NO	cN	lump.	1.911	2.146	1.616	1.616	1.626	1.626	2.612	2.168	2.381	2.381	2.387	2.387	159.48	14814.9
p1	p1	cN	lump.	1.911	2.146	1.998	2.154	1.838	2.160	2.612	2.168	2.007	2.316	2.258	2.316	182.41	15232.6
p1	p2	cN	lump.	1.912	2.146	1.999	2.182	1.951	2.181	2.612	2.168	2.002	2.161	2.105	2.157	219.19	15359.2
p1	p3	cN	lump.	1.912	2.146	1.999	2.196	1.951	2.195	2.612	2.168	2.002	2.177	2.105	2.173	234.14	15489.2
p2	NO	cN	lump.	2.136	2.606	2.191	2.191	2.191	2.191	2.017	2.210	2.153	2.153	2.153	2.153	264.29	15322.3
p2	p1	cN	lump.	2.162	2.652	2.004	2.759	2.104	2.762	2.020	2.248	2.002	2.473	2.032	2.480	275.86	16149.1
p2	p2	cN	lump.	2.242	2.761	2.004	2.969	2.113	2.968	2.029	2.368	2.001	2.798	2.018	2.801	315.04	16358.4
p2	p3	cN	lump.	2.242	2.761	2.004	2.967	2.113	2.966	2.029	2.368	2.001	2.802	2.018	2.805	316.91	16447.8
p3	NO	cN	lump.	2.181	2.661	2.398	2.398	2.400	2.400	2.001	2.061	2.804	2.804	2.798	2.798	317.24	17474.9
p3	p1	cN	lump.	2.307	2.891	2.001	3.077	1.989	3.069	2.016	2.106	2.002	2.766	2.005	2.765	359.13	18915.2
p3	p2	cN	lump.	2.382	3.105	1.999	3.171	1.997	3.163	2.075	2.063	2.001	1.324	2.004	1.308	401.37	19129.4
p3	p3	cN	lump.	2.382	3.105	2.000	3.159	1.998	3.151	2.075	2.063	2.001	1.316	2.004	1.299	409.59	19255.0



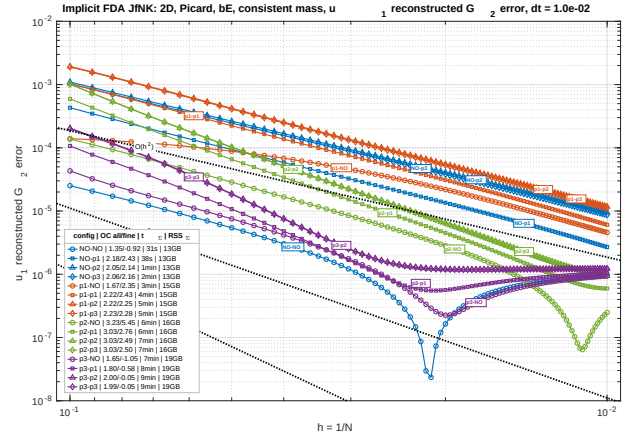
(a) $\|V_m\|_{L_2}$



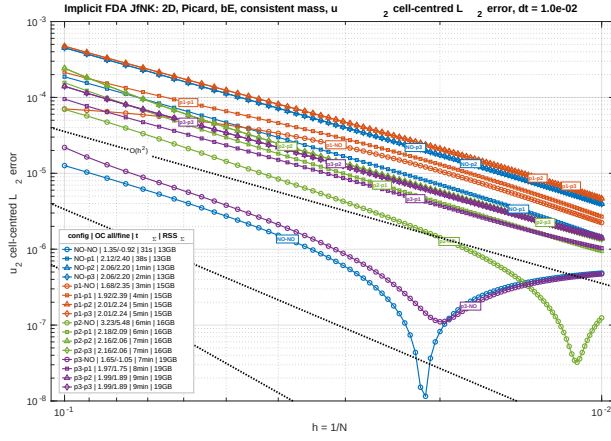
(b) $\|V_m\|_{G_2}$



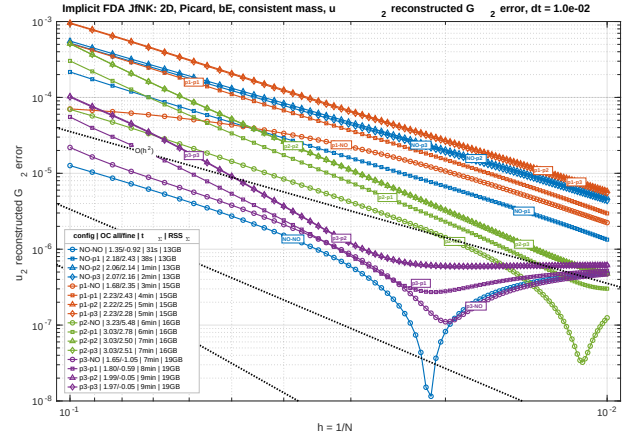
(c) $\|u_1\|_{L_2}$



(d) $\|u_1\|_{G_2}$

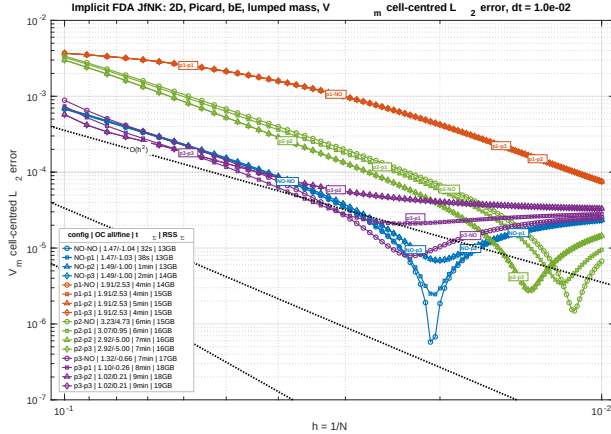


(e) $\|u_2\|_{L_2}$

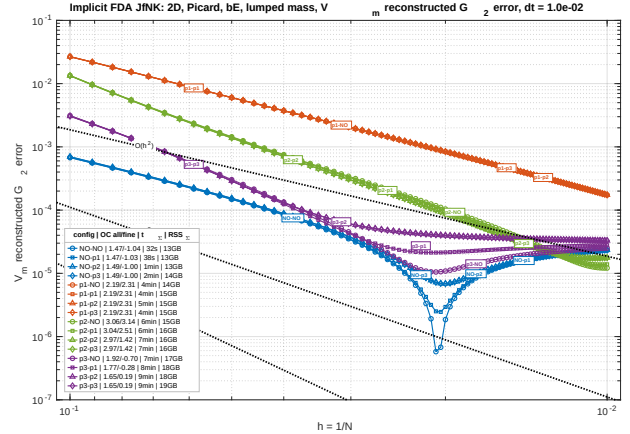


(f) $\|u_2\|_{G_2}$

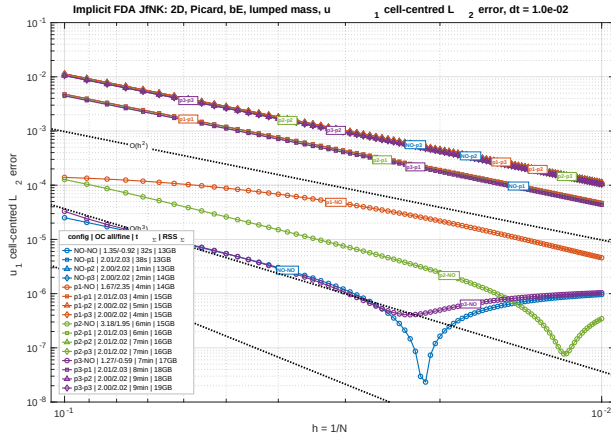
Figure 1: 2D convergence curves for Picard, $\Delta t = 10^{-2}$, bE with cons. mass. The six panels correspond to the six order columns in Table 2 for this scheme/mass block. The plot legends report the two fitted orders and the accumulated runtime/memory cost for each curve.



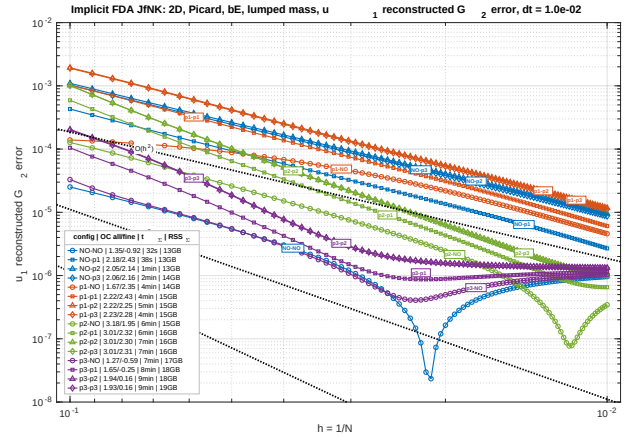
(a) $\|V_m\|_{L_2}$



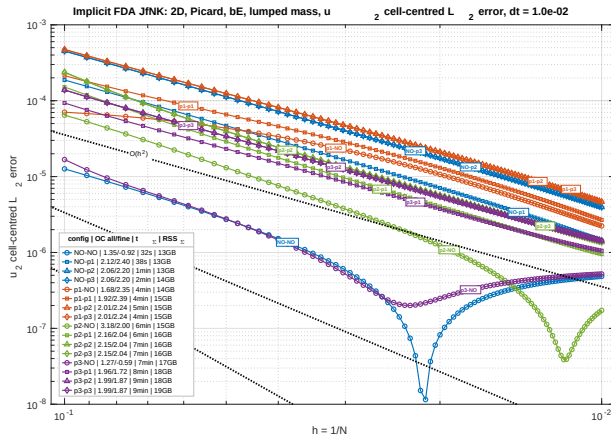
(b) $\|V_m\|_{G_2}$



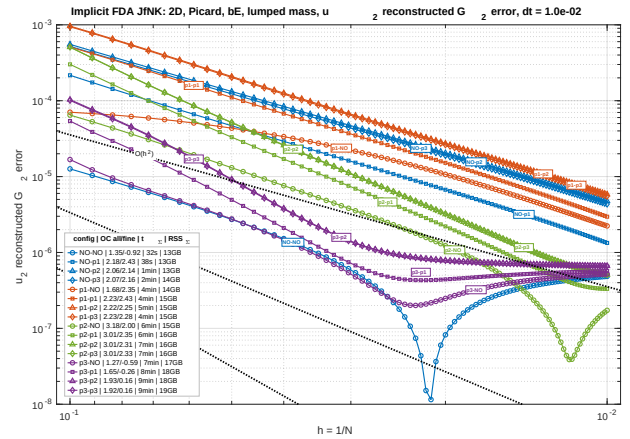
(c) $\|u_1\|_{L_2}$



(d) $\|u_1\|_{G_2}$

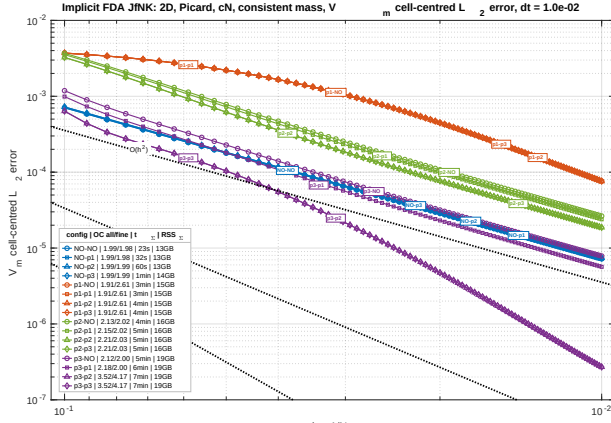


(e) $\|u_2\|_{L_2}$

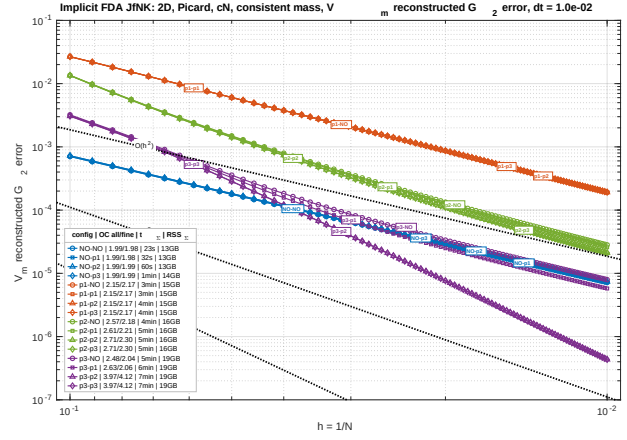


(f) $\|u_2\|_{G_2}$

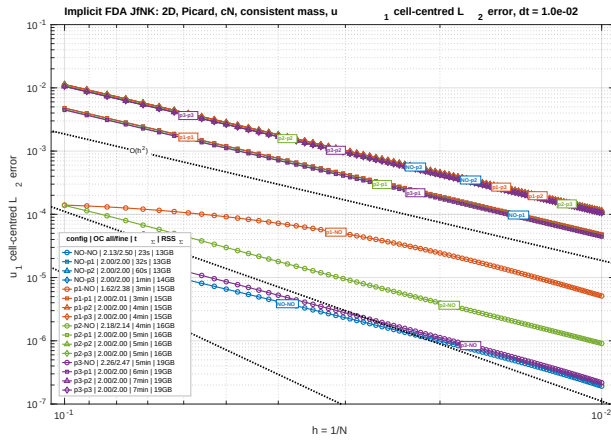
Figure 2: 2D convergence curves for Picard, $\Delta t = 10^{-2}$, bE with lump. mass. The six panels correspond to the six order columns in Table 2 for this scheme/mass block. The plot legends report the two fitted orders and the accumulated runtime/memory cost for each curve.



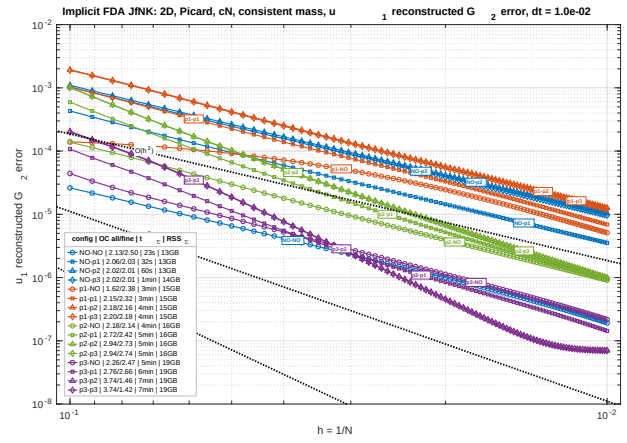
(a) $\|V_m\|_{L_2}$



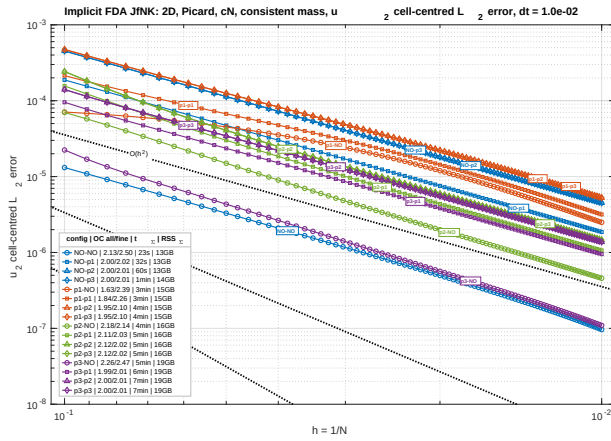
(b) $\|V_m\|_{G_2}$



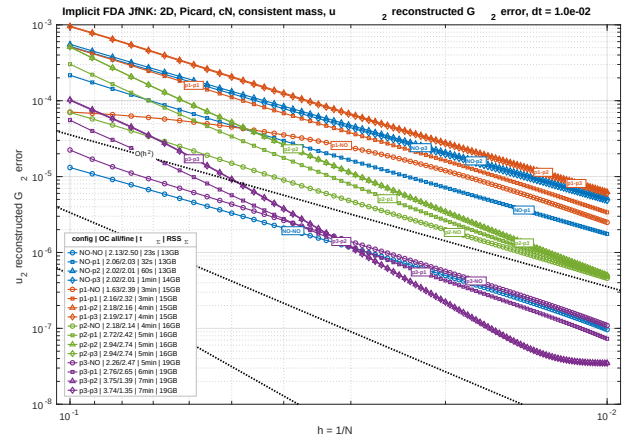
(c) $\|u_1\|_{L_2}$



(d) $\|u_1\|_{G_2}$

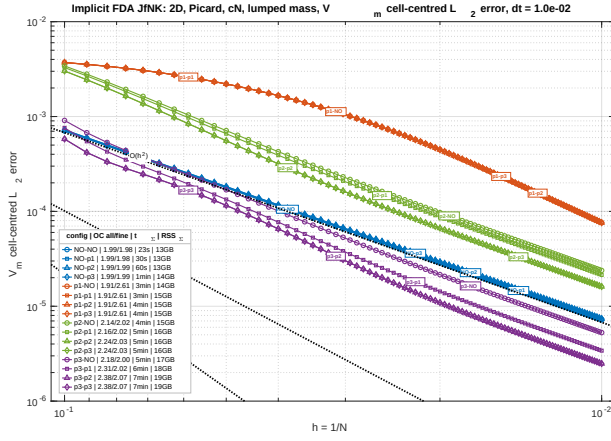


(e) $\|u_2\|_{L_2}$

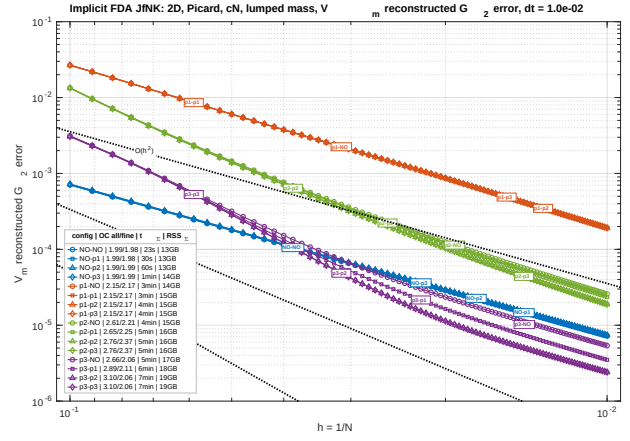


(f) $\|u_2\|_{G_2}$

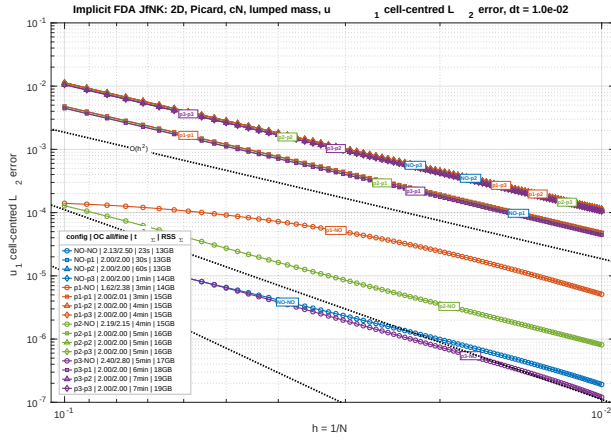
Figure 3: 2D convergence curves for Picard, $\Delta t = 10^{-2}$, cN with cons. mass. The six panels correspond to the six order columns in Table 2 for this scheme/mass block. The plot legends report the two fitted orders and the accumulated runtime/memory cost for each curve.



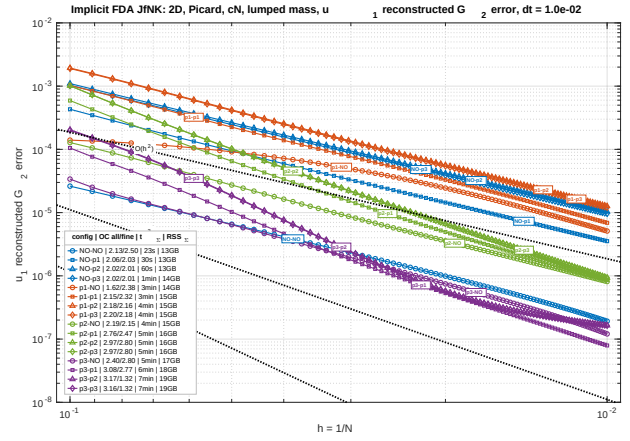
(a) $\|V_m\|_{L_2}$



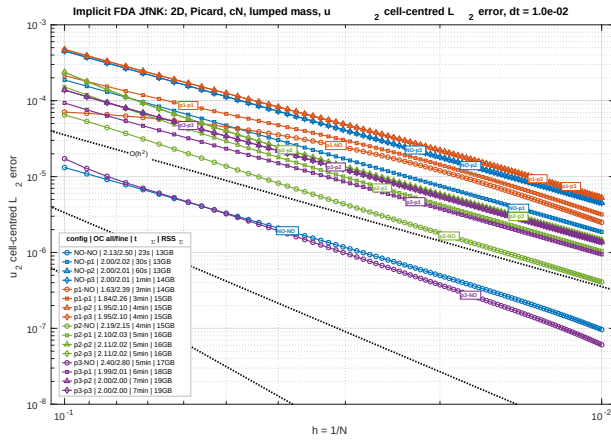
(b) $\|V_m\|_{G_2}$



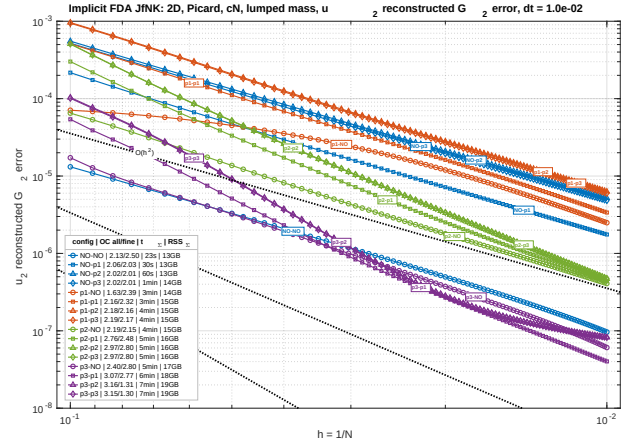
(c) $\|u_1\|_{L_2}$



(d) $\|u_1\|_{G_2}$



(e) $\|u_2\|_{L_2}$



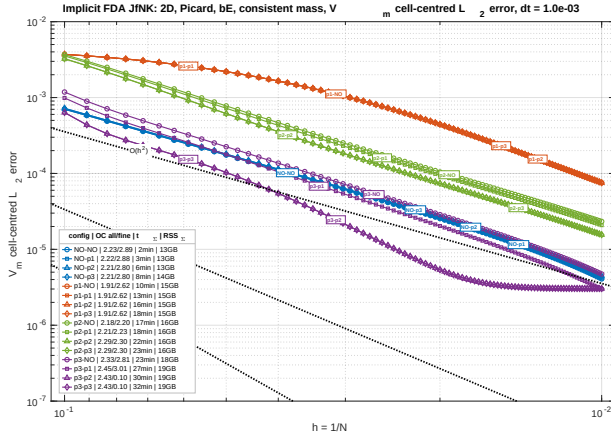
(f) $\|u_2\|_{G_2}$

Figure 4: 2D convergence curves for Picard, $\Delta t = 10^{-2}$, cN with lump. mass. The six panels correspond to the six order columns in Table 2 for this scheme/mass block. The plot legends report the two fitted orders and the accumulated runtime/memory cost for each curve.

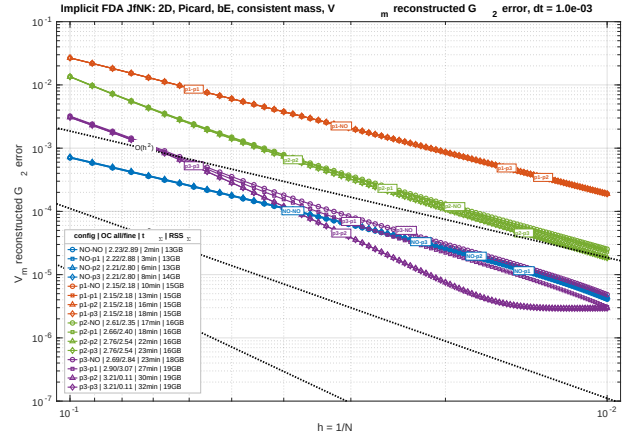
$$\Delta t = 10^{-3}$$

Table 3: 2D manufactured results for Picard with $\Delta t = 10^{-3}$. The first six convergence columns are log-log fits over meshes $N = 10$ –100; the next six columns are fits over $N = 70$ –100. The time column is the sum of the reported wall-clock total over all mesh runs in the row, and the memory column is the sum of the reported peak resident memory in MB over the same mesh runs. The corresponding convergence figures are Figs. 5, 6, 7, 8.

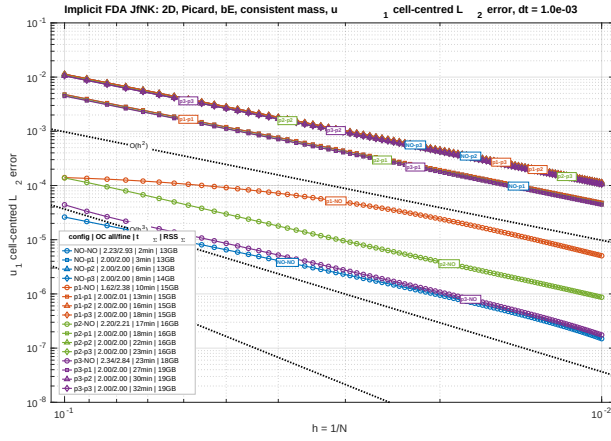
V_m	I_{ion}	Scheme	Mass	Fit $N = 10$ –100						Fit $N = 70$ –100						t_{Σ} [s]	RSS $_{\Sigma}$ [MB]
				V_m	V_m^G	u_1	u_1^G	u_2	u_2^G	V_m	V_m^G	u_1	u_1^G	u_2	u_2^G		
NO	NO	bE	cons.	2.226	2.226	2.233	2.233	2.233	2.233	2.895	2.895	2.926	2.926	2.926	2.926	134.82	13404.6
NO	p1	bE	cons.	2.224	2.224	2.000	2.061	2.009	2.065	2.884	2.884	2.003	2.045	2.036	2.045	185.12	13512.6
NO	p2	bE	cons.	2.208	2.208	1.999	2.017	2.004	2.019	2.799	2.799	2.001	2.014	2.017	2.014	371.86	13733.3
NO	p3	bE	cons.	2.208	2.208	1.999	2.021	2.004	2.023	2.799	2.799	2.001	2.017	2.018	2.017	485.11	13863.6
p1	NO	bE	cons.	1.914	2.151	1.622	1.622	1.632	1.632	2.621	2.183	2.382	2.382	2.388	2.388	620.61	15196.0
p1	p1	bE	cons.	1.914	2.151	1.998	2.158	1.844	2.165	2.621	2.183	2.008	2.320	2.262	2.320	755.10	15347.2
p1	p2	bE	cons.	1.914	2.151	1.999	2.184	1.954	2.182	2.620	2.183	2.002	2.164	2.109	2.160	955.85	15526.5
p1	p3	bE	cons.	1.914	2.151	1.999	2.198	1.954	2.197	2.620	2.183	2.002	2.180	2.109	2.176	1052.8	15666.0
p2	NO	bE	cons.	2.182	2.613	2.196	2.196	2.197	2.197	2.202	2.355	2.211	2.211	2.211	2.211	1036.7	16198.5
p2	p1	bE	cons.	2.209	2.657	2.004	2.732	2.113	2.736	2.225	2.401	2.003	2.483	2.047	2.490	1095.4	16248.3
p2	p2	bE	cons.	2.291	2.764	2.004	2.952	2.121	2.951	2.301	2.538	2.002	2.786	2.027	2.789	1323.7	16492.6
p2	p3	bE	cons.	2.291	2.764	2.004	2.950	2.121	2.949	2.301	2.538	2.002	2.789	2.027	2.793	1359.0	16572.0
p3	NO	bE	cons.	2.332	2.691	2.342	2.342	2.345	2.345	2.807	2.839	2.839	2.839	2.839	2.839	1390.5	18933.1
p3	p1	bE	cons.	2.452	2.900	2.002	2.875	1.993	2.870	3.012	3.067	2.003	3.116	2.012	3.106	1603.1	19039.9
p3	p2	bE	cons.	2.434	3.214	2.000	3.534	2.000	3.532	0.103	0.114	2.002	0.579	2.008	0.519	1828.3	19250.1
p3	p3	bE	cons.	2.434	3.214	2.000	3.525	2.000	3.523	0.103	0.114	2.002	0.549	2.008	0.487	1931.2	19365.8
NO	NO	bE	lump.	2.226	2.226	2.233	2.233	2.233	2.233	2.895	2.895	2.926	2.926	2.926	2.926	137.13	13401.2
NO	p1	bE	lump.	2.224	2.224	2.000	2.061	2.009	2.065	2.884	2.884	2.003	2.045	2.036	2.045	185.71	13506.7
NO	p2	bE	lump.	2.208	2.208	1.999	2.017	2.004	2.019	2.799	2.799	2.001	2.014	2.017	2.014	376.82	13730.3
NO	p3	bE	lump.	2.208	2.208	1.999	2.021	2.004	2.023	2.799	2.799	2.001	2.017	2.018	2.017	477.11	13839.8
p1	NO	bE	lump.	1.914	2.151	1.622	1.622	1.632	1.632	2.621	2.183	2.382	2.382	2.388	2.388	638.75	14736.2
p1	p1	bE	lump.	1.914	2.151	1.998	2.158	1.844	2.165	2.621	2.183	2.008	2.320	2.262	2.320	754.85	15170.7
p1	p2	bE	lump.	1.914	2.151	1.999	2.184	1.954	2.182	2.620	2.183	2.002	2.164	2.109	2.160	944.22	15320.4
p1	p3	bE	lump.	1.914	2.151	1.999	2.198	1.954	2.197	2.620	2.183	2.002	2.180	2.109	2.176	1013.2	15405.2
p2	NO	bE	lump.	2.197	2.656	2.213	2.213	2.214	2.214	2.225	2.402	2.234	2.234	2.235	2.235	1134.8	15201.1
p2	p1	bE	lump.	2.230	2.704	2.004	2.775	2.108	2.779	2.254	2.459	2.003	2.546	2.044	2.552	1216.0	16016.2
p2	p2	bE	lump.	2.335	2.820	2.004	2.979	2.115	2.977	2.355	2.626	2.002	2.847	2.025	2.850	1420.7	16249.9
p2	p3	bE	lump.	2.335	2.820	2.004	2.976	2.116	2.975	2.355	2.626	2.002	2.850	2.025	2.852	1474.5	16341.4
p3	NO	bE	lump.	2.535	3.005	2.554	2.554	2.557	2.557	3.520	3.545	3.591	3.591	3.591	3.591	1352.5	17237.8
p3	p1	bE	lump.	2.577	3.171	2.002	3.148	1.990	3.145	2.412	2.613	2.003	2.546	2.007	2.578	1555.9	18637.7
p3	p2	bE	lump.	1.984	2.690	2.000	3.053	1.998	3.042	1.092	1.046	2.002	1.069	2.006	1.053	1805.3	18832.4
p3	p3	bE	lump.	1.984	2.690	2.000	3.040	1.998	3.030	1.092	1.046	2.002	1.062	2.006	1.046	1892.6	18985.6
NO	NO	cN	cons.	1.997	1.997	1.998	1.998	1.998	1.998	2.000	2.000	2.004	2.004	2.004	2.004	128.25	13418.2
NO	p1	cN	cons.	1.997	1.997	1.999	2.050	1.998	2.053	2.000	2.000	2.000	2.007	2.000	2.007	173.20	13526.4
NO	p2	cN	cons.	1.997	1.997	1.999	2.013	1.998	2.015	1.999	1.999	2.000	2.002	2.000	2.002	356.75	13748.4
NO	p3	cN	cons.	1.997	1.997	1.999	2.016	1.999	2.018	1.999	1.999	2.000	2.002	2.000	2.002	515.80	13884.5
p1	NO	cN	cons.	1.912	2.147	1.616	1.616	1.626	1.626	2.611	2.168	2.380	2.380	2.385	2.385	589.88	15180.3
p1	p1	cN	cons.	1.912	2.147	1.997	2.152	1.836	2.158	2.611	2.168	2.005	2.309	2.249	2.308	654.15	15231.8
p1	p2	cN	cons.	1.913	2.147	1.998	2.181	1.949	2.179	2.611	2.168	2.001	2.155	2.096	2.151	814.26	15455.0
p1	p3	cN	cons.	1.913	2.147	1.999	2.194	1.949	2.193	2.611	2.168	2.001	2.170	2.096	2.166	972.88	15638.6
p2	NO	cN	cons.	2.128	2.568	2.142	2.142	2.143	2.143	2.020	2.183	2.025	2.025	2.026	2.026	837.99	16169.2
p2	p1	cN	cons.	2.150	2.609	2.003	2.689	2.102	2.694	2.023	2.214	2.001	2.304	2.011	2.312	910.80	16202.8
p2	p2	cN	cons.	2.214	2.709	2.003	2.924	2.114	2.923	2.032	2.308	2.000	2.644	2.007	2.650	1123.6	16449.3
p2	p3	cN	cons.	2.214	2.709	2.004	2.923	2.114	2.922	2.032	2.308	2.000	2.650	2.007	2.656	1181.2	16571.0
p3	NO	cN	cons.	2.125	2.485	2.132	2.132	2.135	2.135	2.013	2.050	2.025	2.025	2.024	2.024	1367.5	18973.5
p3	p1	cN	cons.	2.187	2.637	2.001	2.598	1.989	2.596	2.020	2.073	2.000	2.068	1.999	2.066	1470.4	19014.3
p3	p2	cN	cons.	3.538	3.992	1.999	4.003	1.997	4.010	4.323	4.279	2.000	4.194	2.001	4.195	1730.1	19265.3
p3	p3	cN	cons.	3.538	3.992	1.999	4.000	1.998	4.008	4.323	4.279	2.000	4.198	2.001	4.200	1810.1	19373.3
NO	NO	cN	lump.	1.997	1.997	1.998	1.998	1.998	1.998	2.000	2.000	2.004	2.004	2.004	2.004	130.61	13426.7
NO	p1	cN	lump.	1.997	1.997	1.999	2.050	1.998	2.053	2.000	2.000	2.000	2.007	2.000	2.007	168.28	13528.5
NO	p2	cN	lump.	1.997	1.997	1.999	2.013	1.998	2.015	1.999	1.999	2.000	2.002	2.000	2.002	362.16	13753.8
NO	p3	cN	lump.	1.997	1.997	1.999	2.016	1.999	2.018	1.999	1.999	2.000	2.002	2.000	2.002	464.05	13882.0
p1	NO	cN	lump.	1.912	2.147	1.616	1.616	1.626	1.626	2.611	2.168	2.380	2.380	2.385	2.385	548.48	14808.9
p1	p1	cN	lump.	1.912	2.147	1.997	2.152	1.836	2.158	2.611	2.168	2.005	2.309	2.249	2.308	666.75	15214.0
p1	p2	cN	lump.	1.913	2.147	1.998	2.181	1.949	2.179	2.611	2.168	2.001	2.155	2.096	2.151	833.45	15404.0
p1	p3	cN	lump.	1.913	2.147	1.999	2.194	1.949	2.193	2.611	2.168	2.001	2.170	2.096	2.166	983.39	15492.6
p2	NO	cN	lump.	2.137	2.608	2.153	2.153	2.154	2.154	2.021	2.214	2.026	2.026	2.027	2.027	976.72	15305.1
p2	p1	cN	lump.	2.164	2.653	2.003	2.730	2.098	2.734	2.025	2.253	2.000	2.353	2.010	2.361	1062.6	16115.1
p2	p2	cN	lump.	2.244	2.762	2.003	2.952	2.109	2.951	2.036	2.374	2.000	2.710	2.006	2.715	1271.5	16346.0
p2	p3	cN	lump.	2.244	2.762	2.003	2.950	2.110	2.950	2.036	2.374	2.000	2.715	2.006	2.720	1315.9	16423.7
p3	NO	cN	lump.	2.187	2.667	2.201	2.201	2.204	2.204	2.021	2.081	2.042	2.042	2.041	2.041	1237.7	17497.1
p3	p1	cN	lump.	2.315	2.899	2.001	2.866	1.986	2.859	2.041	2.131	2.000	2.124	1.999	2.118	1445.6	18842.9
p3	p2	cN	lump.	2.369	3.091	1.999	3.441	1.996	3.434	2.031	2.018	2.000	2.165	2.000	2.138	1651.5	19099.1
p3	p3	cN	lump.	2.369	3.091	1.999	3.431	1.996	3.424	2.031	2.018	2.000	2.153	2.000	2.125	1769.6	19232.3



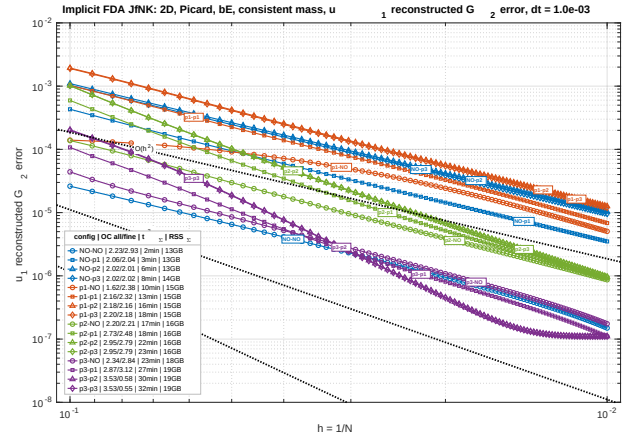
(a) $\|V_m\|_{L_2}$



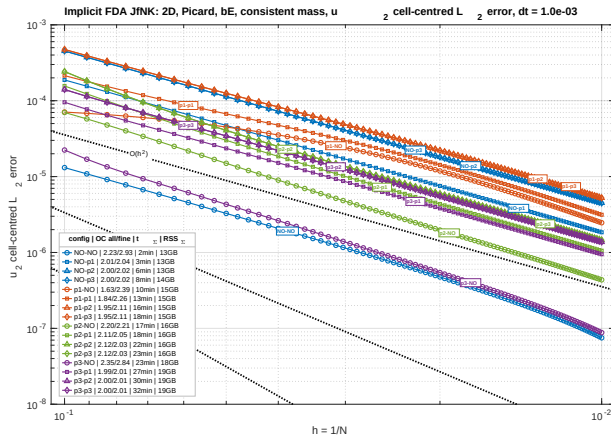
(b) $\|V_m\|_{G_2}$



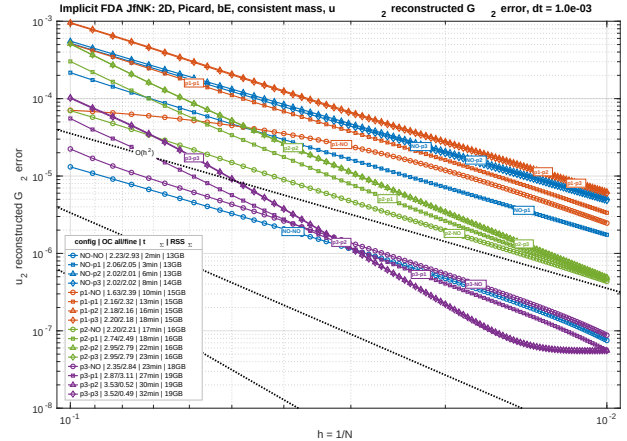
(c) $\|u_1\|_{L_2}$



(d) $\|u_1\|_{G_2}$

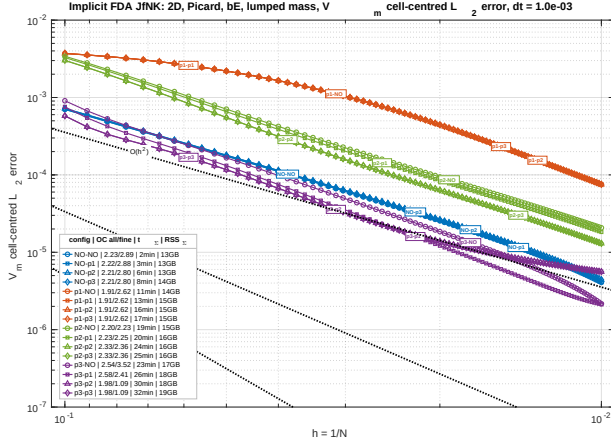


(e) $\|u_2\|_{L_2}$

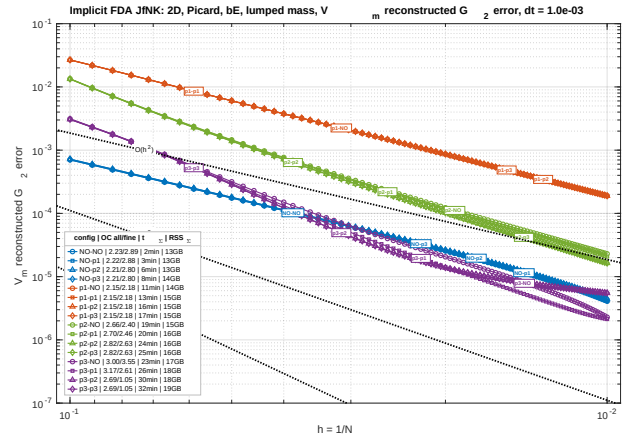


(f) $\|u_2\|_{G_2}$

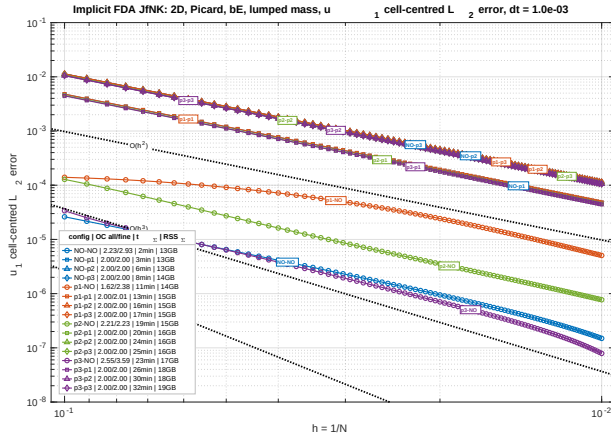
Figure 5: 2D convergence curves for Picard, $\Delta t = 10^{-3}$, bE with cons. mass. The six panels correspond to the six order columns in Table 3 for this scheme/mass block. The plot legends report the two fitted orders and the accumulated runtime/memory cost for each curve.



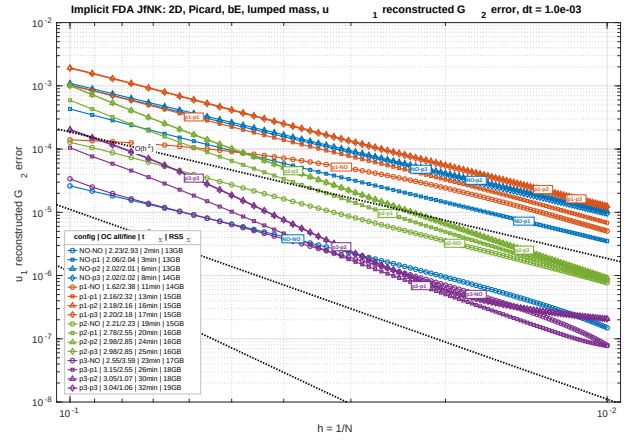
(a) $\|V_m\|_{L_2}$



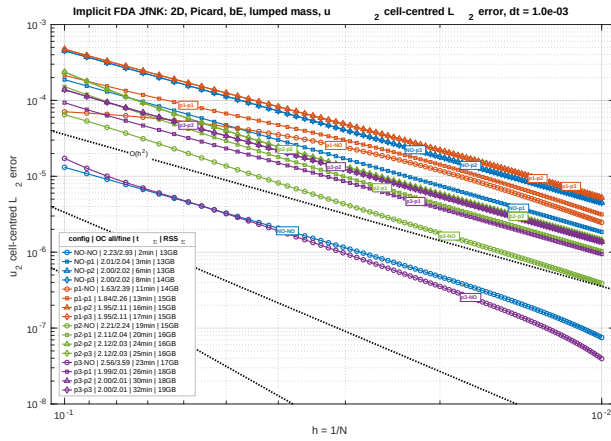
(b) $\|V_m\|_{G_2}$



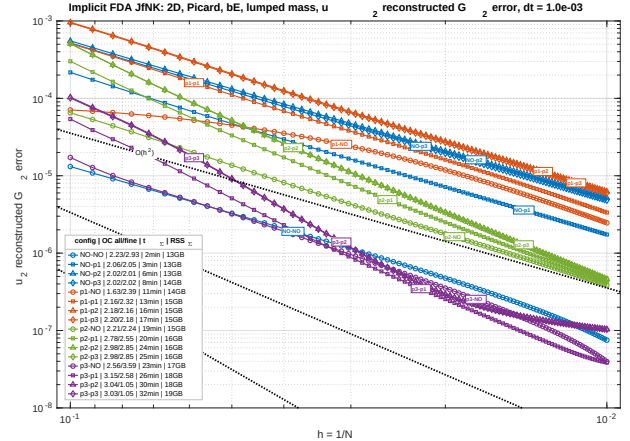
(c) $\|u_1\|_{L_2}$



(d) $\|u_1\|_{G_2}$

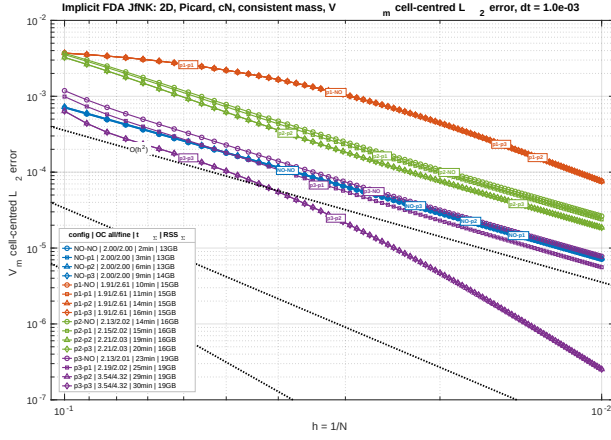


(e) $\|u_2\|_{L_2}$

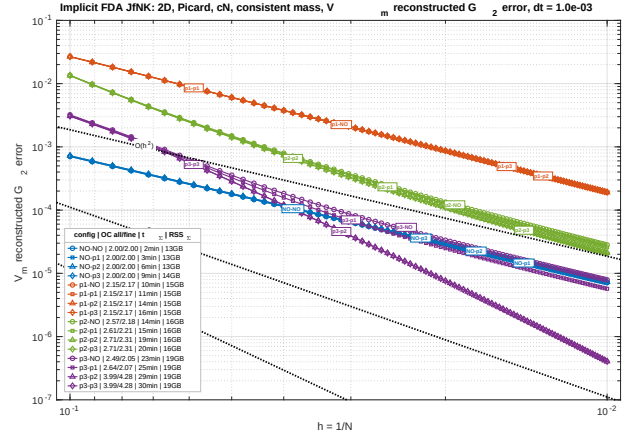


(f) $\|u_2\|_{G_2}$

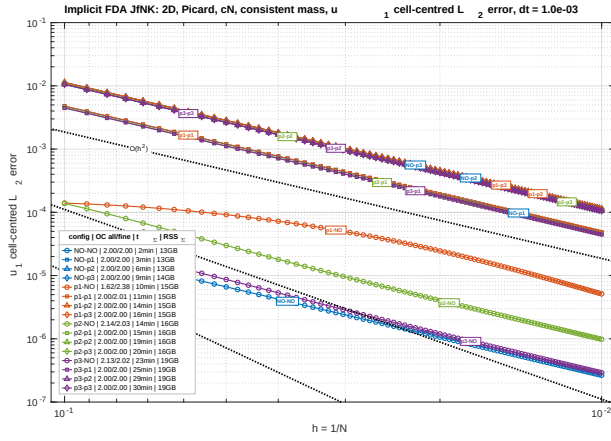
Figure 6: 2D convergence curves for Picard, $\Delta t = 10^{-3}$, bE with lump. mass. The six panels correspond to the six order columns in Table 3 for this scheme/mass block. The plot legends report the two fitted orders and the accumulated runtime/memory cost for each curve.



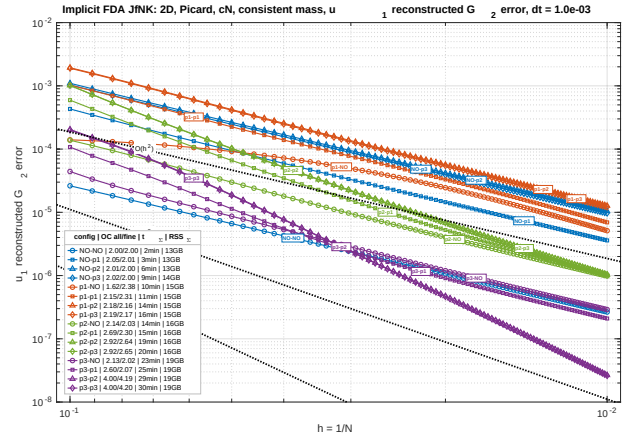
(a) $\|V_m\|_{L_2}$



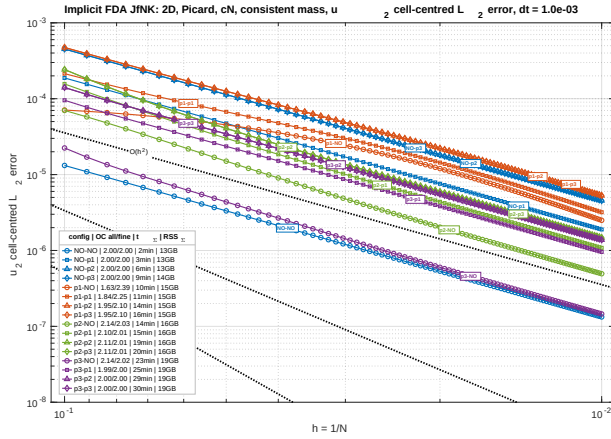
(b) $\|V_m\|_{G_2}$



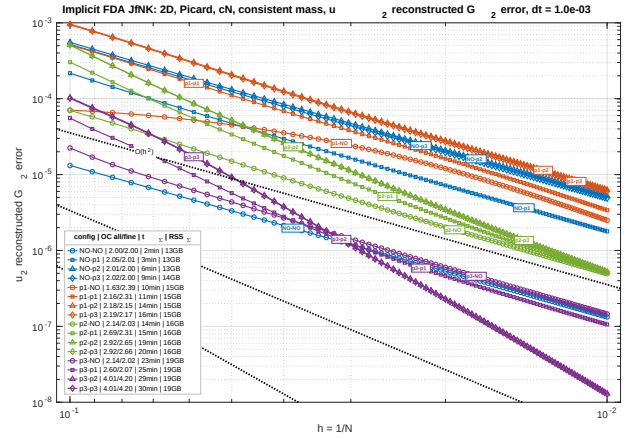
(c) $\|u_1\|_{L_2}$



(d) $\|u_1\|_{G_2}$

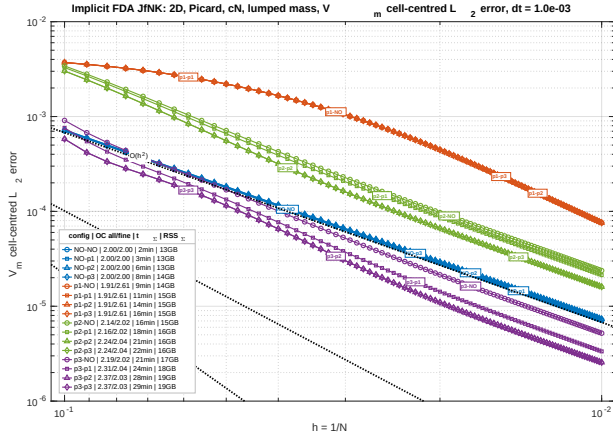


(e) $\|u_2\|_{L_2}$

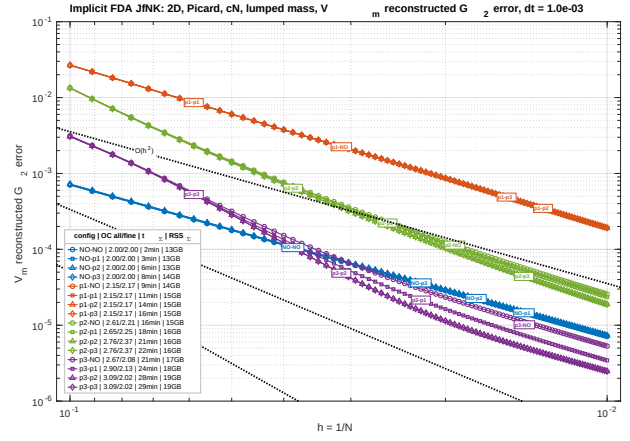


(f) $\|u_2\|_{G_2}$

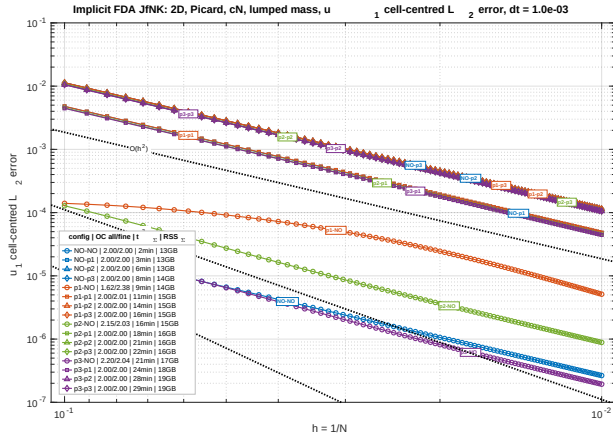
Figure 7: 2D convergence curves for Picard, $\Delta t = 10^{-3}$, cN with cons. mass. The six panels correspond to the six order columns in Table 3 for this scheme/mass block. The plot legends report the two fitted orders and the accumulated runtime/memory cost for each curve.



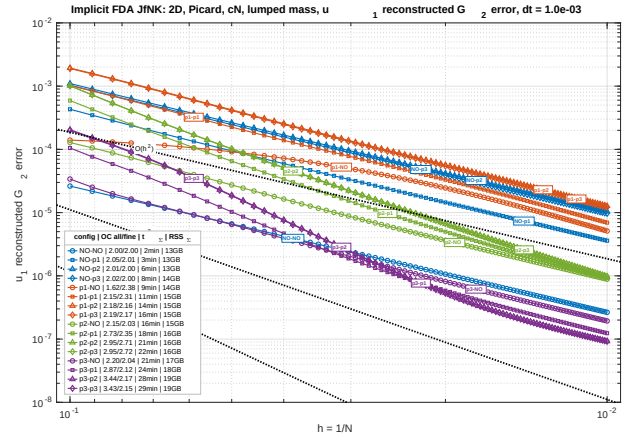
(a) $\|V_m\|_{L_2}$



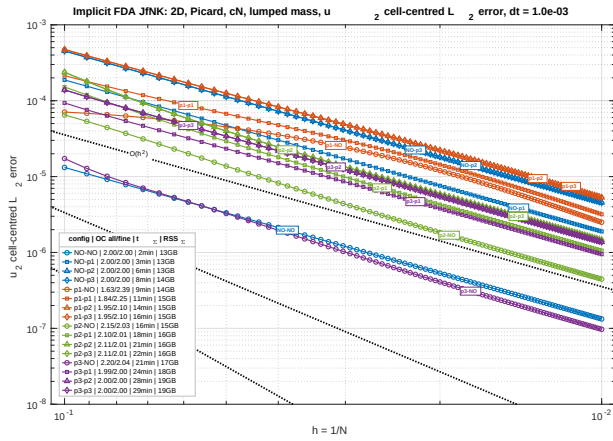
(b) $\|V_m\|_{G_2}$



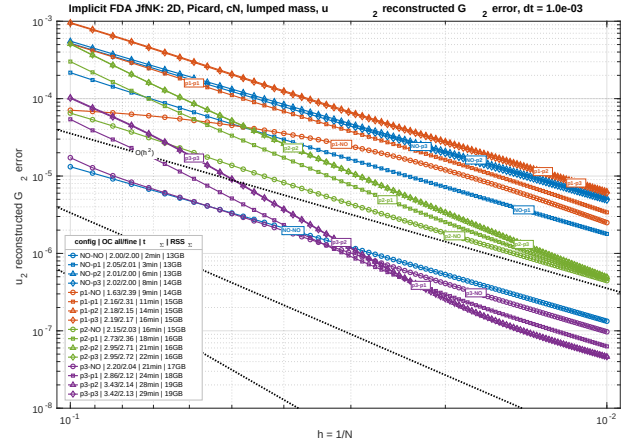
(c) $\|u_1\|_{L_2}$



(d) $\|u_1\|_{G_2}$



(e) $\|u_2\|_{L_2}$



(f) $\|u_2\|_{G_2}$

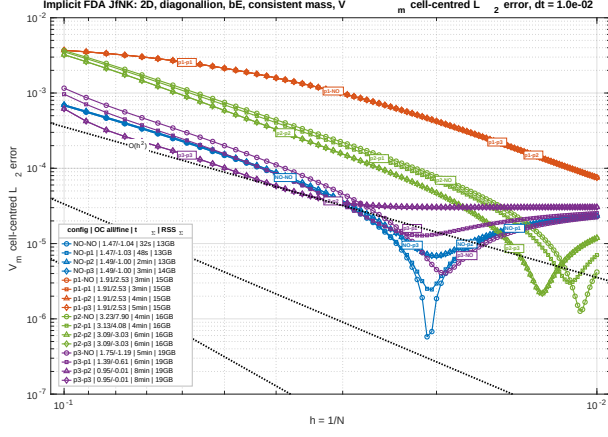
Figure 8: 2D convergence curves for Picard, $\Delta t = 10^{-3}$, cN with lump. mass. The six panels correspond to the six order columns in Table 3 for this scheme/mass block. The plot legends report the two fitted orders and the accumulated runtime/memory cost for each curve.

10.1.2 diagonalIon

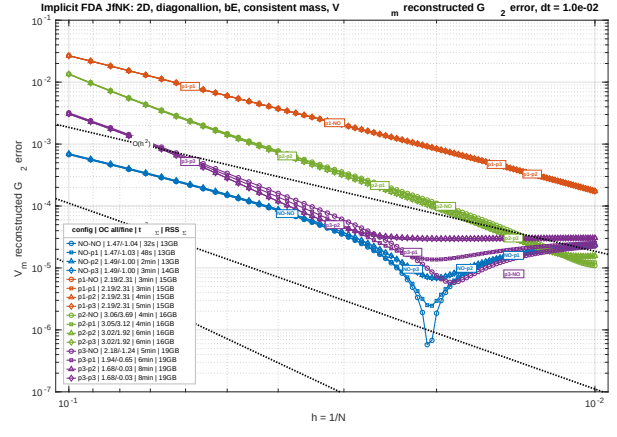
$$\Delta t = 10^{-2}$$

Table 4: 2D manufactured results for `diagonalIon` with $\Delta t = 10^{-2}$. The first six convergence columns are log-log fits over meshes $N = 10$ –100; the next six columns are fits over $N = 70$ –100. The time column is the sum of the reported wall-clock total over all mesh runs in the row, and the memory column is the sum of the reported peak resident memory in MB over the same mesh runs. The corresponding convergence figures are Figs. 9, 10, 11, 12.

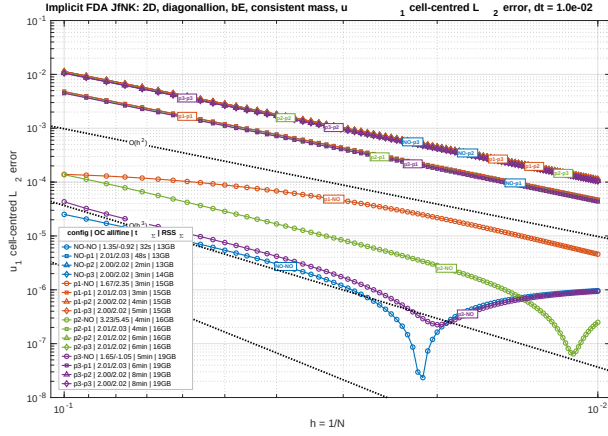
V_m	I_{ion}	Scheme	Mass	Fit $N = 10$ –100						Fit $N = 70$ –100						t_{Σ} [s]	RSS $_{\Sigma}$ [MB]
				V_m	V_m^G	u_1	u_1^G	u_2	u_2^G	V_m	V_m^G	u_1	u_1^G	u_2	u_2^G		
NO	NO	bE	cons.	1.473	1.473	1.349	1.349	1.349	1.349	-1.042	-1.042	-0.923	-0.923	-0.923	-0.923	31.582	13398.3
NO	p1	bE	cons.	1.470	1.470	2.008	2.177	2.118	2.181	-1.035	-1.035	2.029	2.429	2.397	2.429	47.592	13511.2
NO	p2	bE	cons.	1.486	1.486	2.003	2.054	2.058	2.056	-0.995	-0.995	2.015	2.137	2.203	2.138	113.45	13719.6
NO	p3	bE	cons.	1.486	1.486	2.003	2.064	2.058	2.067	-0.995	-0.995	2.015	2.162	2.203	2.164	155.57	13850.8
p1	NO	bE	cons.	1.911	2.188	1.671	1.671	1.681	1.681	2.531	2.308	2.347	2.347	2.351	2.351	170.39	15215.0
p1	p1	bE	cons.	1.911	2.188	2.007	2.219	1.915	2.226	2.531	2.308	2.033	2.426	2.388	2.430	191.40	15269.2
p1	p2	bE	cons.	1.912	2.188	2.003	2.218	2.008	2.216	2.531	2.308	2.016	2.251	2.242	2.249	245.48	15459.8
p1	p3	bE	cons.	1.912	2.188	2.003	2.235	2.008	2.234	2.531	2.308	2.016	2.278	2.242	2.277	309.17	15602.9
p2	NO	bE	cons.	3.232	3.056	3.225	3.225	3.225	3.225	7.896	3.694	5.447	5.447	5.484	5.484	250.97	16246.0
p2	p1	bE	cons.	3.125	3.053	2.013	3.035	2.179	3.034	4.083	3.116	2.032	2.760	2.090	2.783	265.57	16307.7
p2	p2	bE	cons.	3.086	3.017	2.008	3.032	2.158	3.029	-3.034	1.918	2.016	2.489	2.061	2.502	349.56	16516.8
p2	p3	bE	cons.	3.086	3.017	2.009	3.028	2.158	3.026	-3.034	1.918	2.017	2.501	2.061	2.513	367.15	16621.9
p3	NO	bE	cons.	1.749	2.178	1.650	1.650	1.654	1.654	-1.193	-1.240	-1.051	-1.051	-1.052	-1.052	306.14	18995.4
p3	p1	bE	cons.	1.394	1.937	2.011	1.801	1.972	1.804	-0.612	-0.646	2.031	-0.582	1.754	-0.590	354.38	19100.3
p3	p2	bE	cons.	0.948	1.680	2.004	1.999	1.994	1.987	-0.005	-0.028	2.016	-0.045	1.894	-0.048	454.12	19246.5
p3	p3	bE	cons.	0.948	1.680	2.004	1.986	1.995	1.974	-0.005	-0.028	2.016	-0.046	1.894	-0.048	473.80	19380.6
NO	NO	bE	lump.	1.473	1.473	1.349	1.349	1.349	1.349	-1.042	-1.042	-0.923	-0.923	-0.923	-0.923	32.308	13385.2
NO	p1	bE	lump.	1.470	1.470	2.008	2.177	2.118	2.181	-1.035	-1.035	2.029	2.429	2.397	2.429	48.281	13505.6
NO	p2	bE	lump.	1.486	1.486	2.003	2.054	2.058	2.056	-0.995	-0.995	2.015	2.137	2.203	2.138	116.77	13718.1
NO	p3	bE	lump.	1.486	1.486	2.003	2.064	2.058	2.067	-0.995	-0.995	2.015	2.162	2.203	2.164	150.47	13854.6
p1	NO	bE	lump.	1.911	2.188	1.671	1.671	1.681	1.681	2.531	2.308	2.347	2.347	2.351	2.351	171.38	14751.3
p1	p1	bE	lump.	1.911	2.188	2.007	2.219	1.915	2.226	2.531	2.308	2.033	2.426	2.388	2.430	188.26	15114.9
p1	p2	bE	lump.	1.912	2.188	2.003	2.218	2.008	2.216	2.531	2.308	2.016	2.251	2.242	2.249	259.22	15269.5
p1	p3	bE	lump.	1.912	2.188	2.003	2.235	2.008	2.234	2.531	2.308	2.016	2.278	2.242	2.277	296.77	15428.9
p2	NO	bE	lump.	3.227	3.058	3.176	3.176	3.177	3.177	4.728	3.144	1.952	1.952	1.998	1.998	244.74	15257.2
p2	p1	bE	lump.	3.068	3.037	2.013	3.014	2.163	3.014	0.954	2.507	2.032	2.316	2.042	2.351	270.52	16024.6
p2	p2	bE	lump.	2.922	2.970	2.008	3.011	2.147	3.010	-5.003	1.425	2.016	2.298	2.038	2.313	348.46	16198.9
p2	p3	bE	lump.	2.922	2.970	2.008	3.008	2.147	3.007	-5.003	1.425	2.017	2.314	2.038	2.329	372.47	16331.3
p3	NO	bE	lump.	1.324	1.925	1.267	1.267	1.268	1.268	-0.661	-0.697	-0.592	-0.592	-0.593	-0.593	301.96	17222.4
p3	p1	bE	lump.	1.102	1.774	2.010	1.651	1.956	1.649	-0.256	-0.284	2.031	-0.254	1.717	-0.260	353.66	18717.2
p3	p2	bE	lump.	1.018	1.647	2.004	1.942	1.986	1.929	0.212	0.192	2.016	0.164	1.873	0.162	445.73	18876.0
p3	p3	bE	lump.	1.018	1.647	2.004	1.929	1.986	1.917	0.212	0.192	2.016	0.163	1.873	0.162	484.59	19005.4
NO	NO	cN	cons.	1.992	1.992	2.135	2.135	2.134	2.134	1.985	1.985	2.500	2.500	2.497	2.497	29.753	13411.3
NO	p1	cN	cons.	1.992	1.992	2.000	2.057	2.005	2.061	1.985	1.985	2.002	2.031	2.023	2.031	47.031	13512.7
NO	p2	cN	cons.	1.992	1.992	1.999	2.015	2.002	2.017	1.986	1.986	2.001	2.010	2.011	2.010	115.87	13762.7
NO	p3	cN	cons.	1.992	1.992	1.999	2.019	2.002	2.021	1.986	1.986	2.001	2.011	2.011	2.011	147.37	13873.1
p1	NO	cN	cons.	1.911	2.146	1.616	1.616	1.626	1.626	2.612	2.168	2.381	2.381	2.387	2.387	159.85	15210.1
p1	p1	cN	cons.	1.911	2.146	1.998	2.154	1.838	2.160	2.612	2.168	2.007	2.316	2.258	2.316	179.80	15261.0
p1	p2	cN	cons.	1.912	2.146	1.999	2.182	1.951	2.181	2.612	2.168	2.002	2.161	2.105	2.157	255.83	15471.2
p1	p3	cN	cons.	1.912	2.146	1.999	2.196	1.951	2.195	2.612	2.168	2.002	2.177	2.105	2.173	292.17	15649.1
p2	NO	cN	cons.	2.127	2.567	2.176	2.176	2.176	2.176	2.016	2.179	2.139	2.139	2.139	2.139	222.25	16196.7
p2	p1	cN	cons.	2.149	2.608	2.004	2.716	2.109	2.720	2.019	2.210	2.002	2.416	2.034	2.422	252.27	16254.3
p2	p2	cN	cons.	2.212	2.708	2.004	2.942	2.118	2.941	2.026	2.303	2.001	2.734	2.020	2.738	319.78	16520.2
p2	p3	cN	cons.	2.212	2.708	2.004	2.940	2.118	2.940	2.026	2.303	2.001	2.739	2.020	2.743	339.35	16616.3
p3	NO	cN	cons.	2.121	2.481	2.256	2.256	2.258	2.258	1.999	2.037	2.472	2.472	2.469	2.469	293.15	19029.5
p3	p1	cN	cons.	2.181	2.631	2.002	2.762	1.992	2.757	2.003	2.056	2.002	2.661	2.008	2.650	323.21	19080.2
p3	p2	cN	cons.	3.520	3.971	2.000	3.744	1.999	3.748	4.171	4.124	2.001	1.458	2.006	1.396	395.69	19267.8
p3	p3	cN	cons.	3.520	3.971	2.000	3.738	1.999	3.742	4.171	4.124	2.001	1.420	2.006	1.355	440.47	19404.5
NO	NO	cN	lump.	1.992	1.992	2.135	2.135	2.134	2.134	1.985	1.985	2.500	2.500	2.497	2.497	30.642	13415.3
NO	p1	cN	lump.	1.992	1.992	2.000	2.057	2.005	2.061	1.985	1.985	2.002	2.031	2.023	2.031	47.562	13522.1
NO	p2	cN	lump.	1.992	1.992	1.999	2.015	2.002	2.017	1.986	1.986	2.001	2.010	2.011	2.010	114.54	13736.6
NO	p3	cN	lump.	1.992	1.992	1.999	2.019	2.002	2.021	1.986	1.986	2.001	2.011	2.011	2.011	148.23	13875.4
p1	NO	cN	lump.	1.911	2.146	1.616	1.616	1.626	1.626	2.612	2.168	2.381	2.381	2.387	2.387	162.54	14837.5
p1	p1	cN	lump.	1.911	2.146	1.998	2.154	1.838	2.160	2.612	2.168	2.007	2.316	2.258	2.316	177.04	15212.4
p1	p2	cN	lump.	1.912	2.146	1.999	2.182	1.951	2.181	2.612	2.168	2.002	2.161	2.105	2.157	237.86	15356.1
p1	p3	cN	lump.	1.912	2.146	1.999	2.196	1.951	2.195	2.612	2.168	2.002	2.177	2.105	2.173	296.97	15480.4
p2	NO	cN	lump.	2.136	2.606	2.191	2.191	2.191	2.191	2.017	2.210	2.153	2.153	2.153	2.153	230.89	15326.5
p2	p1	cN	lump.	2.162	2.652	2.004	2.759	2.104	2.762	2.020	2.248	2.002	2.473	2.032	2.480	249.18	16162.1
p2	p2	cN	lump.	2.242	2.761	2.004	2.969	2.113	2.968	2.029	2.368	2.001	2.798	2.018	2.801	333.56	16349.6
p2	p3	cN	lump.	2.242	2.761	2.004	2.967	2.113	2.966	2.029	2.368	2.001	2.802	2.018	2.805	353.27	16455.1
p3	NO	cN	lump.	2.181	2.661	2.398	2.398	2.399	2.399	2.001	2.061	2.804	2.804	2.797	2.797	280.34	17469.9
p3	p1	cN	lump.	2.307	2.891	2.001	3.077	1.989	3.069	2.016	2.106	2.002	2.766	2.005	2.765	328.01	18914.1
p3	p2	cN	lump.	2.382	3.105	1.999	3.171	1.997	3.163	2.075	2.063	2.001	1.325	2.004	1.308	397.22	19136.5
p3	p3	cN	lump.	2.382	3.105	2.000	3.159	1.998	3.151	2.075	2.063	2.001	1.316	2.004	1.299	448.87	19271.0



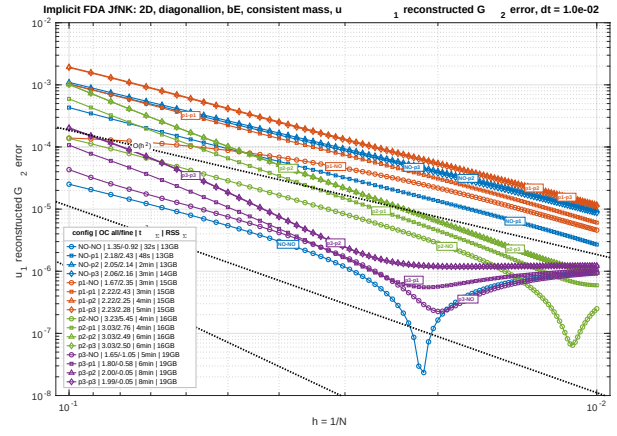
(a) $\|V_m\|_{L_2}$



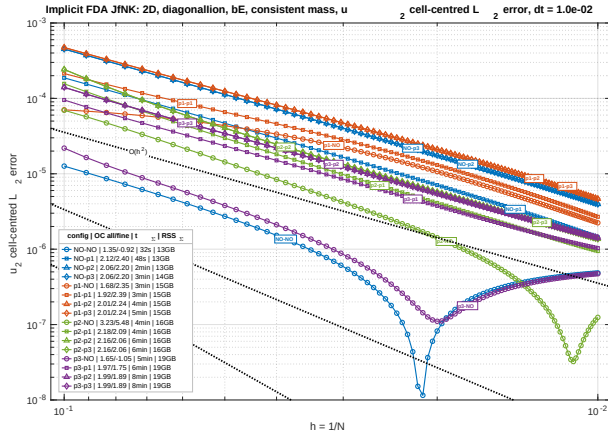
(b) $\|V_m\|_{G_2}$



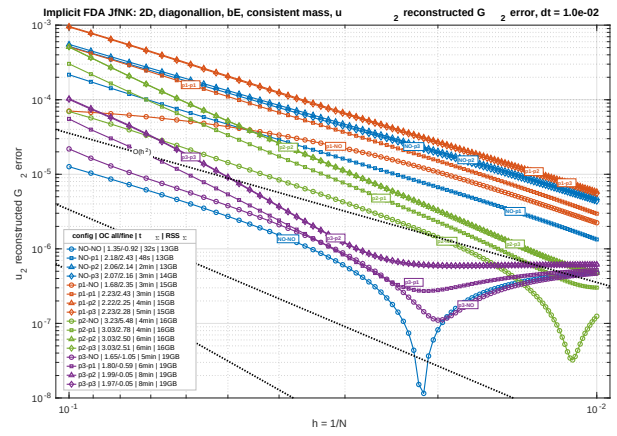
(c) $\|u_1\|_{L_2}$



(d) $\|u_1\|_{G_2}$

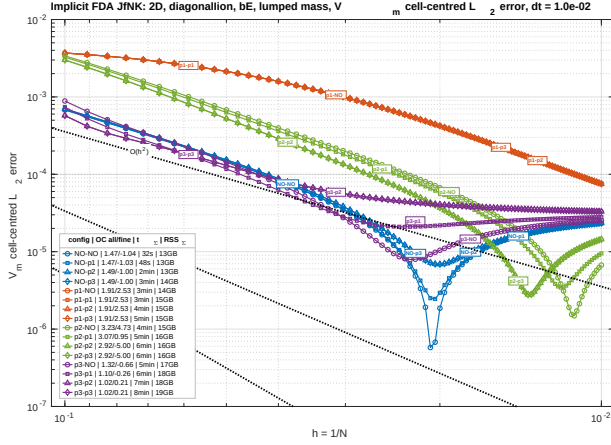


(e) $\|u_2\|_{L_2}$

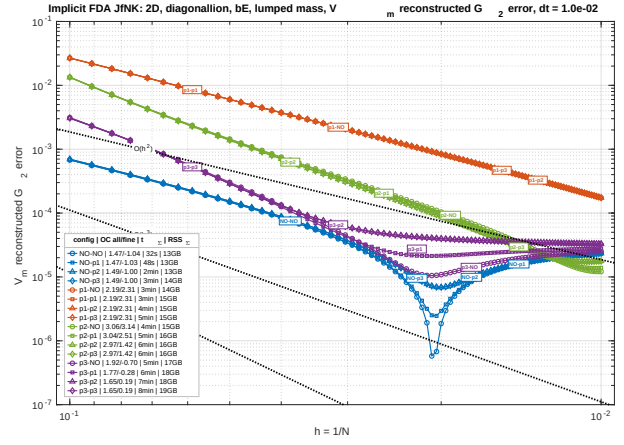


(f) $\|u_2\|_{G_2}$

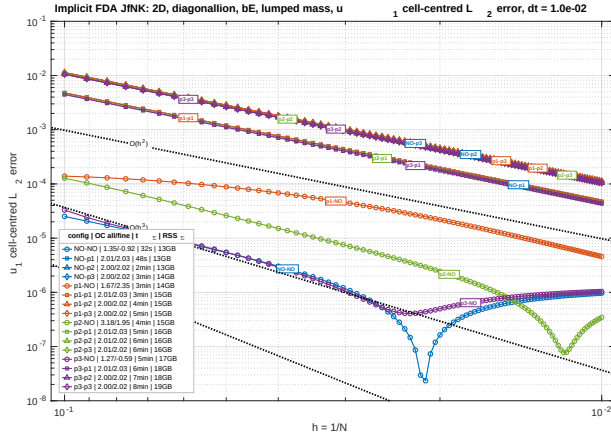
Figure 9: 2D convergence curves for diagonalIion, $\Delta t = 10^{-2}$, bE with cons. mass. The six panels correspond to the six order columns in Table 4 for this scheme/mass block. The plot legends report the two fitted orders and the accumulated runtime/memory cost for each curve.



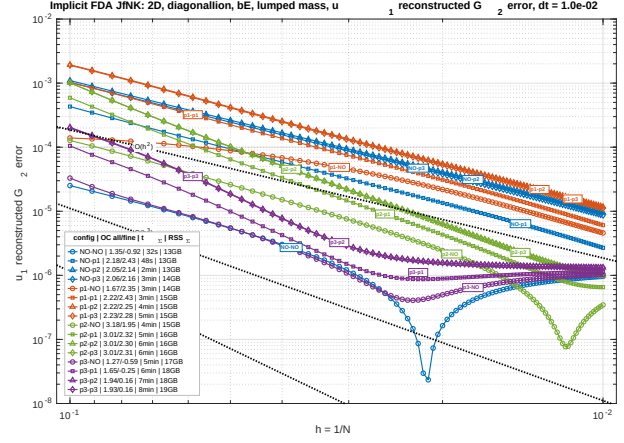
(a) $\|V_m\|_{L_2}$



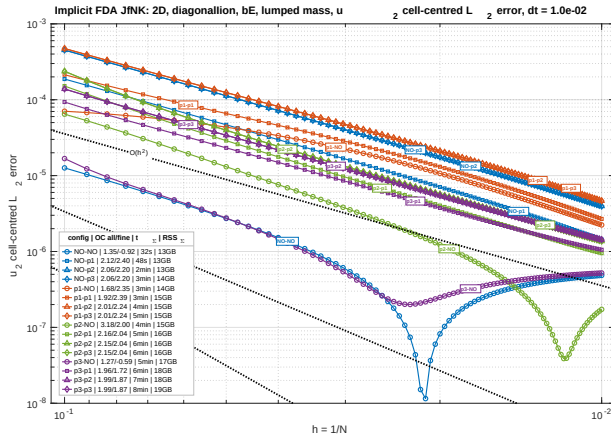
(b) $\|V_m\|_{G_2}$



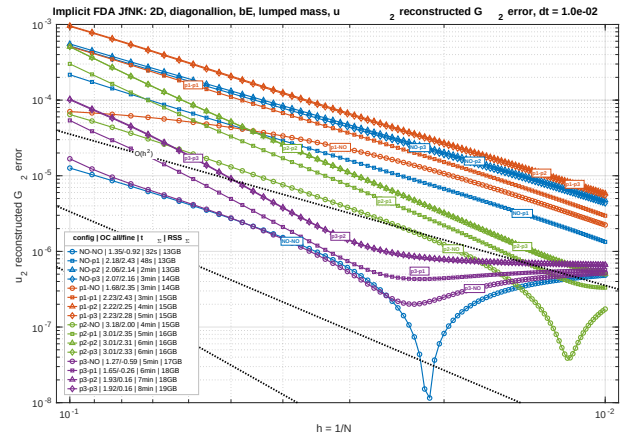
(c) $\|u_1\|_{L_2}$



(d) $\|u_1\|_{G_2}$

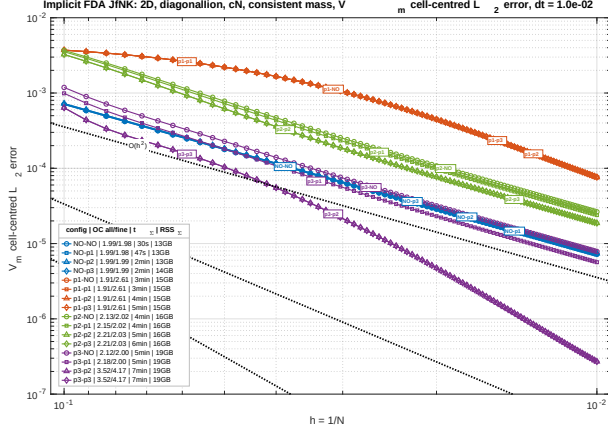


(e) $\|u_2\|_{L_2}$

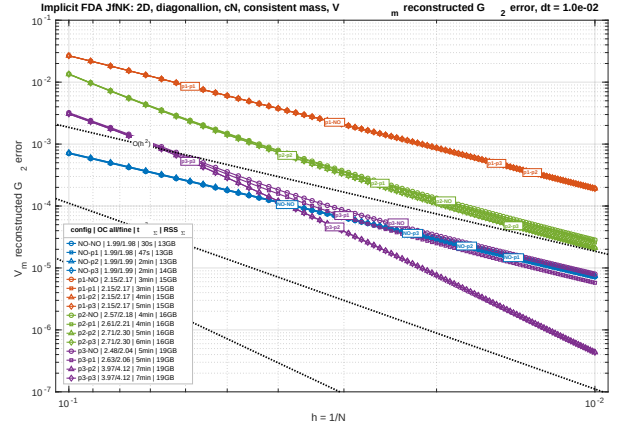


(f) $\|u_2\|_{G_2}$

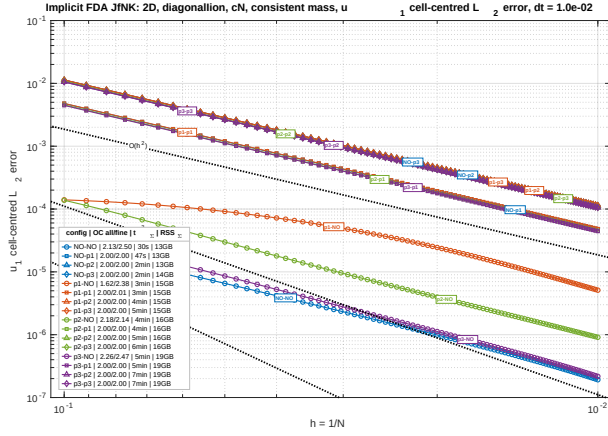
Figure 10: 2D convergence curves for diagonalion, $\Delta t = 10^{-2}$, bE with lump. mass. The six panels correspond to the six order columns in Table 4 for this scheme/mass block. The plot legends report the two fitted orders and the accumulated runtime/memory cost for each curve.



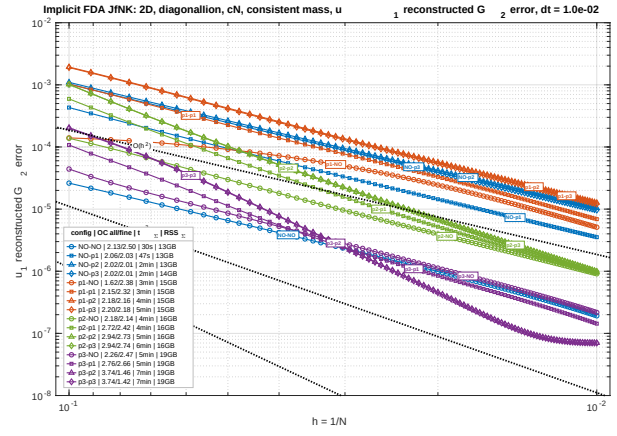
(a) $\|V_m\|_{L_2}$



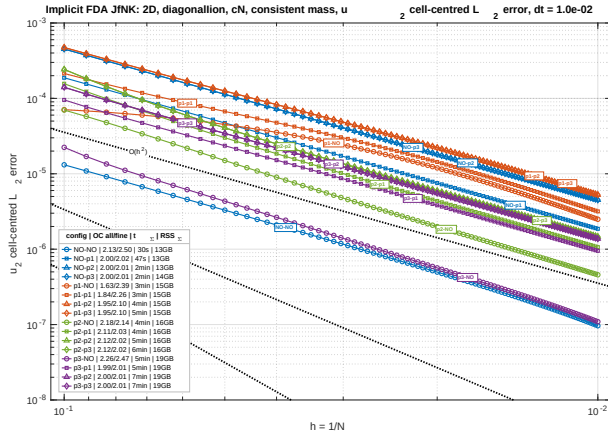
(b) $\|V_m\|_{G_2}$



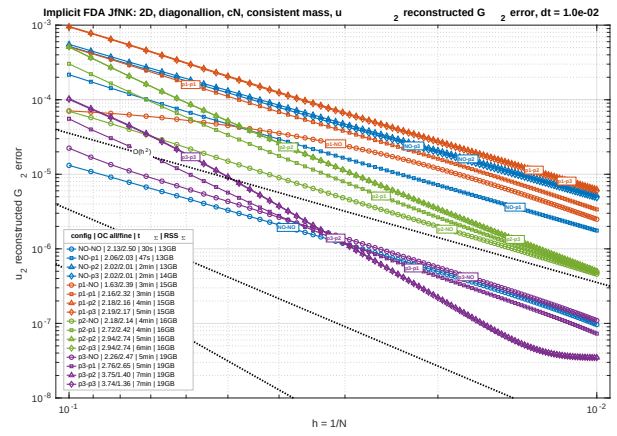
(c) $\|u_1\|_{L_2}$



(d) $\|u_1\|_{G_2}$

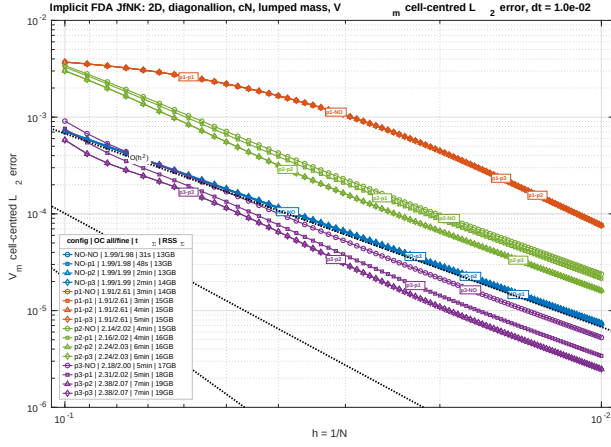


(e) $\|u_2\|_{L_2}$

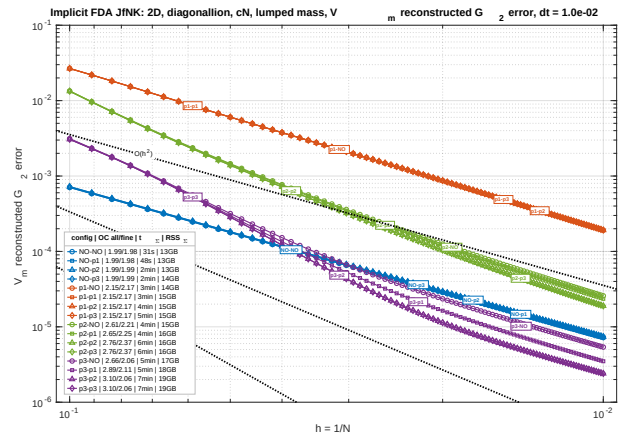


(f) $\|u_2\|_{G_2}$

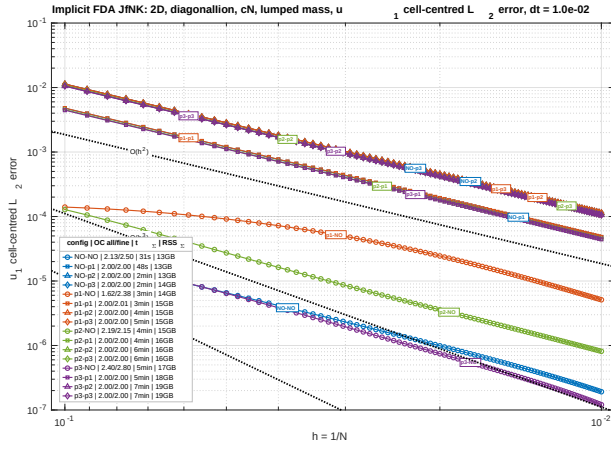
Figure 11: 2D convergence curves for diagonalion, $\Delta t = 10^{-2}$, cN with cons. mass. The six panels correspond to the six order columns in Table 4 for this scheme/mass block. The plot legends report the two fitted orders and the accumulated runtime/memory cost for each curve.



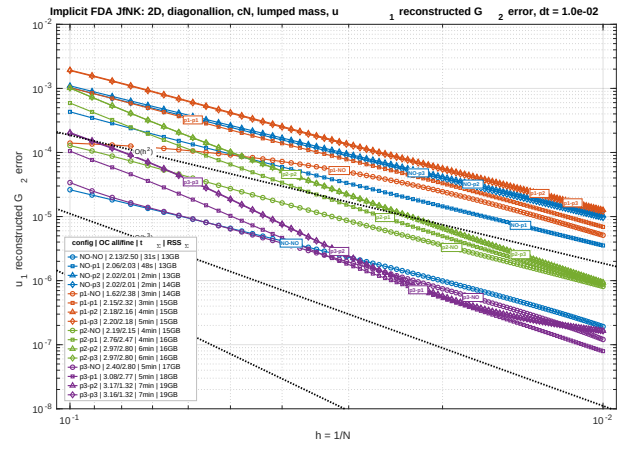
(a) $\|V_m\|_{L_2}$



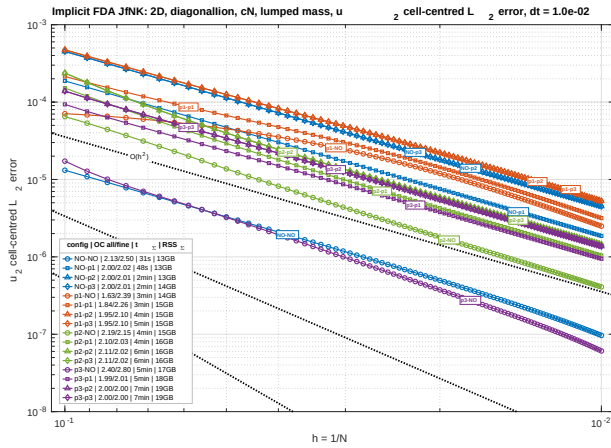
(b) $\|V_m\|_{G_2}$



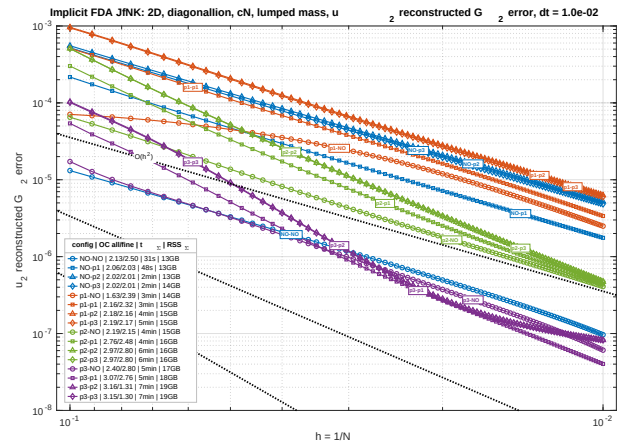
(c) $\|u_1\|_{L_2}$



(d) $\|u_1\|_{G_2}$



(e) $\|u_2\|_{L_2}$



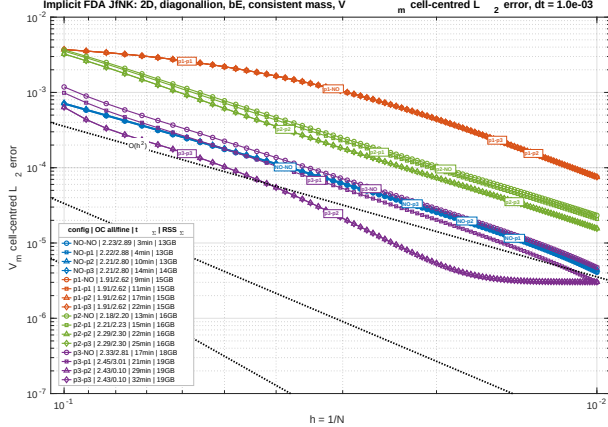
(f) $\|u_2\|_{G_2}$

Figure 12: 2D convergence curves for diagonalIion, $\Delta t = 10^{-2}$, cN with lump. mass. The six panels correspond to the six order columns in Table 4 for this scheme/mass block. The plot legends report the two fitted orders and the accumulated runtime/memory cost for each curve.

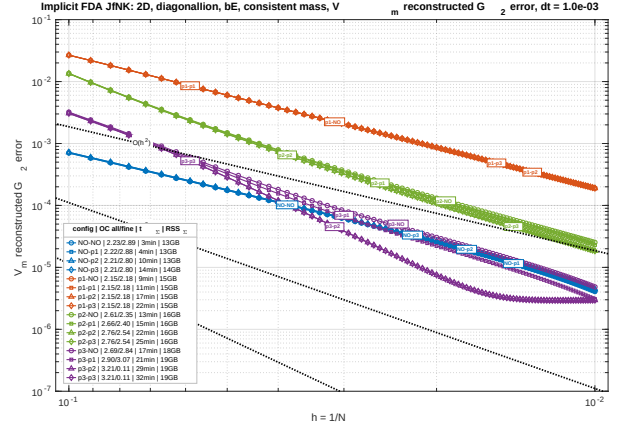
$$\Delta t = 10^{-3}$$

Table 5: 2D manufactured results for `diagonalIion` with $\Delta t = 10^{-3}$. The first six convergence columns are log-log fits over meshes $N = 10$ –100; the next six columns are fits over $N = 70$ –100. The time column is the sum of the reported wall-clock total over all mesh runs in the row, and the memory column is the sum of the reported peak resident memory in MB over the same mesh runs. The corresponding convergence figures are Figs. 13, 14, 15, 16.

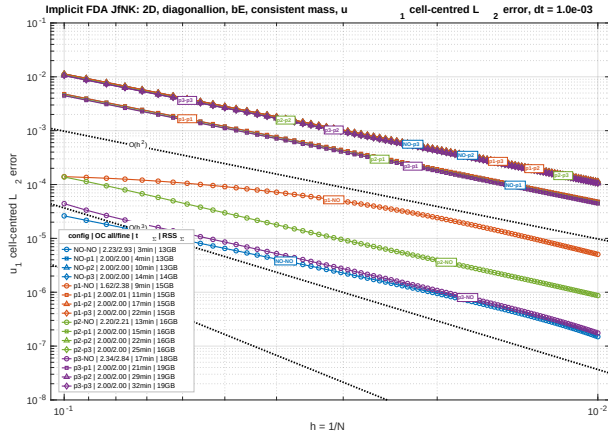
V_m	I_{ion}	Scheme	Mass	Fit $N = 10$ –100						Fit $N = 70$ –100						t_{Σ} [s]	RSS $_{\Sigma}$ [MB]
				V_m	V_m^G	u_1	u_1^G	u_2	u_2^G	V_m	V_m^G	u_1	u_1^G	u_2	u_2^G		
NO	NO	bE	cons.	2.226	2.226	2.233	2.233	2.233	2.233	2.895	2.895	2.926	2.926	2.926	2.926	159.22	13408.0
NO	p1	bE	cons.	2.224	2.224	2.000	2.061	2.009	2.065	2.884	2.884	2.003	2.045	2.036	2.045	253.20	13504.8
NO	p2	bE	cons.	2.208	2.208	1.999	2.017	2.004	2.019	2.799	2.799	2.001	2.014	2.017	2.014	629.33	13727.4
NO	p3	bE	cons.	2.208	2.208	1.999	2.021	2.004	2.023	2.799	2.799	2.001	2.017	2.018	2.017	821.01	13867.6
p1	NO	bE	cons.	1.914	2.151	1.622	1.622	1.632	1.632	2.621	2.183	2.382	2.382	2.388	2.388	552.01	15219.4
p1	p1	bE	cons.	1.914	2.151	1.998	2.158	1.844	2.165	2.621	2.183	2.008	2.320	2.262	2.320	652.06	15311.9
p1	p2	bE	cons.	1.914	2.151	1.999	2.184	1.954	2.182	2.620	2.183	2.002	2.164	2.109	2.160	1000.3	15499.2
p1	p3	bE	cons.	1.914	2.151	1.999	2.198	1.954	2.197	2.620	2.183	2.002	2.180	2.109	2.176	1294.6	15621.1
p2	NO	bE	cons.	2.182	2.613	2.196	2.196	2.197	2.197	2.202	2.355	2.211	2.211	2.211	2.211	793.69	16184.7
p2	p1	bE	cons.	2.209	2.657	2.004	2.732	2.113	2.736	2.225	2.401	2.003	2.483	2.047	2.490	916.91	16250.6
p2	p2	bE	cons.	2.291	2.764	2.004	2.952	2.121	2.951	2.301	2.538	2.002	2.786	2.027	2.789	1339.4	16479.5
p2	p3	bE	cons.	2.291	2.764	2.004	2.950	2.121	2.949	2.301	2.538	2.002	2.789	2.027	2.793	1483.3	16563.5
p3	NO	bE	cons.	2.332	2.691	2.342	2.342	2.345	2.345	2.807	2.839	2.839	2.839	2.838	2.838	1028.3	18919.7
p3	p1	bE	cons.	2.452	2.900	2.002	2.875	1.993	2.870	3.012	3.067	2.003	3.115	2.012	3.105	1287.0	19050.1
p3	p2	bE	cons.	2.434	3.214	2.000	3.534	2.000	3.532	0.103	0.114	2.002	0.580	2.008	0.519	1754.4	19238.8
p3	p3	bE	cons.	2.434	3.214	2.000	3.525	2.000	3.524	0.103	0.114	2.002	0.549	2.008	0.488	1942.1	19364.6
NO	NO	bE	lump.	2.226	2.226	2.233	2.233	2.233	2.233	2.895	2.895	2.926	2.926	2.926	2.926	161.39	13405.1
NO	p1	bE	lump.	2.224	2.224	2.000	2.061	2.009	2.065	2.884	2.884	2.003	2.045	2.036	2.045	256.62	13514.6
NO	p2	bE	lump.	2.208	2.208	1.999	2.017	2.004	2.019	2.799	2.799	2.001	2.014	2.017	2.014	634.84	13707.8
NO	p3	bE	lump.	2.208	2.208	1.999	2.021	2.004	2.023	2.799	2.799	2.001	2.017	2.018	2.017	923.19	13864.1
p1	NO	bE	lump.	1.914	2.151	1.622	1.622	1.632	1.632	2.621	2.183	2.382	2.382	2.388	2.388	555.55	14746.5
p1	p1	bE	lump.	1.914	2.151	1.998	2.158	1.844	2.165	2.621	2.183	2.008	2.320	2.262	2.320	658.88	15135.6
p1	p2	bE	lump.	1.914	2.151	1.999	2.184	1.954	2.182	2.620	2.183	2.002	2.164	2.109	2.160	1039.7	15303.1
p1	p3	bE	lump.	1.914	2.151	1.999	2.198	1.954	2.197	2.620	2.183	2.002	2.180	2.109	2.176	1286.7	15418.5
p2	NO	bE	lump.	2.197	2.656	2.213	2.213	2.214	2.214	2.225	2.402	2.234	2.234	2.235	2.235	847.77	15233.3
p2	p1	bE	lump.	2.230	2.704	2.004	2.775	2.108	2.779	2.254	2.459	2.003	2.546	2.044	2.552	990.18	15979.0
p2	p2	bE	lump.	2.335	2.820	2.004	2.979	2.115	2.977	2.355	2.626	2.002	2.847	2.025	2.850	1417.1	16249.6
p2	p3	bE	lump.	2.335	2.820	2.004	2.976	2.116	2.975	2.355	2.626	2.002	2.850	2.025	2.852	1545.4	16293.6
p3	NO	bE	lump.	2.535	3.005	2.554	2.554	2.557	2.557	3.520	3.545	3.591	3.591	3.590	3.590	991.73	17122.1
p3	p1	bE	lump.	2.577	3.171	2.002	3.148	1.990	3.145	2.412	2.613	2.003	2.546	2.007	2.578	1248.4	18646.9
p3	p2	bE	lump.	1.984	2.690	2.000	3.053	1.998	3.042	1.092	1.046	2.002	1.069	2.006	1.053	1710.2	18851.9
p3	p3	bE	lump.	1.984	2.690	2.000	3.040	1.998	3.030	1.092	1.046	2.002	1.062	2.006	1.046	1913.1	18989.3
NO	NO	cN	cons.	1.997	1.997	1.998	1.998	1.998	1.998	2.000	2.000	2.004	2.004	2.004	2.004	149.66	13424.6
NO	p1	cN	cons.	1.997	1.997	1.999	2.050	1.998	2.053	2.000	2.000	2.000	2.007	2.000	2.007	244.05	13528.4
NO	p2	cN	cons.	1.997	1.997	1.999	2.013	1.998	2.015	1.999	1.999	2.000	2.002	2.000	2.002	640.39	13774.4
NO	p3	cN	cons.	1.997	1.997	1.999	2.016	1.999	2.018	1.999	1.999	2.000	2.002	2.000	2.002	876.95	13879.3
p1	NO	cN	cons.	1.912	2.147	1.616	1.616	1.626	1.626	2.611	2.168	2.380	2.380	2.385	2.385	495.64	15202.4
p1	p1	cN	cons.	1.912	2.147	1.997	2.152	1.836	2.158	2.611	2.168	2.005	2.309	2.249	2.308	607.77	15237.8
p1	p2	cN	cons.	1.913	2.147	1.998	2.181	1.949	2.179	2.611	2.168	2.001	2.155	2.096	2.151	1026.6	15466.3
p1	p3	cN	cons.	1.913	2.147	1.999	2.194	1.949	2.193	2.611	2.168	2.001	2.170	2.096	2.166	1239.2	15664.5
p2	NO	cN	cons.	2.128	2.568	2.142	2.142	2.143	2.143	2.020	2.183	2.025	2.025	2.026	2.026	656.90	16196.8
p2	p1	cN	cons.	2.150	2.609	2.003	2.689	2.102	2.694	2.023	2.214	2.001	2.304	2.011	2.312	797.10	16195.7
p2	p2	cN	cons.	2.214	2.709	2.003	2.924	2.114	2.923	2.032	2.308	2.000	2.644	2.007	2.650	1156.2	16445.5
p2	p3	cN	cons.	2.214	2.709	2.004	2.923	2.114	2.922	2.032	2.308	2.000	2.650	2.007	2.656	1321.2	16561.5
p3	NO	cN	cons.	2.125	2.485	2.132	2.132	2.135	2.135	2.013	2.050	2.025	2.025	2.024	2.024	1025.7	18970.0
p3	p1	cN	cons.	2.187	2.637	2.001	2.598	1.989	2.596	2.020	2.073	2.000	2.068	1.999	2.066	1184.7	19022.5
p3	p2	cN	cons.	3.538	3.992	1.999	4.003	1.997	4.010	4.323	4.279	2.000	4.194	2.001	4.195	1661.8	19310.4
p3	p3	cN	cons.	3.538	3.992	1.999	4.000	1.998	4.008	4.323	4.279	2.000	4.198	2.001	4.200	1843.3	19375.8
NO	NO	cN	lump.	1.997	1.997	1.998	1.998	1.998	1.998	2.000	2.000	2.004	2.004	2.004	2.004	156.11	13419.8
NO	p1	cN	lump.	1.997	1.997	1.999	2.050	1.998	2.053	2.000	2.000	2.000	2.007	2.000	2.007	251.67	13523.6
NO	p2	cN	lump.	1.997	1.997	1.999	2.013	1.998	2.015	1.999	1.999	2.000	2.002	2.000	2.002	651.57	13747.8
NO	p3	cN	lump.	1.997	1.997	1.999	2.016	1.999	2.018	1.999	1.999	2.000	2.002	2.000	2.002	819.07	13881.2
p1	NO	cN	lump.	1.912	2.147	1.616	1.616	1.626	1.626	2.611	2.168	2.380	2.380	2.385	2.385	499.23	14822.1
p1	p1	cN	lump.	1.912	2.147	1.997	2.152	1.836	2.158	2.611	2.168	2.005	2.309	2.249	2.308	609.85	15190.0
p1	p2	cN	lump.	1.913	2.147	1.998	2.181	1.949	2.179	2.611	2.168	2.001	2.155	2.096	2.151	1028.5	15386.9
p1	p3	cN	lump.	1.913	2.147	1.999	2.194	1.949	2.193	2.611	2.168	2.001	2.170	2.096	2.166	1243.6	15469.5
p2	NO	cN	lump.	2.137	2.608	2.153	2.153	2.154	2.154	2.021	2.214	2.026	2.026	2.027	2.027	741.96	15312.9
p2	p1	cN	lump.	2.164	2.653	2.003	2.730	2.098	2.734	2.025	2.253	2.000	2.353	2.010	2.361	873.94	16107.6
p2	p2	cN	lump.	2.244	2.762	2.003	2.952	2.109	2.951	2.036	2.374	2.000	2.710	2.006	2.715	1302.4	16345.2
p2	p3	cN	lump.	2.244	2.762	2.003	2.950	2.110	2.950	2.036	2.374	2.000	2.715	2.006	2.720	1446.5	16417.1
p3	NO	cN	lump.	2.187	2.667	2.201	2.201	2.204	2.204	2.021	2.081	2.042	2.042	2.041	2.041	987.88	17504.9
p3	p1	cN	lump.	2.315	2.899	2.001	2.866	1.986	2.859	2.041	2.131	2.000	2.124	1.999	2.118	1176.2	18833.6
p3	p2	cN	lump.	2.369	3.091	1.999	3.441	1.996	3.435	2.031	2.018	2.000	2.165	2.000	2.138	1604.4	19112.6
p3	p3	cN	lump.	2.369	3.091	1.999	3.431	1.996	3.424	2.031	2.018	2.000	2.153	2.000	2.126	1818.9	19229.9



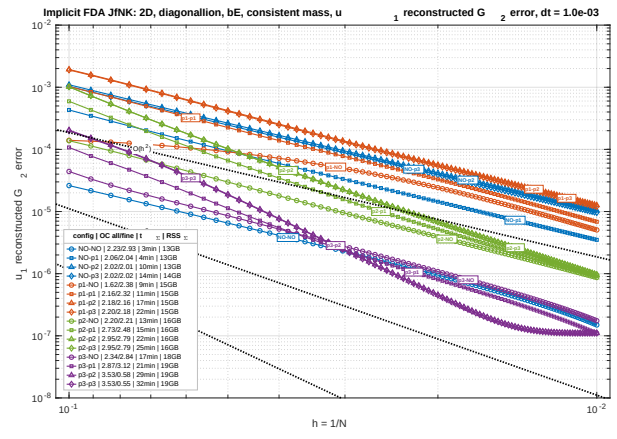
(a) $\|V_m\|_{L_2}$



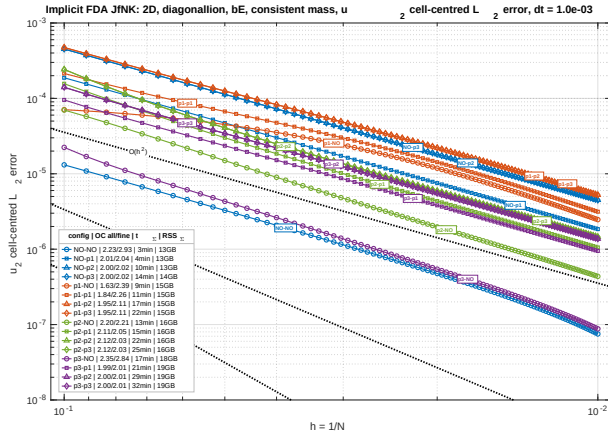
(b) $\|V_m\|_{G_2}$



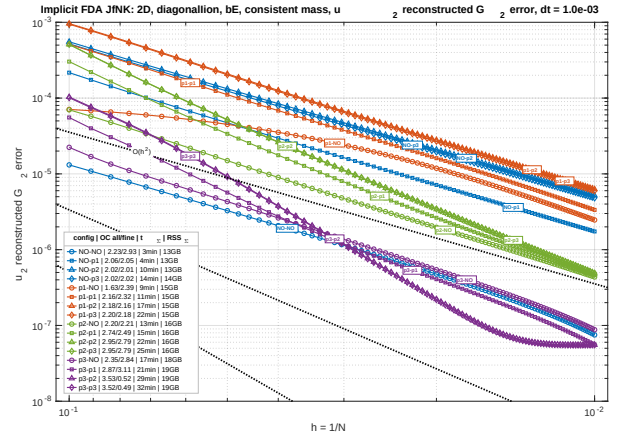
(c) $\|u_1\|_{L_2}$



(d) $\|u_1\|_{G_2}$

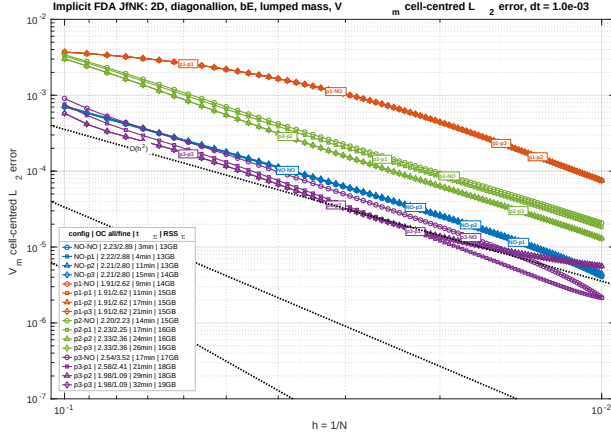


(e) $\|u_2\|_{L_2}$

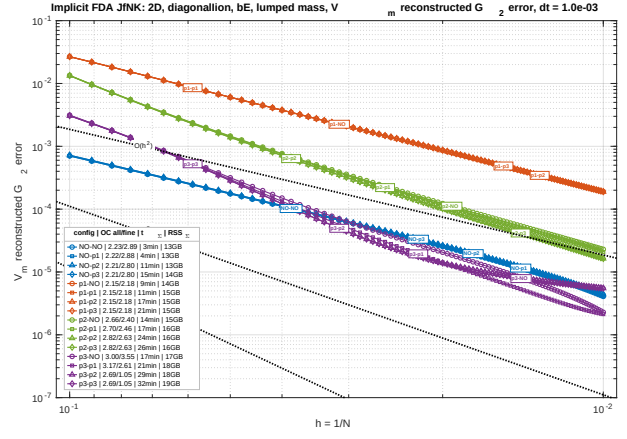


(f) $\|u_2\|_{G_2}$

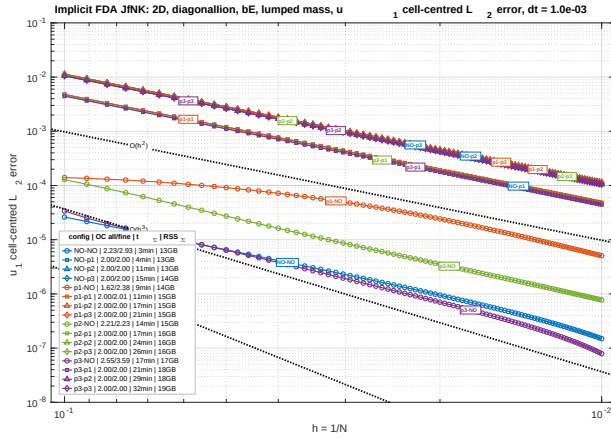
Figure 13: 2D convergence curves for diagonallion, $\Delta t = 10^{-3}$, bE with cons. mass. The six panels correspond to the six order columns in Table 5 for this scheme/mass block. The plot legends report the two fitted orders and the accumulated runtime/memory cost for each curve.



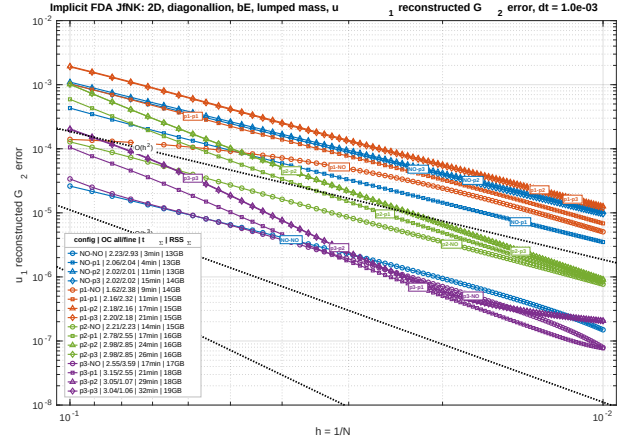
(a) $\|V_m\|_{L_2}$



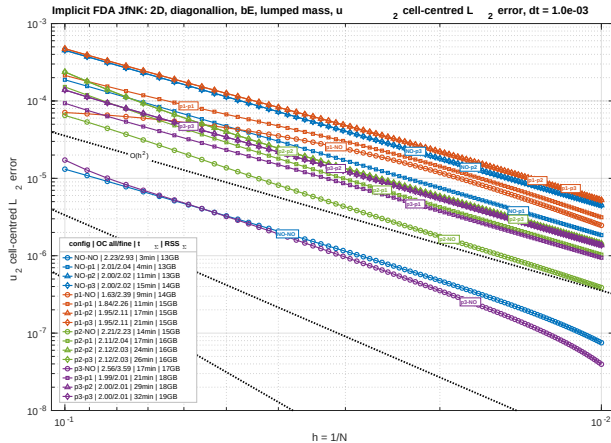
(b) $\|V_m\|_{G_2}$



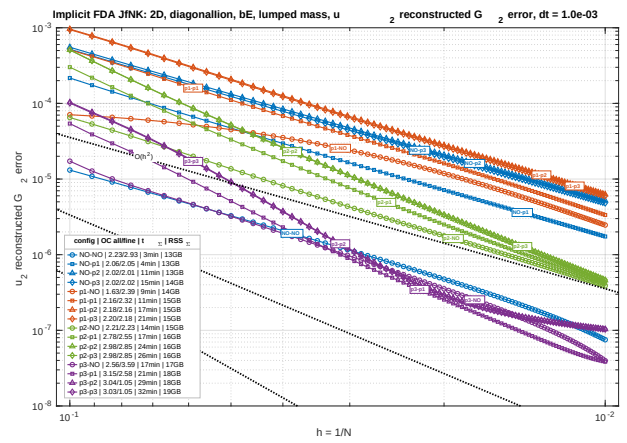
(c) $\|u_1\|_{L_2}$



(d) $\|u_1\|_{G_2}$

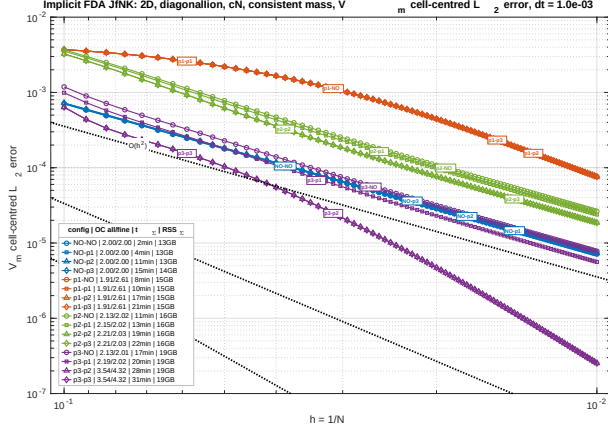


(e) $\|u_2\|_{L_2}$

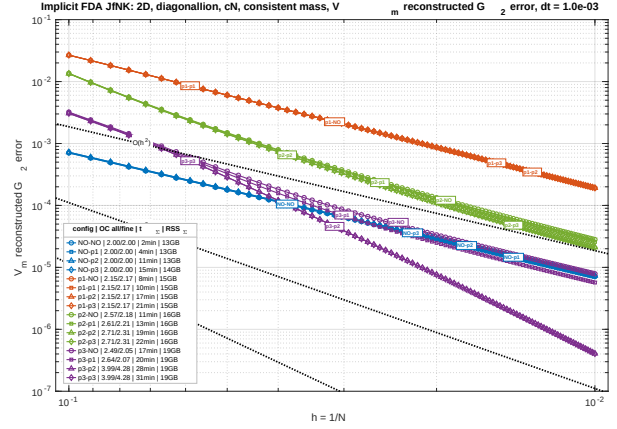


(f) $\|u_2\|_{G_2}$

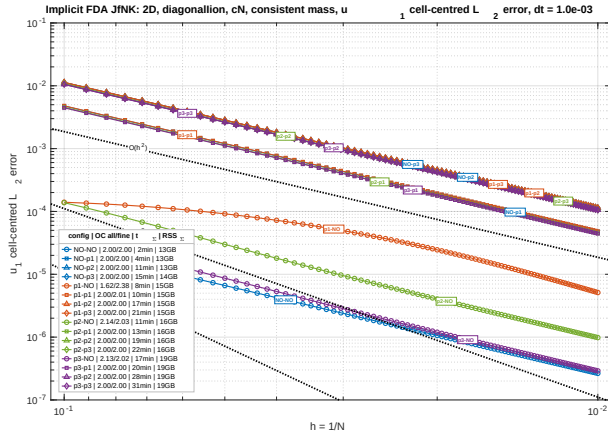
Figure 14: 2D convergence curves for diagonalion, $\Delta t = 10^{-3}$, bE with lump. mass. The six panels correspond to the six order columns in Table 5 for this scheme/mass block. The plot legends report the two fitted orders and the accumulated runtime/memory cost for each curve.



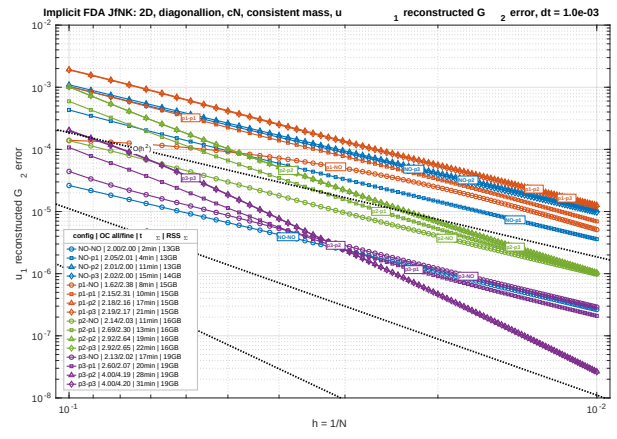
(a) $\|V_m\|_{L_2}$



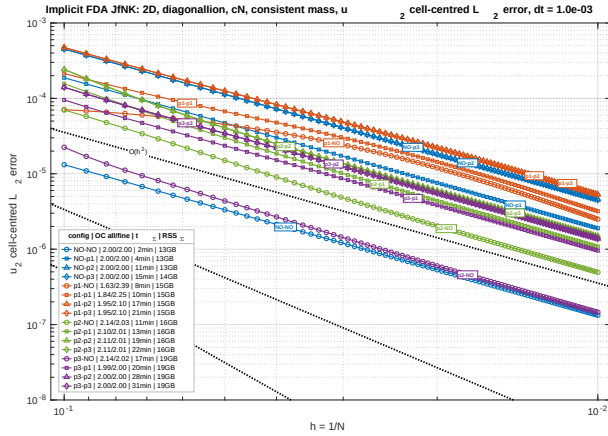
(b) $\|V_m\|_{G_2}$



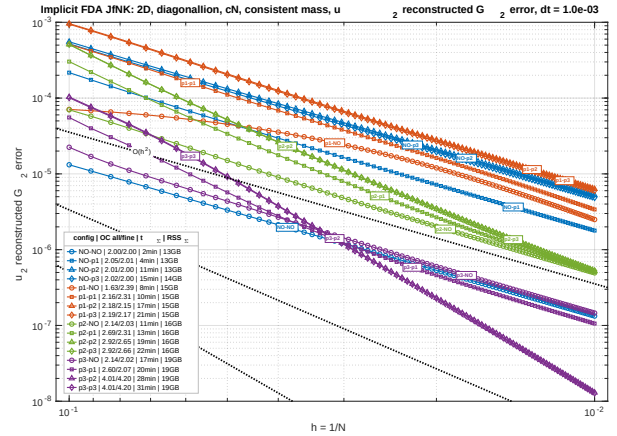
(c) $\|u_1\|_{L_2}$



(d) $\|u_1\|_{G_2}$

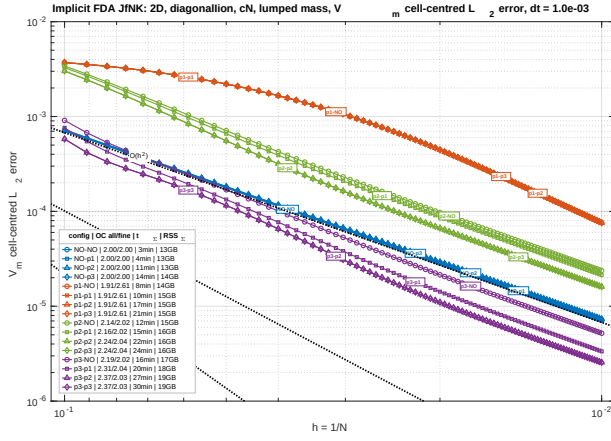


(e) $\|u_2\|_{L_2}$

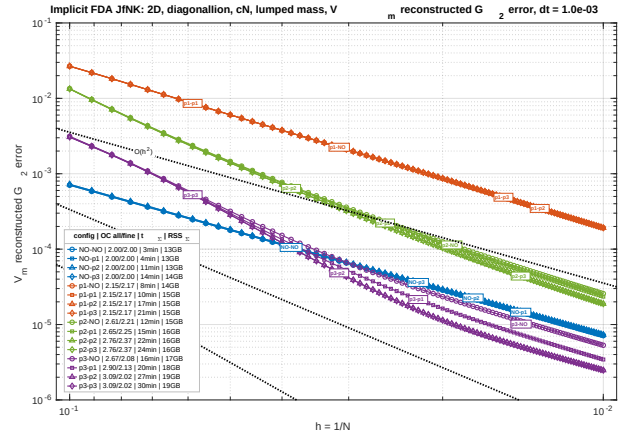


(f) $\|u_2\|_{G_2}$

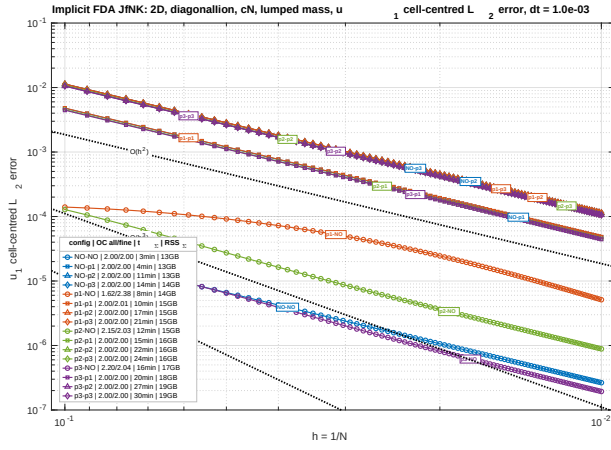
Figure 15: 2D convergence curves for diagonalion, $\Delta t = 10^{-3}$, cN with cons. mass. The six panels correspond to the six order columns in Table 5 for this scheme/mass block. The plot legends report the two fitted orders and the accumulated runtime/memory cost for each curve.



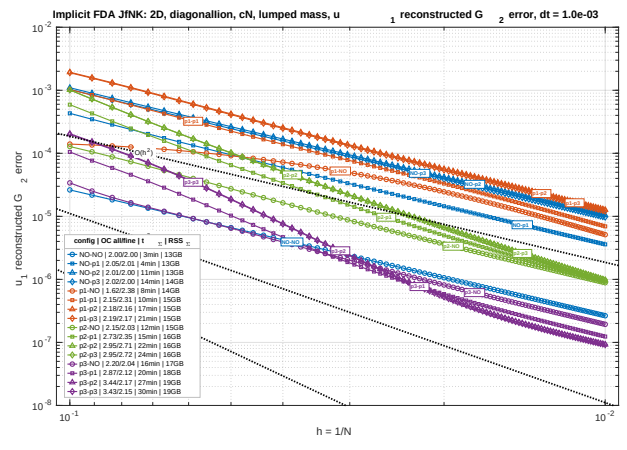
(a) $\|V_m\|_{L_2}$



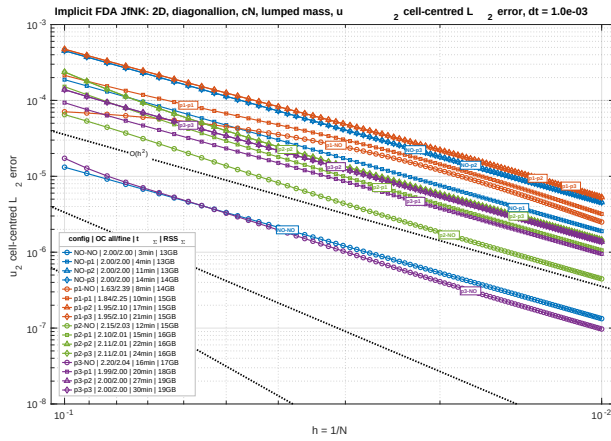
(b) $\|V_m\|_{G_2}$



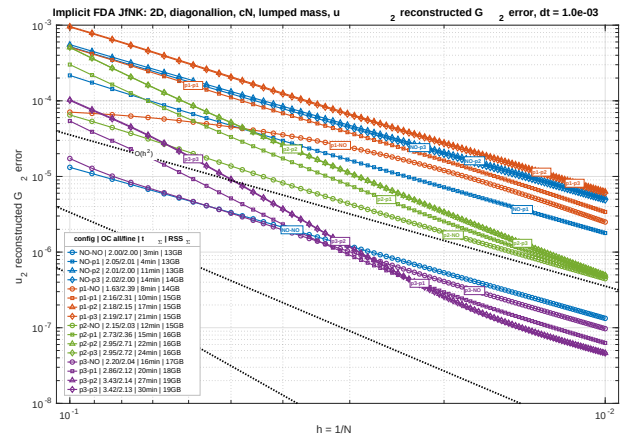
(c) $\|u_1\|_{L_2}$



(d) $\|u_1\|_{G_2}$



(e) $\|u_2\|_{L_2}$



(f) $\|u_2\|_{G_2}$

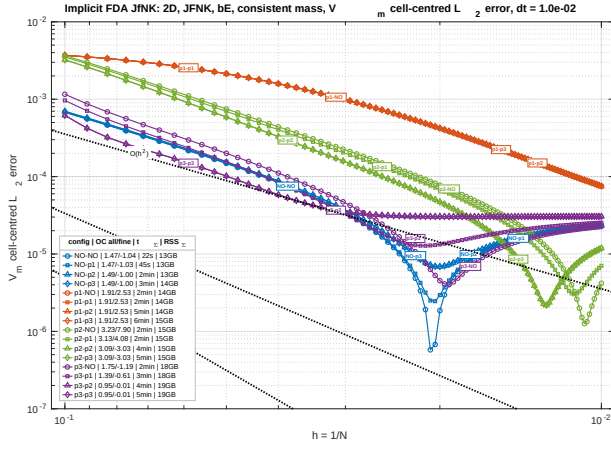
Figure 16: 2D convergence curves for diagonalion, $\Delta t = 10^{-3}$, cN with lump. mass. The six panels correspond to the six order columns in Table 5 for this scheme/mass block. The plot legends report the two fitted orders and the accumulated runtime/memory cost for each curve.

10.1.3 JFNK

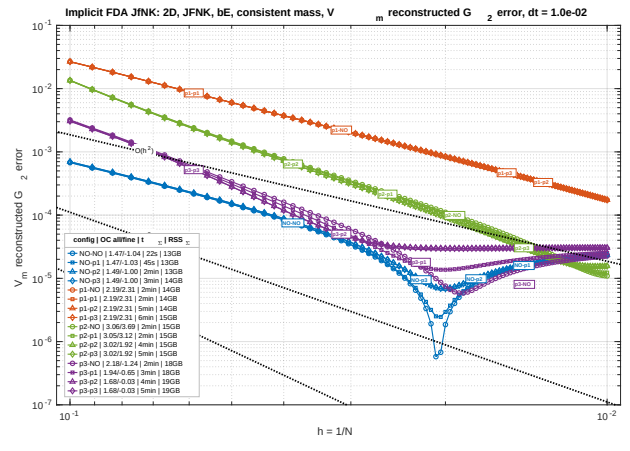
$$\Delta t = 10^{-2}$$

Table 6: 2D manufactured results for JFNK with $\Delta t = 10^{-2}$. The first six convergence columns are log-log fits over meshes $N = 10$ –100; the next six columns are fits over $N = 70$ –100. The time column is the sum of the reported wall-clock total over all mesh runs in the row, and the memory column is the sum of the reported peak resident memory in MB over the same mesh runs. The corresponding convergence figures are Figs. 17, 18, 19, 20.

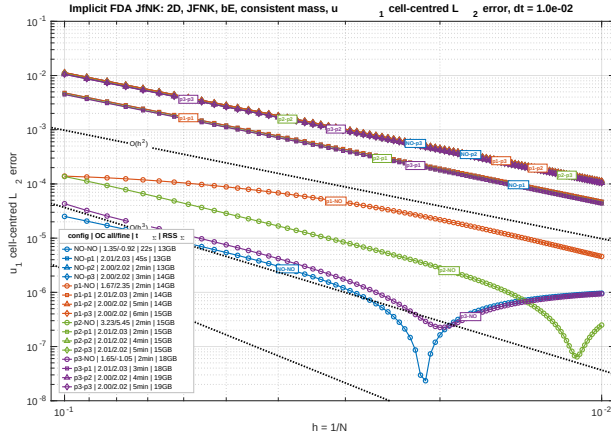
V_m	I_{ion}	Scheme	Mass	Fit $N = 10$ –100						Fit $N = 70$ –100						t_{Σ} [s]	RSS $_{\Sigma}$ [MB]
				V_m	V_m^G	u_1	u_1^G	u_2	u_2^G	V_m	V_m^G	u_1	u_1^G	u_2	u_2^G		
NO	NO	bE	cons.	1.473	1.473	1.349	1.349	1.349	1.349	-1.042	-1.042	-0.923	-0.923	-0.923	-0.923	22.362	13364.9
NO	p1	bE	cons.	1.470	1.470	2.008	2.177	2.118	2.181	-1.035	-1.035	2.029	2.429	2.397	2.429	44.714	13469.6
NO	p2	bE	cons.	1.486	1.486	2.003	2.054	2.058	2.056	-0.995	-0.995	2.015	2.137	2.203	2.138	123.62	13713.0
NO	p3	bE	cons.	1.486	1.486	2.003	2.064	2.058	2.067	-0.995	-0.995	2.015	2.162	2.203	2.164	189.96	13871.1
p1	NO	bE	cons.	1.911	2.188	1.671	1.671	1.681	1.681	2.531	2.308	2.347	2.347	2.351	2.351	100.14	14534.3
p1	p1	bE	cons.	1.911	2.188	2.007	2.219	1.915	2.226	2.531	2.308	2.033	2.426	2.388	2.430	141.11	14580.1
p1	p2	bE	cons.	1.912	2.188	2.003	2.218	2.008	2.216	2.531	2.308	2.016	2.251	2.242	2.249	297.03	14802.6
p1	p3	bE	cons.	1.912	2.188	2.003	2.235	2.008	2.234	2.531	2.308	2.016	2.278	2.242	2.277	343.91	14915.8
p2	NO	bE	cons.	3.232	3.056	3.225	3.225	3.225	3.225	7.895	3.694	5.447	5.447	5.484	5.484	95.284	15398.1
p2	p1	bE	cons.	3.125	3.053	2.013	3.035	2.179	3.034	4.083	3.116	2.032	2.760	2.090	2.783	123.50	15441.4
p2	p2	bE	cons.	3.086	3.017	2.008	3.032	2.158	3.029	-3.034	1.918	2.016	2.489	2.061	2.502	217.67	15566.9
p2	p3	bE	cons.	3.086	3.017	2.009	3.028	2.158	3.026	-3.034	1.918	2.017	2.501	2.061	2.513	274.96	15680.9
p3	NO	bE	cons.	1.749	2.178	1.650	1.650	1.654	1.654	-1.193	-1.240	-1.051	-1.051	-1.052	-1.052	123.49	18808.9
p3	p1	bE	cons.	1.394	1.937	2.011	1.801	1.972	1.804	-0.612	-0.646	2.031	-0.582	1.754	-0.590	155.78	18854.2
p3	p2	bE	cons.	0.948	1.680	2.004	1.999	1.994	1.987	-0.005	-0.028	2.016	-0.045	1.894	-0.048	266.70	18966.6
p3	p3	bE	cons.	0.948	1.680	2.004	1.986	1.995	1.974	-0.005	-0.028	2.016	-0.046	1.894	-0.048	303.83	19018.8
NO	NO	bE	lump.	1.473	1.473	1.349	1.349	1.349	1.349	-1.042	-1.042	-0.923	-0.923	-0.923	-0.923	23.559	13356.5
NO	p1	bE	lump.	1.470	1.470	2.008	2.177	2.118	2.181	-1.035	-1.035	2.029	2.429	2.397	2.429	45.776	13461.2
NO	p2	bE	lump.	1.486	1.486	2.003	2.054	2.058	2.056	-0.995	-0.995	2.015	2.137	2.203	2.138	127.89	13718.4
NO	p3	bE	lump.	1.486	1.486	2.003	2.064	2.058	2.067	-0.995	-0.995	2.015	2.162	2.203	2.164	167.34	13873.7
p1	NO	bE	lump.	1.911	2.188	1.671	1.671	1.681	1.681	2.531	2.308	2.347	2.347	2.351	2.351	90.101	14131.3
p1	p1	bE	lump.	1.911	2.188	2.007	2.219	1.915	2.226	2.531	2.308	2.033	2.426	2.388	2.430	140.11	14374.4
p1	p2	bE	lump.	1.912	2.188	2.003	2.218	2.008	2.216	2.531	2.308	2.016	2.251	2.242	2.249	294.55	14623.6
p1	p3	bE	lump.	1.912	2.188	2.003	2.235	2.008	2.234	2.531	2.308	2.016	2.278	2.242	2.277	358.55	14745.8
p2	NO	bE	lump.	3.227	3.058	3.176	3.176	3.177	3.177	4.727	3.144	1.951	1.951	1.998	1.998	90.842	14786.7
p2	p1	bE	lump.	3.068	3.037	2.013	3.014	2.163	3.014	0.954	2.507	2.032	2.316	2.042	2.351	122.37	15059.8
p2	p2	bE	lump.	2.922	2.970	2.008	3.011	2.147	3.010	-5.003	1.425	2.016	2.298	2.038	2.313	221.43	15304.3
p2	p3	bE	lump.	2.922	2.970	2.008	3.008	2.147	3.007	-5.003	1.425	2.017	2.314	2.038	2.329	273.46	15430.0
p3	NO	bE	lump.	1.324	1.925	1.267	1.267	1.268	1.268	-0.661	-0.697	-0.592	-0.592	-0.593	-0.593	118.60	17127.1
p3	p1	bE	lump.	1.102	1.774	2.010	1.651	1.956	1.649	-0.256	-0.284	2.031	-0.254	1.717	-0.260	157.42	17227.1
p3	p2	bE	lump.	1.018	1.647	2.004	1.942	1.986	1.929	0.212	0.192	2.016	0.164	1.873	0.162	260.08	17484.8
p3	p3	bE	lump.	1.018	1.647	2.004	1.929	1.986	1.917	0.212	0.192	2.016	0.163	1.873	0.162	304.55	17603.0
NO	NO	cN	cons.	1.992	1.992	2.135	2.135	2.134	2.134	1.985	1.985	2.500	2.500	2.497	2.497	19.770	13371.7
NO	p1	cN	cons.	1.992	1.992	2.000	2.057	2.005	2.061	1.985	1.985	2.002	2.031	2.023	2.031	37.604	13493.4
NO	p2	cN	cons.	1.992	1.992	1.999	2.015	2.002	2.017	1.986	1.986	2.001	2.010	2.011	2.010	106.95	13731.0
NO	p3	cN	cons.	1.992	1.992	1.999	2.019	2.002	2.021	1.986	1.986	2.001	2.011	2.011	2.011	137.89	13899.1
p1	NO	cN	cons.	1.911	2.146	1.616	1.616	1.626	1.626	2.612	2.168	2.381	2.381	2.387	2.387	94.212	14535.7
p1	p1	cN	cons.	1.911	2.146	1.998	2.154	1.838	2.160	2.612	2.168	2.007	2.316	2.258	2.316	127.47	14575.8
p1	p2	cN	cons.	1.912	2.146	1.999	2.182	1.951	2.181	2.612	2.168	2.002	2.161	2.105	2.157	252.80	14813.5
p1	p3	cN	cons.	1.912	2.146	1.999	2.196	1.951	2.195	2.612	2.168	2.002	2.177	2.105	2.173	308.98	14934.7
p2	NO	cN	cons.	2.127	2.567	2.176	2.176	2.176	2.176	2.016	2.179	2.139	2.139	2.139	2.139	88.693	15396.6
p2	p1	cN	cons.	2.149	2.608	2.004	2.716	2.109	2.720	2.019	2.210	2.002	2.416	2.034	2.422	113.04	15443.1
p2	p2	cN	cons.	2.212	2.708	2.004	2.942	2.118	2.941	2.026	2.303	2.001	2.734	2.020	2.738	202.26	15571.3
p2	p3	cN	cons.	2.212	2.708	2.004	2.940	2.118	2.940	2.026	2.303	2.001	2.739	2.020	2.743	241.61	15694.8
p3	NO	cN	cons.	2.121	2.481	2.256	2.256	2.258	2.258	1.999	2.037	2.472	2.472	2.469	2.469	117.12	18809.5
p3	p1	cN	cons.	2.181	2.631	2.002	2.762	1.992	2.757	2.003	2.056	2.002	2.661	2.008	2.650	151.50	18850.8
p3	p2	cN	cons.	3.520	3.971	2.000	3.744	1.999	3.748	4.169	4.123	2.001	1.457	2.006	1.396	223.70	18963.1
p3	p3	cN	cons.	3.520	3.971	2.000	3.738	1.999	3.742	4.169	4.123	2.001	1.419	2.006	1.355	268.17	19018.0
NO	NO	cN	lump.	1.992	1.992	2.135	2.135	2.134	2.134	1.985	1.985	2.500	2.500	2.497	2.497	18.653	13373.5
NO	p1	cN	lump.	1.992	1.992	2.000	2.057	2.005	2.061	1.985	1.985	2.002	2.031	2.023	2.031	37.825	13487.1
NO	p2	cN	lump.	1.992	1.992	1.999	2.015	2.002	2.017	1.986	1.986	2.001	2.010	2.011	2.010	104.42	13736.8
NO	p3	cN	lump.	1.992	1.992	1.999	2.019	2.002	2.021	1.986	1.986	2.001	2.011	2.011	2.011	137.54	13898.6
p1	NO	cN	lump.	1.911	2.146	1.616	1.616	1.626	1.626	2.612	2.168	2.381	2.381	2.387	2.387	92.871	14192.5
p1	p1	cN	lump.	1.911	2.146	1.998	2.154	1.838	2.160	2.612	2.168	2.007	2.316	2.258	2.316	127.98	14466.1
p1	p2	cN	lump.	1.912	2.146	1.999	2.182	1.951	2.181	2.612	2.168	2.002	2.161	2.105	2.157	251.60	14712.6
p1	p3	cN	lump.	1.912	2.146	1.999	2.196	1.951	2.195	2.612	2.168	2.002	2.177	2.105	2.173	314.57	14842.8
p2	NO	cN	lump.	2.136	2.606	2.191	2.191	2.191	2.191	2.017	2.209	2.153	2.153	2.153	2.153	88.700	14879.1
p2	p1	cN	lump.	2.162	2.652	2.004	2.759	2.104	2.762	2.020	2.248	2.002	2.473	2.032	2.480	112.08	15170.1
p2	p2	cN	lump.	2.242	2.761	2.004	2.969	2.113	2.968	2.029	2.368	2.001	2.798	2.018	2.801	199.92	15419.6
p2	p3	cN	lump.	2.242	2.761	2.004	2.967	2.113	2.966	2.029	2.368	2.001	2.802	2.018	2.805	240.03	15545.4
p3	NO	cN	lump.	2.181	2.661	2.398	2.398	2.399	2.399	2.001	2.061	2.804	2.804	2.797	2.797	116.23	17213.2
p3	p1	cN	lump.	2.307	2.891	2.001	3.077	1.989	3.069	2.016	2.106	2.002	2.766	2.005	2.765	149.01	17482.5
p3	p2	cN	lump.	2.382	3.105	1.999	3.171	1.997	3.163	2.075	2.063	2.001	1.325	2.004	1.308	232.09	17727.9
p3	p3	cN	lump.	2.382	3.105	2.000	3.159	1.998	3.151	2.075	2.063	2.001	1.316	2.004	1.299	265.23	17849.6



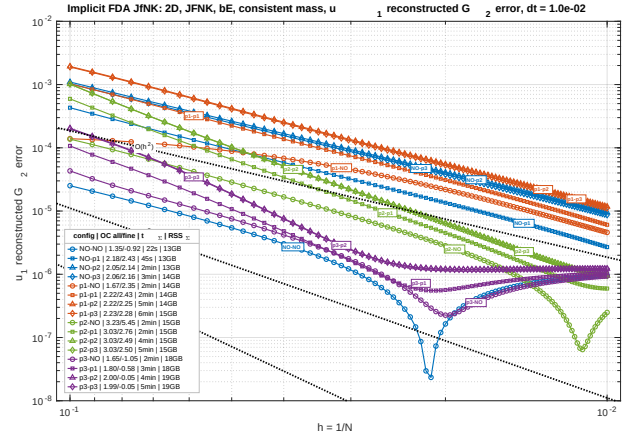
(a) $\|V_m\|_{L_2}$



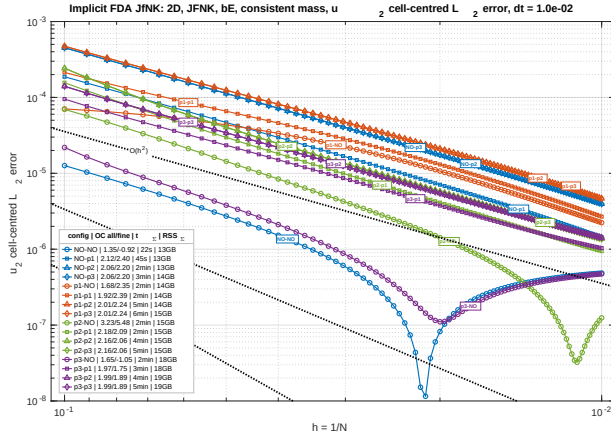
(b) $\|V_m\|_{G_2}$



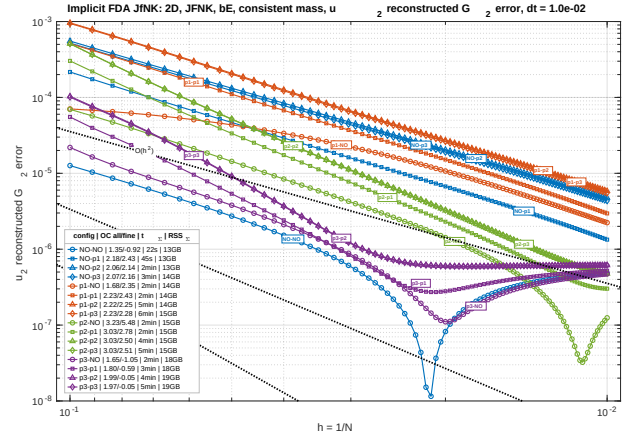
(c) $\|u_1\|_{L_2}$



(d) $\|u_1\|_{G_2}$

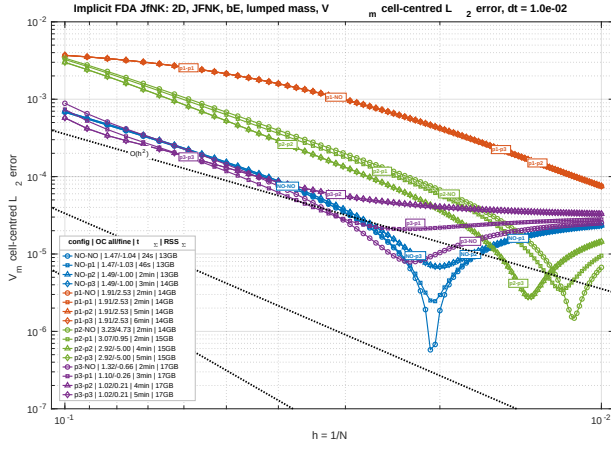


(e) $\|u_2\|_{L_2}$

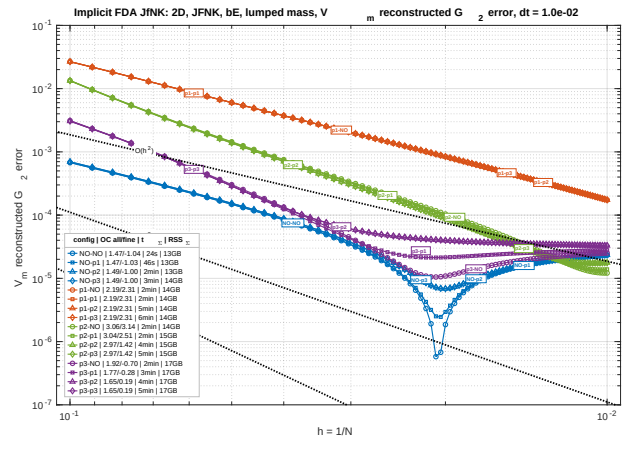


(f) $\|u_2\|_{G_2}$

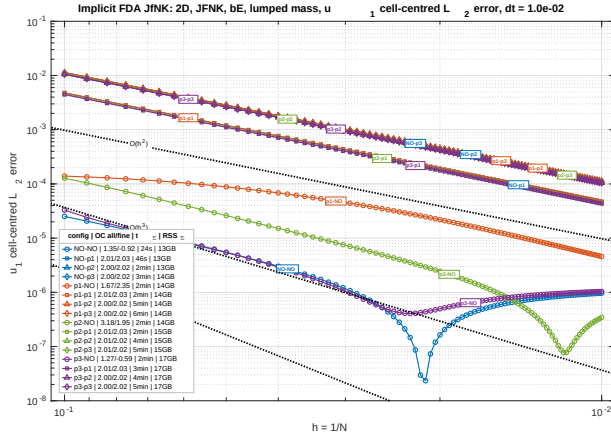
Figure 17: 2D convergence curves for JFNK, $\Delta t = 10^{-2}$, bE with cons. mass. The six panels correspond to the six order columns in Table 6 for this scheme/mass block. The plot legends report the two fitted orders and the accumulated runtime/memory cost for each curve.



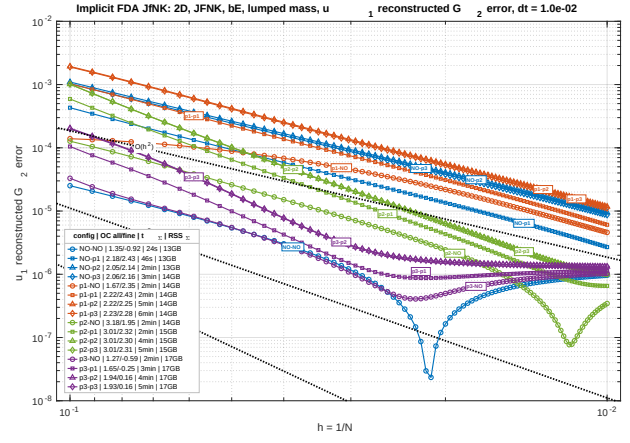
(a) $\|V_m\|_{L_2}$



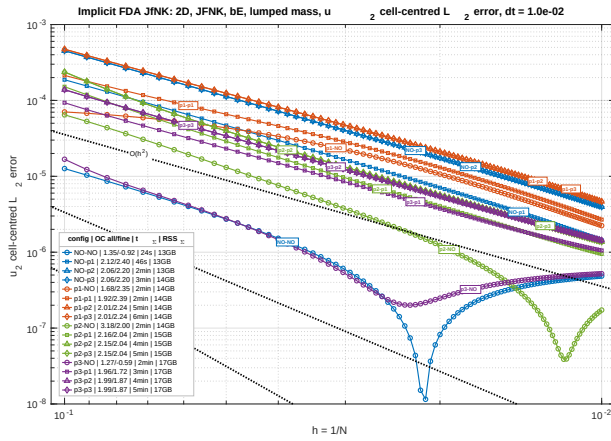
(b) $\|V_m\|_{G_2}$



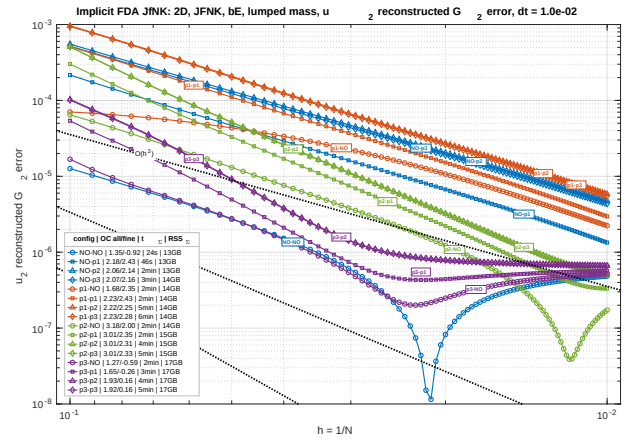
(c) $\|u_1\|_{L_2}$



(d) $\|u_1\|_{G_2}$

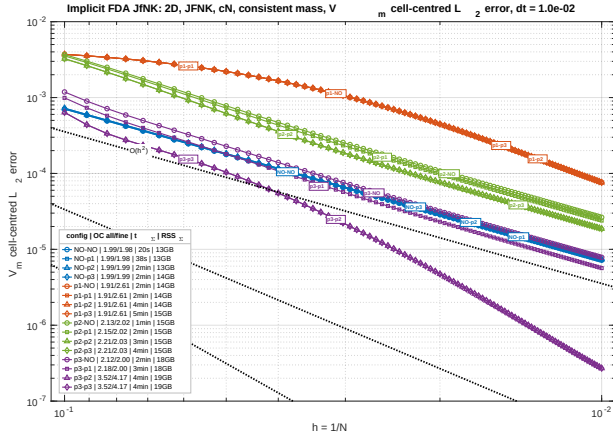


(e) $\|u_2\|_{L_2}$

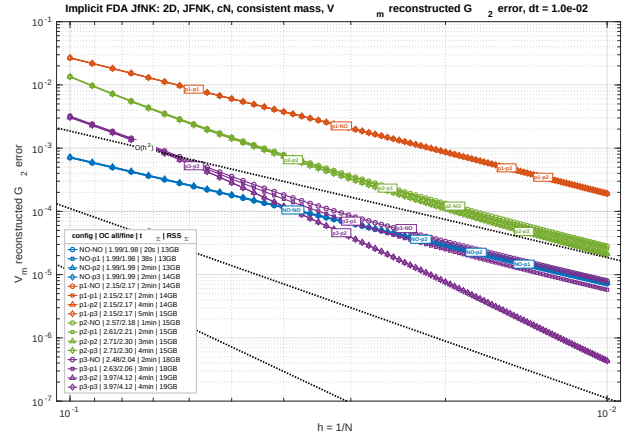


(f) $\|u_2\|_{G_2}$

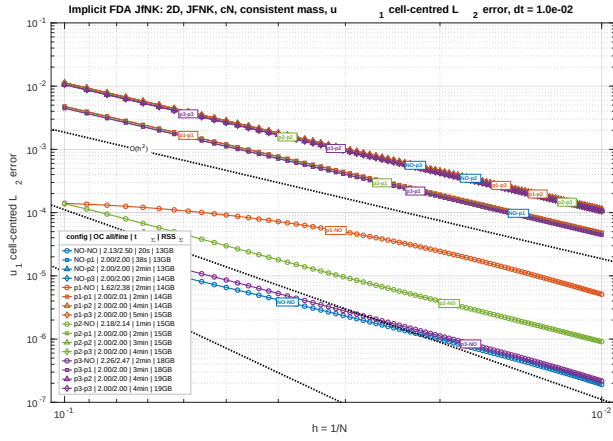
Figure 18: 2D convergence curves for JFNK, $\Delta t = 10^{-2}$, bE with lump. mass. The six panels correspond to the six order columns in Table 6 for this scheme/mass block. The plot legends report the two fitted orders and the accumulated runtime/memory cost for each curve.



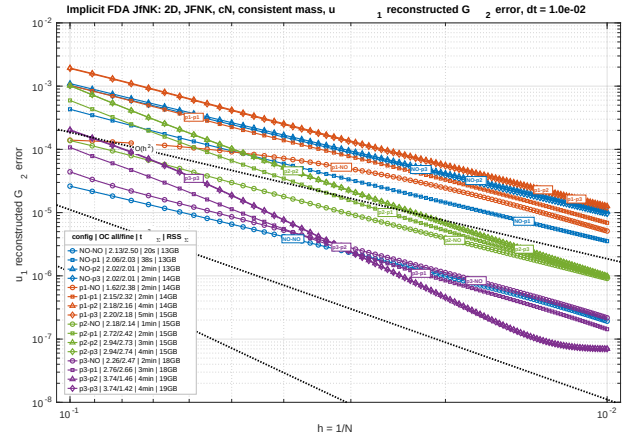
(a) $\|V_m\|_{L_2}$



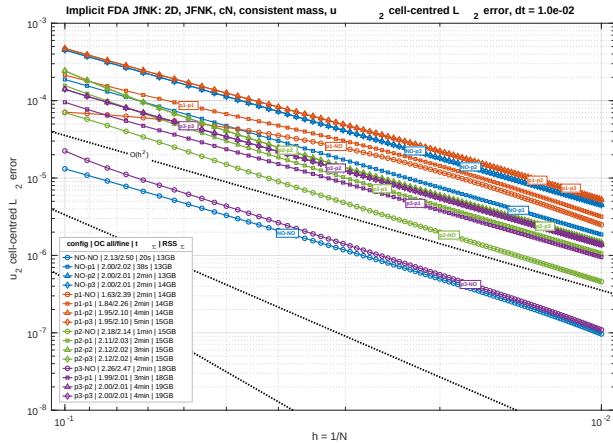
(b) $\|V_m\|_{G_2}$



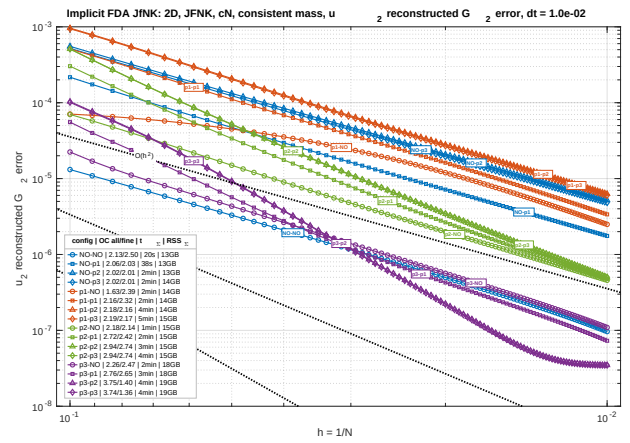
(c) $\|u_1\|_{L_2}$



(d) $\|u_1\|_{G_2}$

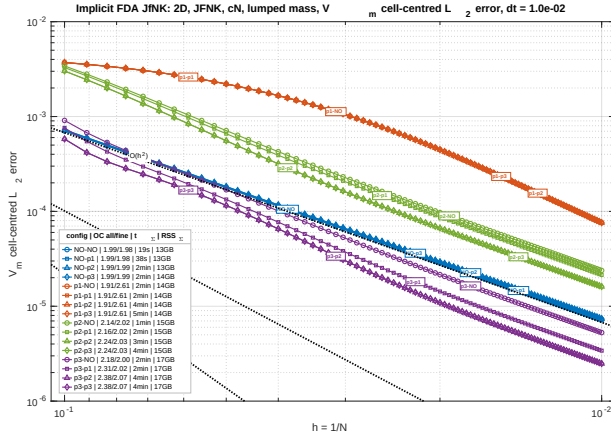


(e) $\|u_2\|_{L_2}$

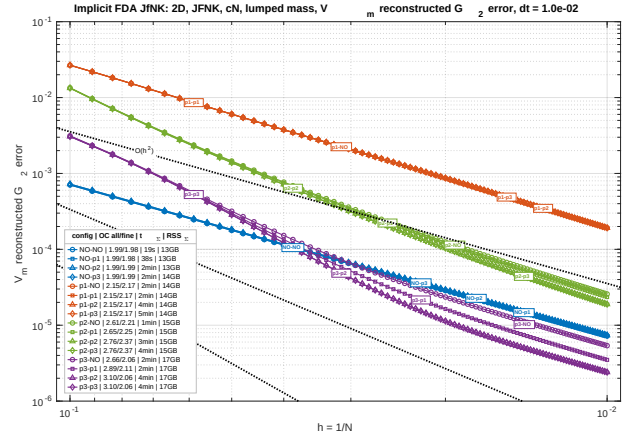


(f) $\|u_2\|_{G_2}$

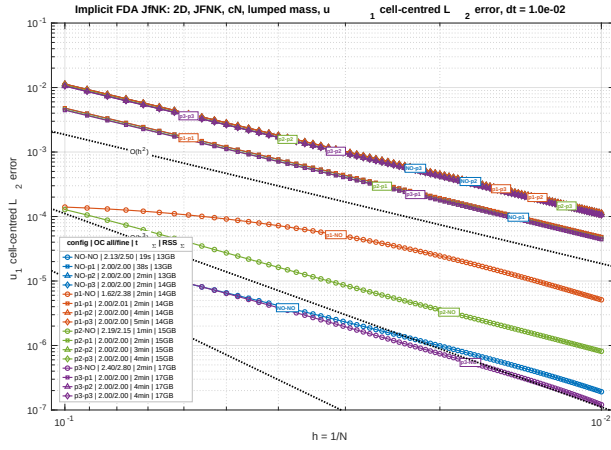
Figure 19: 2D convergence curves for JFNK, $\Delta t = 10^{-2}$, cN with cons. mass. The six panels correspond to the six order columns in Table 6 for this scheme/mass block. The plot legends report the two fitted orders and the accumulated runtime/memory cost for each curve.



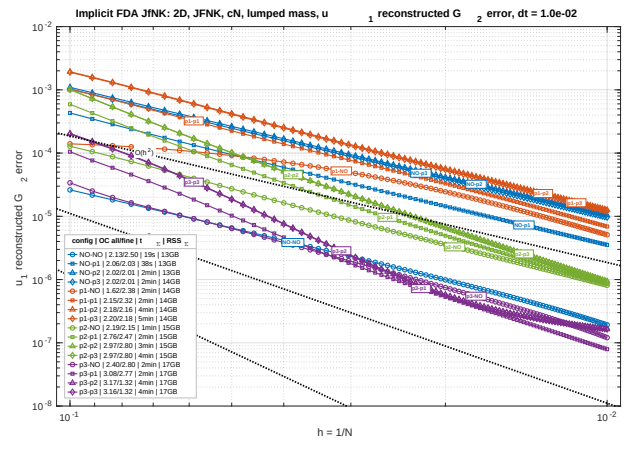
(a) $\|V_m\|_{L_2}$



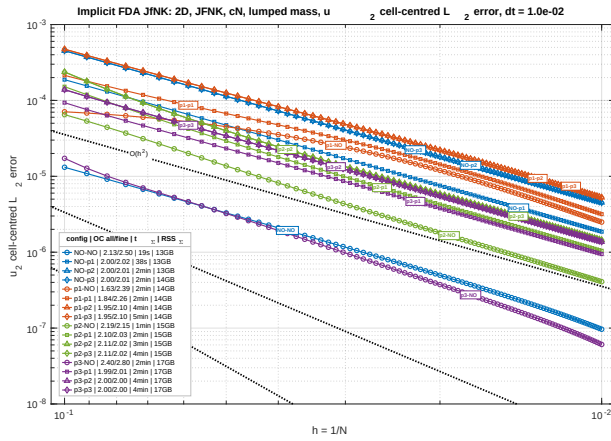
(b) $\|V_m\|_{G_2}$



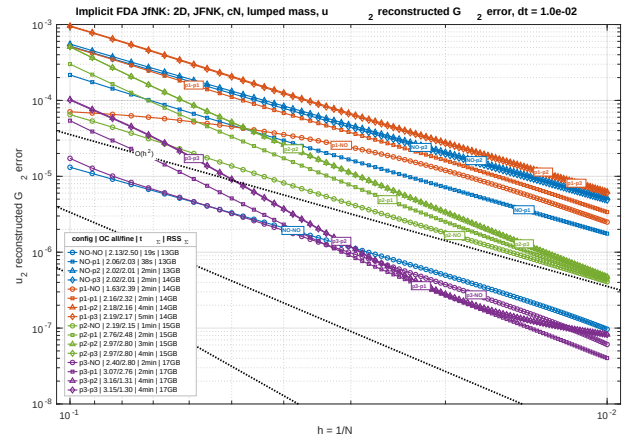
(c) $\|u_1\|_{L_2}$



(d) $\|u_1\|_{G_2}$



(e) $\|u_2\|_{L_2}$



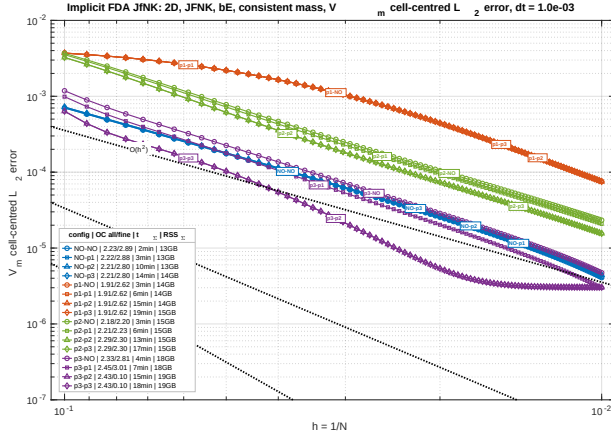
(f) $\|u_2\|_{G_2}$

Figure 20: 2D convergence curves for JfNK, $\Delta t = 10^{-2}$, cN with lump. mass. The six panels correspond to the six order columns in Table 6 for this scheme/mass block. The plot legends report the two fitted orders and the accumulated runtime/memory cost for each curve.

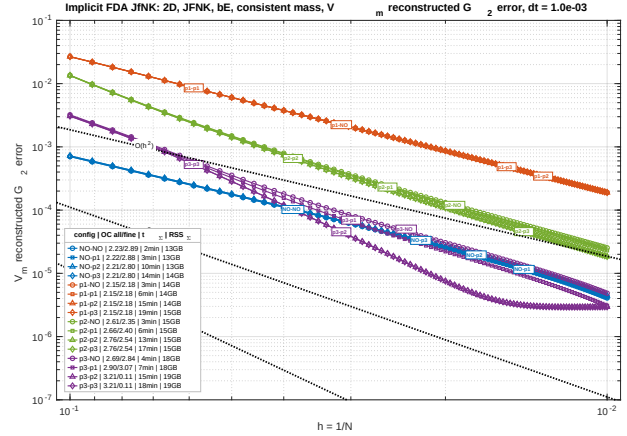
$$\Delta t = 10^{-3}$$

Table 7: 2D manufactured results for JFNK with $\Delta t = 10^{-3}$. The first six convergence columns are log-log fits over meshes $N = 10$ –100; the next six columns are fits over $N = 70$ –100. The time column is the sum of the reported wall-clock total over all mesh runs in the row, and the memory column is the sum of the reported peak resident memory in MB over the same mesh runs. The corresponding convergence figures are Figs. 21, 22, 23, 24.

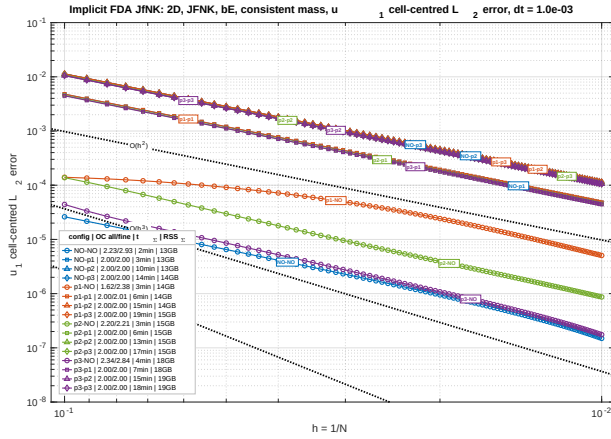
V_m	I_{ion}	Scheme	Mass	Fit $N = 10$ –100						Fit $N = 70$ –100						t_{Σ} [s]	RSS $_{\Sigma}$ [MB]
				V_m	V_m^G	u_1	u_1^G	u_2	u_2^G	V_m	V_m^G	u_1	u_1^G	u_2	u_2^G		
NO	NO	bE	cons.	2.226	2.226	2.233	2.233	2.233	2.233	2.895	2.895	2.926	2.926	2.926	2.926	110.31	13363.5
NO	p1	bE	cons.	2.224	2.224	2.000	2.061	2.009	2.065	2.884	2.884	2.003	2.045	2.036	2.045	207.73	13474.4
NO	p2	bE	cons.	2.208	2.208	1.999	2.017	2.004	2.019	2.799	2.799	2.001	2.014	2.017	2.014	596.45	13723.6
NO	p3	bE	cons.	2.208	2.208	1.999	2.021	2.004	2.023	2.799	2.799	2.001	2.017	2.018	2.017	864.18	13875.4
p1	NO	bE	cons.	1.914	2.151	1.622	1.622	1.632	1.632	2.621	2.183	2.382	2.382	2.388	2.388	203.73	14533.6
p1	p1	bE	cons.	1.914	2.151	1.998	2.158	1.844	2.165	2.621	2.183	2.008	2.320	2.262	2.320	351.78	14577.0
p1	p2	bE	cons.	1.914	2.151	1.999	2.184	1.954	2.182	2.620	2.183	2.002	2.164	2.109	2.160	881.80	14813.4
p1	p3	bE	cons.	1.914	2.151	1.999	2.198	1.954	2.197	2.620	2.183	2.002	2.180	2.109	2.176	1139.7	14928.1
p2	NO	bE	cons.	2.182	2.613	2.196	2.196	2.197	2.197	2.202	2.355	2.211	2.211	2.211	2.211	195.75	15400.6
p2	p1	bE	cons.	2.209	2.657	2.004	2.732	2.113	2.736	2.225	2.401	2.003	2.483	2.047	2.490	334.97	15443.0
p2	p2	bE	cons.	2.291	2.764	2.004	2.952	2.121	2.951	2.301	2.538	2.002	2.786	2.027	2.789	797.44	15566.4
p2	p3	bE	cons.	2.291	2.764	2.004	2.950	2.121	2.949	2.301	2.538	2.002	2.789	2.027	2.793	1039.2	15681.6
p3	NO	bE	cons.	2.332	2.691	2.342	2.342	2.345	2.345	2.807	2.839	2.839	2.839	2.838	2.838	219.01	18806.1
p3	p1	bE	cons.	2.452	2.900	2.002	2.875	1.993	2.870	3.012	3.067	2.003	3.115	2.012	3.105	406.83	18853.4
p3	p2	bE	cons.	2.434	3.214	2.000	3.534	2.000	3.532	0.103	0.114	2.002	0.579	2.008	0.519	870.16	18967.8
p3	p3	bE	cons.	2.434	3.214	2.000	3.525	2.000	3.524	0.103	0.114	2.002	0.549	2.008	0.488	1056.1	19020.2
NO	NO	bE	lump.	2.226	2.226	2.233	2.233	2.233	2.233	2.895	2.895	2.926	2.926	2.926	2.926	107.15	13358.5
NO	p1	bE	lump.	2.224	2.224	2.000	2.061	2.009	2.065	2.884	2.884	2.003	2.045	2.036	2.045	209.88	13478.9
NO	p2	bE	lump.	2.208	2.208	1.999	2.017	2.004	2.019	2.799	2.799	2.001	2.014	2.017	2.014	601.04	13717.9
NO	p3	bE	lump.	2.208	2.208	1.999	2.021	2.004	2.023	2.799	2.799	2.001	2.017	2.018	2.017	812.63	13871.1
p1	NO	bE	lump.	1.914	2.151	1.622	1.622	1.632	1.632	2.621	2.183	2.382	2.382	2.388	2.388	197.04	14144.4
p1	p1	bE	lump.	1.914	2.151	1.998	2.158	1.844	2.165	2.621	2.183	2.008	2.320	2.262	2.320	347.92	14376.8
p1	p2	bE	lump.	1.914	2.151	1.999	2.184	1.954	2.182	2.620	2.183	2.002	2.164	2.109	2.160	883.53	14626.6
p1	p3	bE	lump.	1.914	2.151	1.999	2.198	1.954	2.197	2.620	2.183	2.002	2.180	2.109	2.176	1068.6	14744.5
p2	NO	bE	lump.	2.197	2.656	2.213	2.213	2.214	2.214	2.225	2.402	2.234	2.234	2.235	2.235	187.13	14782.8
p2	p1	bE	lump.	2.230	2.704	2.004	2.775	2.108	2.779	2.254	2.459	2.003	2.546	2.044	2.552	318.34	15059.9
p2	p2	bE	lump.	2.335	2.820	2.004	2.979	2.115	2.977	2.355	2.626	2.002	2.847	2.025	2.850	784.23	15302.7
p2	p3	bE	lump.	2.335	2.820	2.004	2.976	2.116	2.975	2.355	2.626	2.002	2.850	2.025	2.852	1003.6	15431.6
p3	NO	bE	lump.	2.535	3.005	2.554	2.554	2.557	2.557	3.520	3.545	3.591	3.591	3.590	3.590	216.69	17106.3
p3	p1	bE	lump.	2.577	3.171	2.002	3.148	1.990	3.145	2.411	2.613	2.003	2.546	2.007	2.578	386.57	17238.5
p3	p2	bE	lump.	1.984	2.690	2.000	3.053	1.998	3.042	1.092	1.046	2.002	1.068	2.006	1.053	845.14	17486.4
p3	p3	bE	lump.	1.984	2.690	2.000	3.040	1.998	3.030	1.092	1.046	2.002	1.061	2.006	1.046	1021.6	17605.4
NO	NO	cN	cons.	1.997	1.997	1.998	1.998	1.998	1.998	2.000	2.000	2.004	2.004	2.004	2.004	105.81	13380.8
NO	p1	cN	cons.	1.997	1.997	1.999	2.050	1.998	2.053	2.000	2.000	2.000	2.007	2.000	2.007	205.61	13497.8
NO	p2	cN	cons.	1.997	1.997	1.999	2.013	1.998	2.015	1.999	1.999	2.000	2.002	2.000	2.002	608.97	13748.6
NO	p3	cN	cons.	1.997	1.997	1.999	2.016	1.999	2.018	1.999	1.999	2.000	2.002	2.000	2.002	775.78	13893.8
p1	NO	cN	cons.	1.912	2.147	1.616	1.616	1.626	1.626	2.611	2.168	2.380	2.380	2.385	2.385	194.01	14537.5
p1	p1	cN	cons.	1.912	2.147	1.997	2.152	1.836	2.158	2.611	2.168	2.005	2.309	2.249	2.308	326.97	14580.5
p1	p2	cN	cons.	1.913	2.147	1.998	2.181	1.949	2.179	2.611	2.168	2.001	2.155	2.096	2.151	784.26	14810.2
p1	p3	cN	cons.	1.913	2.147	1.999	2.194	1.949	2.193	2.611	2.168	2.001	2.170	2.096	2.166	1057.0	14941.0
p2	NO	cN	cons.	2.128	2.568	2.142	2.142	2.143	2.143	2.020	2.183	2.025	2.025	2.026	2.026	183.76	15399.6
p2	p1	cN	cons.	2.150	2.609	2.003	2.689	2.102	2.694	2.023	2.214	2.001	2.304	2.011	2.312	320.67	15443.0
p2	p2	cN	cons.	2.214	2.709	2.003	2.924	2.114	2.923	2.032	2.308	2.000	2.644	2.007	2.650	773.14	15573.6
p2	p3	cN	cons.	2.214	2.709	2.004	2.923	2.114	2.922	2.032	2.308	2.000	2.650	2.007	2.656	972.07	15696.3
p3	NO	cN	cons.	2.125	2.485	2.132	2.132	2.135	2.135	2.013	2.050	2.024	2.024	2.024	2.024	214.78	18807.2
p3	p1	cN	cons.	2.187	2.637	2.001	2.598	1.989	2.596	2.020	2.073	2.000	2.068	1.999	2.066	391.42	18852.4
p3	p2	cN	cons.	3.539	3.993	1.999	4.003	1.997	4.010	4.323	4.279	2.000	4.195	2.001	4.196	798.27	18963.3
p3	p3	cN	cons.	3.539	3.993	1.999	4.000	1.998	4.008	4.323	4.279	2.000	4.199	2.001	4.201	1012.1	19020.8
NO	NO	cN	lump.	1.997	1.997	1.998	1.998	1.998	1.998	2.000	2.000	2.004	2.004	2.004	2.004	106.65	13377.9
NO	p1	cN	lump.	1.997	1.997	1.999	2.050	1.998	2.053	2.000	2.000	2.000	2.007	2.000	2.007	200.19	13499.9
NO	p2	cN	lump.	1.997	1.997	1.999	2.013	1.998	2.015	1.999	1.999	2.000	2.002	2.000	2.002	581.26	13745.1
NO	p3	cN	lump.	1.997	1.997	1.999	2.016	1.999	2.018	1.999	1.999	2.000	2.002	2.000	2.002	769.68	13901.8
p1	NO	cN	lump.	1.912	2.147	1.616	1.616	1.626	1.626	2.611	2.168	2.380	2.380	2.385	2.385	195.64	14188.6
p1	p1	cN	lump.	1.912	2.147	1.997	2.152	1.836	2.158	2.611	2.168	2.005	2.309	2.249	2.308	326.69	14467.2
p1	p2	cN	lump.	1.913	2.147	1.998	2.181	1.949	2.179	2.611	2.168	2.001	2.155	2.096	2.151	818.28	14718.8
p1	p3	cN	lump.	1.913	2.147	1.999	2.194	1.949	2.193	2.611	2.168	2.001	2.170	2.096	2.166	1063.2	14845.4
p2	NO	cN	lump.	2.137	2.608	2.153	2.153	2.154	2.154	2.021	2.214	2.026	2.026	2.027	2.027	181.87	14878.8
p2	p1	cN	lump.	2.164	2.653	2.003	2.730	2.098	2.734	2.025	2.253	2.000	2.353	2.010	2.361	311.84	15174.8
p2	p2	cN	lump.	2.244	2.762	2.003	2.952	2.109	2.951	2.036	2.374	2.000	2.710	2.006	2.715	749.34	15420.8
p2	p3	cN	lump.	2.244	2.762	2.003	2.950	2.110	2.950	2.036	2.374	2.000	2.715	2.006	2.720	969.82	15542.1
p3	NO	cN	lump.	2.187	2.667	2.201	2.201	2.204	2.204	2.021	2.081	2.042	2.042	2.041	2.041	213.69	17216.3
p3	p1	cN	lump.	2.315	2.899	2.001	2.866	1.986	2.859	2.041	2.131	2.000	2.123	1.999	2.118	385.77	17483.0
p3	p2	cN	lump.	2.369	3.091	1.999	3.441	1.996	3.435	2.031	2.018	2.000	2.166	2.000	2.138	802.98	17728.5
p3	p3	cN	lump.	2.369	3.091	1.999	3.431	1.996	3.424	2.031	2.018	2.000	2.153	2.000	2.126	994.03	17847.9



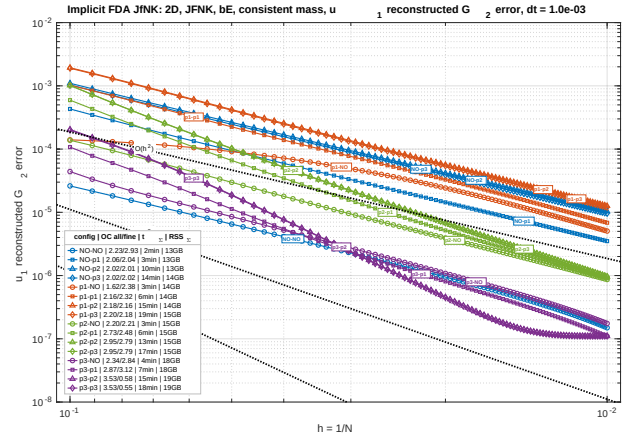
(a) $\|V_m\|_{L_2}$



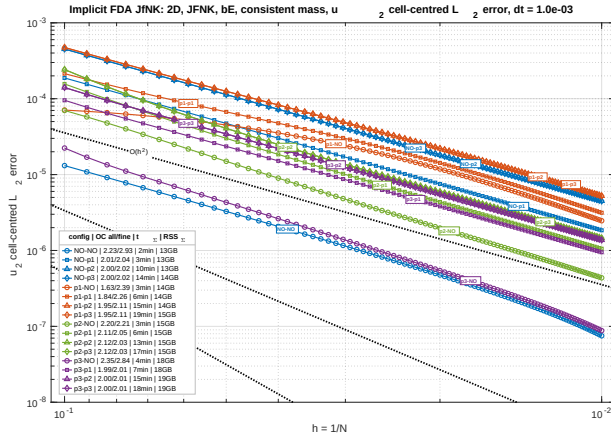
(b) $\|V_m\|_{G_2}$



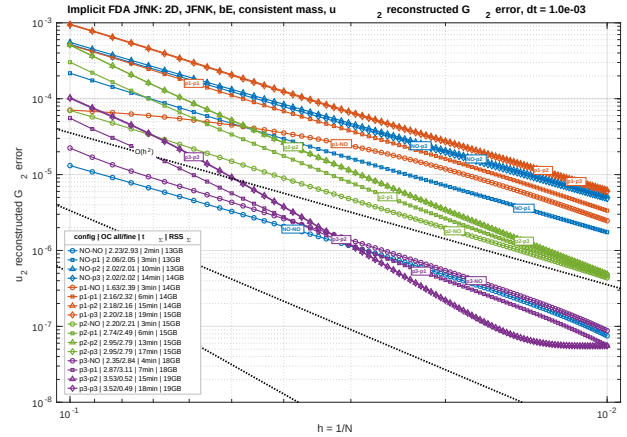
(c) $\|u_1\|_{L_2}$



(d) $\|u_1\|_{G_2}$

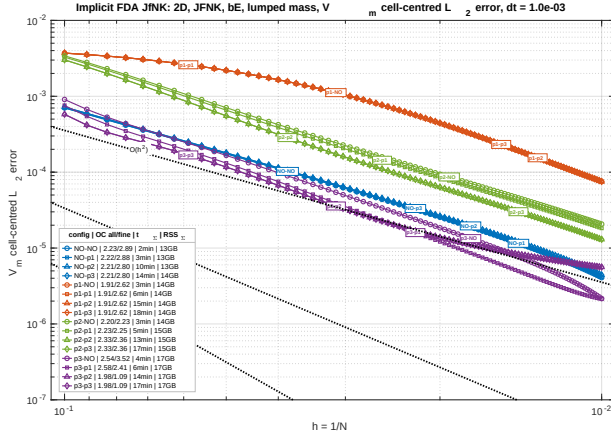


(e) $\|u_2\|_{L_2}$

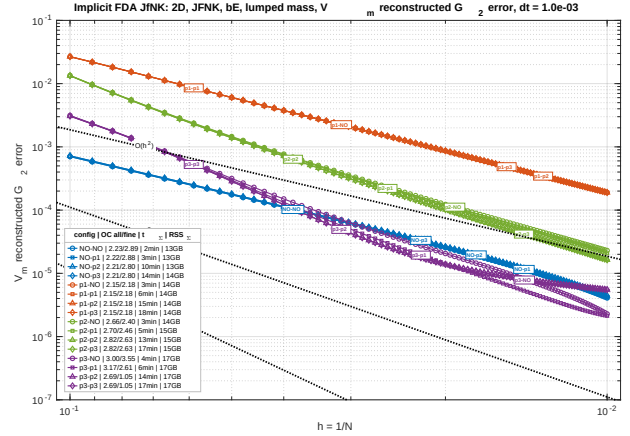


(f) $\|u_2\|_{G_2}$

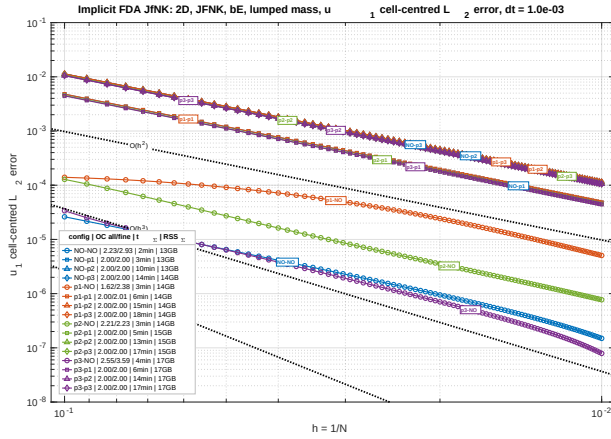
Figure 21: 2D convergence curves for JFNK, $\Delta t = 10^{-3}$, bE with cons. mass. The six panels correspond to the six order columns in Table 7 for this scheme/mass block. The plot legends report the two fitted orders and the accumulated runtime/memory cost for each curve.



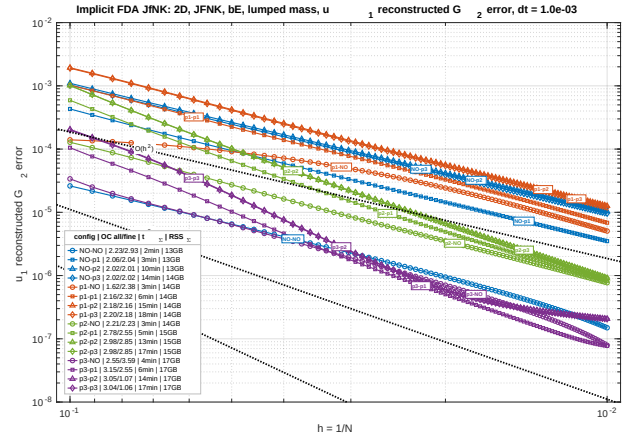
(a) $\|V_m\|_{L_2}$



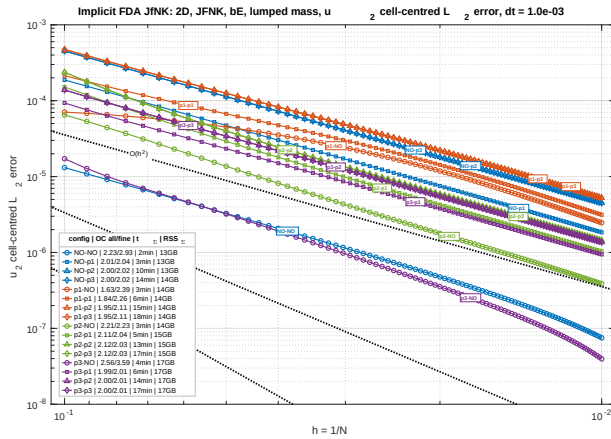
(b) $\|V_m\|_{G_2}$



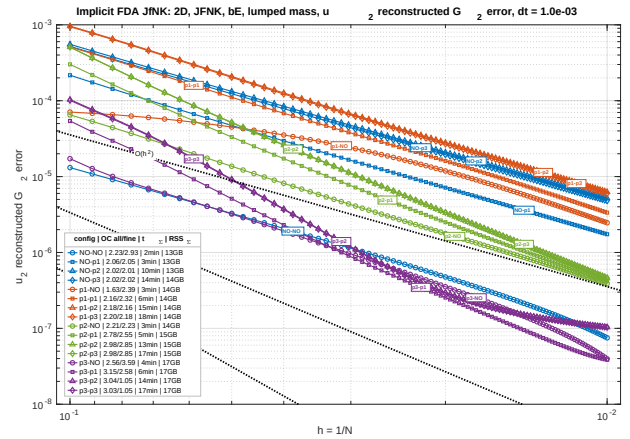
(c) $\|u_1\|_{L_2}$



(d) $\|u_1\|_{G_2}$

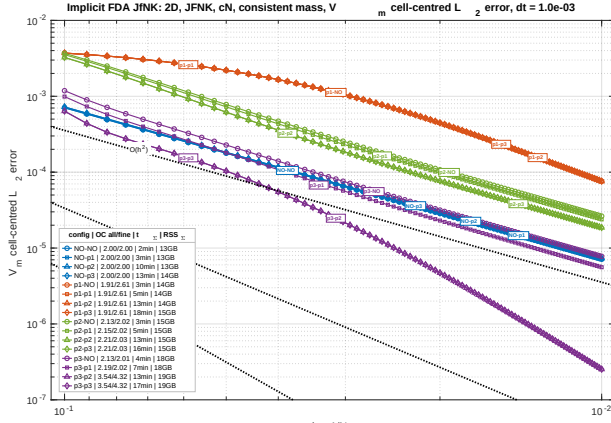


(e) $\|u_2\|_{L_2}$

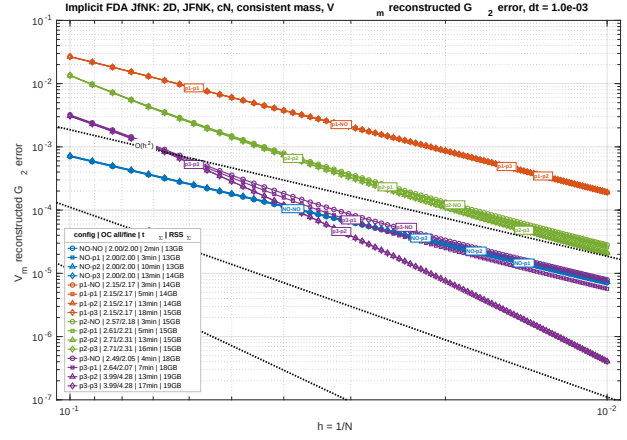


(f) $\|u_2\|_{G_2}$

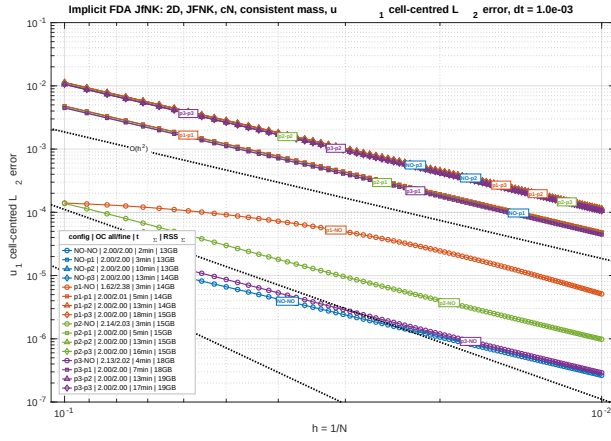
Figure 22: 2D convergence curves for JFNK, $\Delta t = 10^{-3}$, bE with lump. mass. The six panels correspond to the six order columns in Table 7 for this scheme/mass block. The plot legends report the two fitted orders and the accumulated runtime/memory cost for each curve.



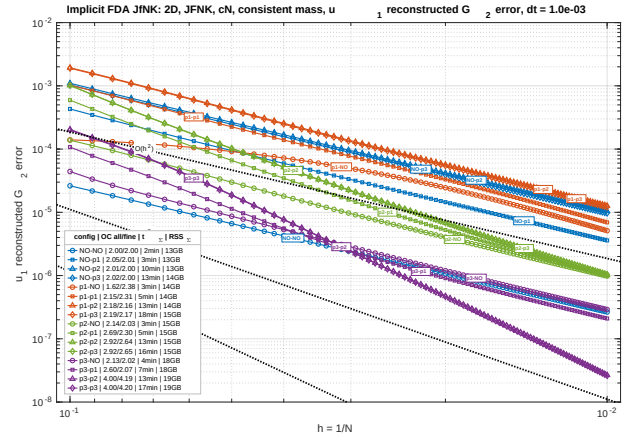
(a) $\|V_m\|_{L_2}$



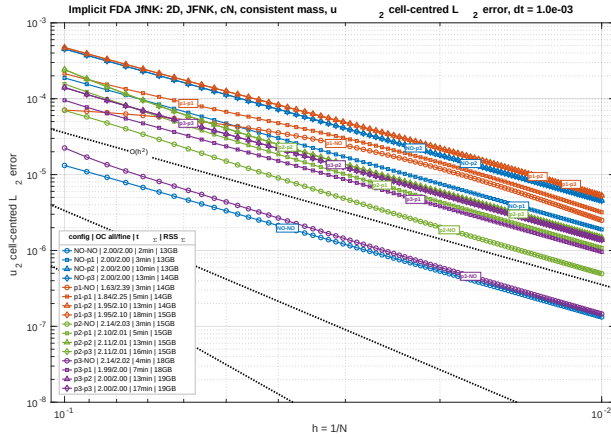
(b) $\|V_m\|_{G_2}$



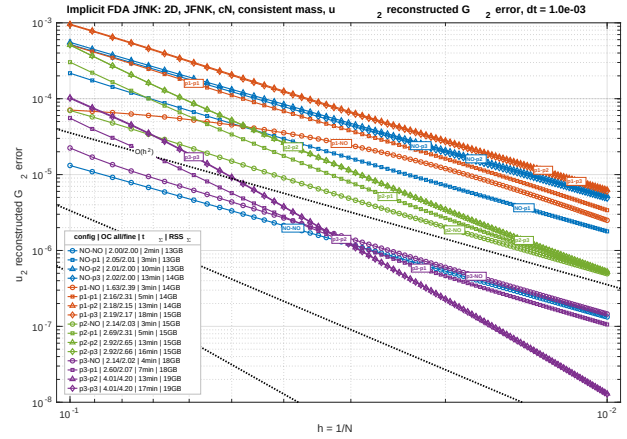
(c) $\|u_1\|_{L_2}$



(d) $\|u_1\|_{G_2}$

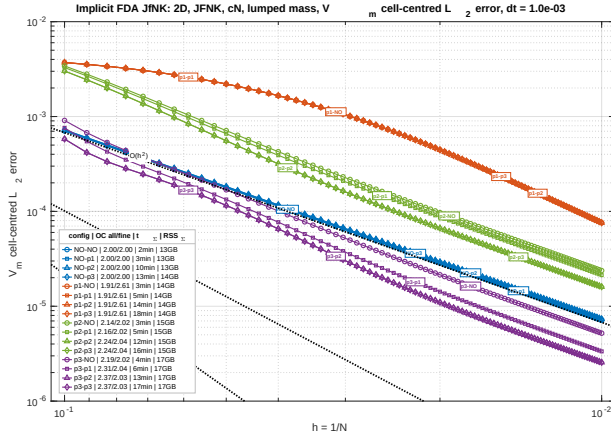


(e) $\|u_2\|_{L_2}$

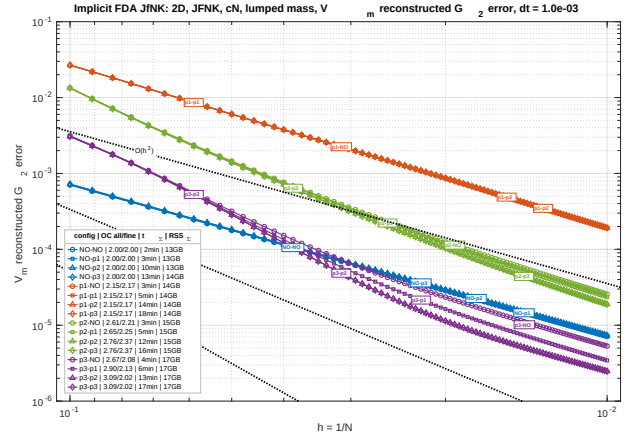


(f) $\|u_2\|_{G_2}$

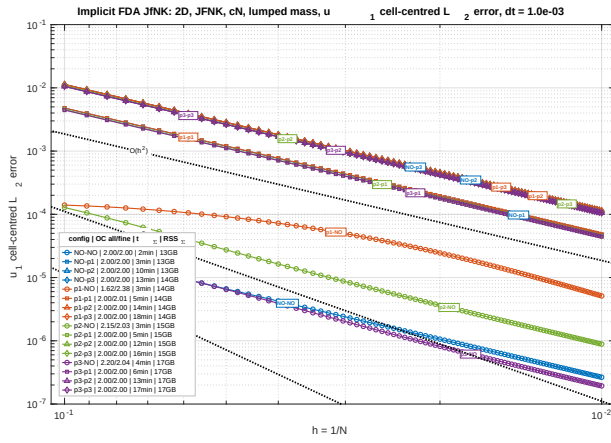
Figure 23: 2D convergence curves for JFNK, $\Delta t = 10^{-3}$, cN with cons. mass. The six panels correspond to the six order columns in Table 7 for this scheme/mass block. The plot legends report the two fitted orders and the accumulated runtime/memory cost for each curve.



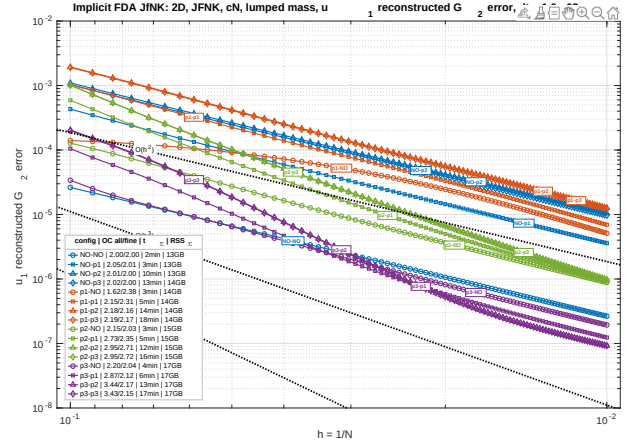
(a) $\|V_m\|_{L_2}$



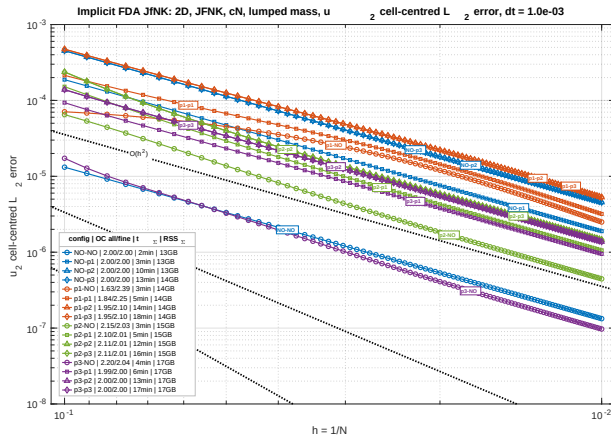
(b) $\|V_m\|_{G_2}$



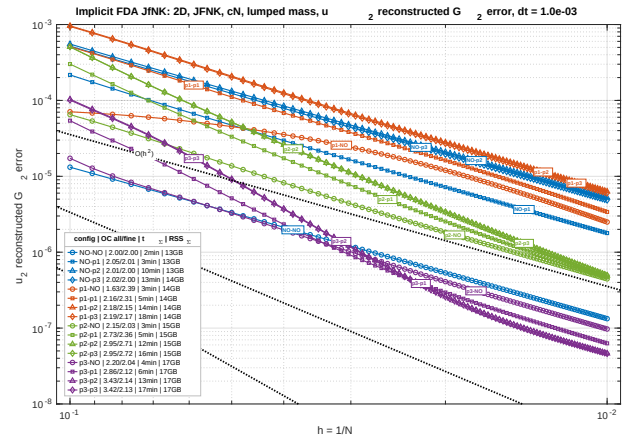
(c) $\|u_1\|_{L_2}$



(d) $\|u_1\|_{G_2}$



(e) $\|u_2\|_{L_2}$



(f) $\|u_2\|_{G_2}$

Figure 24: 2D convergence curves for JFNK, $\Delta t = 10^{-3}$, cN with lump. mass. The six panels correspond to the six order columns in Table 7 for this scheme/mass block. The plot legends report the two fitted orders and the accumulated runtime/memory cost for each curve.

10.2 Two-dimensional results: conclusions

The three nonlinear strategies give almost identical convergence orders for a fixed spatial, temporal, and mass-matrix configuration. Therefore, in this MMS case the main distinction between `Picard`, `diagonalIion`, and `JFNK` is computational cost rather than asymptotic accuracy. The clearest spatial orders are obtained with `crankNicolson`, `consistent` mass, and $\Delta t = 10^{-3}$; for $\Delta t = 10^{-2}$ the finest-mesh slopes of the high-order state diagnostics are still visibly affected by time error.

Picard. `Picard` is the most direct baseline and reproduces the expected high-order trends when the rest of the discretisation is chosen appropriately. For example, with `p3` voltage reconstruction, `p2` source/state quadrature, `crankNicolson`, `consistent` mass, and $\Delta t = 10^{-3}$, the fitted orders are 3.538 for the cell-centred V_m , 3.992 for V_m^G , and about 4.0 for the reconstructed state errors u_1^G and u_2^G (Table 3). Increasing the source/state level from `p2` to `p3` changes these orders negligibly but increases the runtime. The disadvantage is efficiency: the same `p3/p2`, `crankNicolson`, `consistent`, $\Delta t = 10^{-3}$ sweep takes 1730.1 s with `Picard`, compared with 798.27 s with `JFNK`.

diagonalIion. `diagonalIion` produces essentially the same orders as `Picard`. Its diagonal treatment of the local ionic-current sensitivity can help the nonlinear voltage equation behave more implicitly, but for the present manufactured problem this does not translate into a systematic gain in either accuracy or runtime. In the same `p3/p2`, `crankNicolson`, `consistent`, $\Delta t = 10^{-3}$ case, the orders again are 3.538 for V_m , 3.992 for V_m^G , and approximately 4.0 for u_1^G, u_2^G , while the accumulated runtime is 1661.8 s (Table 5). Thus, for these 2D MMS runs, `diagonalIion` is mainly a useful robustness option, not the best cost–accuracy choice.

JFNK. `JFNK` gives the same convergence orders as the other methods but is the most efficient nonlinear coupling in these results. For the recommended `p3/p2`, `crankNicolson`, `consistent`, $\Delta t = 10^{-3}$ configuration, it obtains 3.539 for V_m , 3.993 for V_m^G , 4.003 for u_1^G , and 4.010 for u_2^G over $N = 10$ –100; over $N = 70$ –100, the reconstructed diagnostics remain close to fourth order (Table 7). The accumulated runtime is 798.27 s, less than half the corresponding `Picard` and `diagonalIion` runs, while the summed peak-RSS cost is comparable. This makes `JFNK` the best nonlinear method for the 2D manufactured study.

Temporal scheme. For high-order spatial verification, the second-order `crankNicolson` scheme is preferable. The first-order time error of `backwardEuler` masks the high-order spatial behaviour on the fine meshes; for example, the `p3/p2`, `consistent`, $\Delta t = 10^{-3}$ `backwardEuler` case has good full-range reconstructed slopes but the $N = 70$ –100 slopes of V_m^G , u_1^G , and u_2^G collapse to values below one in Table 7. With `crankNicolson`, the same spatial configuration recovers fourth-order reconstructed behaviour. `backwardEuler` is useful as a robustness or comparison run, but it should not be used as the primary evidence for high-order spatial convergence.

Mass matrix. The `consistent` mass matrix is the better choice for the high-order configurations. The effect is especially clear for `p3`: with `JFNK`, `crankNicolson`, $\Delta t = 10^{-3}$, and `p3/p2`, the `consistent` mass case gives full-range orders 3.539, 3.993, 4.003, and 4.010 for V_m , V_m^G , u_1^G , and u_2^G , whereas the `lumped` case gives 2.369, 3.091, 3.441, and 3.435 (Table 7). The `lumped` matrix remains adequate for low-order or second-order comparisons, but it degrades the high-order `p3` regime.

Matching the voltage and source/state orders. The source/state quadrature should usually be matched to the voltage reconstruction level, but the data suggest one important practical refinement:

- `N0` voltage should be paired with `N0` source/states. Higher source quadrature does not improve the second-order voltage result and only adds cost.
- `p1` voltage should be paired with at least `p1` source/states. The `p1/N0` cases under-resolve the state coupling, whereas `p1/p1` gives the expected second-order reconstructed state behaviour.

- p2 voltage should be paired with p2 source/states. Although the formal interpolation expectation for p2 is third order, the cell-centred finite-volume unknown often converges with order near two or slightly above two; in these data the reconstructed Gauss-point diagnostics and source/state quantities show the clearer benefit of the quadratic reconstruction.
- p3 voltage should be paired with p2 source/states for the best cost–accuracy balance. The p3/p3 option is more symmetric and remains a conservative verification choice, but in the present 2D results it gives essentially the same orders as p3/p2 at higher runtime.

Recommended 2D configuration. Considering both convergence and efficiency, the best overall configuration in the current 2D MMS data is

$$\boxed{\text{JFNK} + \text{crankNicolson} + \text{consistent mass} + \Delta t = 10^{-3} + V_m : \text{p3} + I_{\text{ion}}/\mathbf{u} : \text{p2}}$$

This choice reaches fourth-order behaviour in the reconstructed voltage and state diagnostics, keeps the cell-centred voltage close to fourth order on the fine meshes, and is substantially faster than the equivalent Picard and `diagonalIion` runs. For a cheaper second-order study, the consistent recommendation is JFNK with `crankNicolson` and matched N0/N0 or p1/p1 reconstruction, depending on whether only the standard finite-volume method or the first high-order reconstruction level is being tested.

11 Notes

Backward Euler is first-order in time and Crank–Nicolson is second-order for the smooth manufactured problem. Comparing the same spatial configuration across the bE/cN figures indicates whether the spatial error or time discretization dominates. The mass comparison isolates the effect of replacing the diagonal/lumped finite-volume mass by the consistent high-order mass matrix in the implicit potential solve. The reconstructed quantities V_m^G , u_1^G , and u_2^G should be read as quadrature-point/reconstructed diagnostics; the cell-centred fields are the values used in the main finite-volume update and in the L_2 convergence table.